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North Carolina Department of Transportation
Division of Highways
Statewide Planning Branch
Systems Planning Unit

OUTER BANKS THOROUGHFARE PLAN

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OUTER BANKS THOROUGHFARE PLAN

Prepared by:

Statewide Planning Branch
Division of Highways
North Carolina Department of Transportation

In cooperation with:

Town of Kill Devil Hills
Town of Kitty Hawk
Town of Nags Head
Town of Southern Shores
Dare County
Federal Highway Administration

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TECHNICAL WORDS AND PHRASES

ADT - Average daily traffic. Unless otherwise noted, in this report, ADT would refer to the *average summer weekday* traffic.

Capacity - The formal definition of capacity is given on page V-1. Maximum capacity occurs at LOS E.

Dwelling Unit - A dwelling unit is a place where one can live. Examples of a single dwelling unit include: a single family house, one hotel room, one apartment in an apartment building, or one half of a duplex.

DU - Dwelling unit.

Level of Service - The relationship of traffic volumes to the capacity of the roadway will determine the level of service. It is a rating system for such a relationship, similar to a report card. The table on page V-2 defines the various levels of service A through F.

LOS - Level of Service

NCDOT - North Carolina Department of Transportation

Population - Unless otherwise stated, the term "population" in this report refers to the *average summer weekday* population.

Practical Capacity - This is the capacity at a LOS D.

TIP - Transportation Improvement Program. A document produced yearly by the NCDOT which reflects the spending schedule for Transportation projects.

vpd - vehicles per day, unless otherwise stated, the term "vpd" in this report refers to the *average summer weekday* vpd.

I. INTRODUCTION

Transportation system efficiency, today more than ever, plays a vital role in our economy and way of life. It is used every day by the business, industrial and residential sectors. To insure that individual transportation projects eventually form a cohesive, co-ordinated system, a **thoroughfare plan** is developed.

Recognizing the importance of such a system, the towns of Nags Head and Kill Devil Hills have used a Thoroughfare Plan to assist in planning transportation improvements for many years. In 1972 a Thoroughfare Plan was prepared for the area, and was mutually adopted by Nags Head, Kill Devil Hills, and the North Carolina Department of Transportation.

The purpose of this study was to re-examine the present and future transportation needs of the area, and from this derive a revised Thoroughfare Plan. The system of thoroughfares proposed was developed following the principles of thoroughfare planning outlined in Chapter II of this report.

The recommended cross-sections are based on existing conditions and the expected volume of traffic in the design year. Before a project is implemented a more detailed evaluation will be performed. Every effort was made to use as much of the existing street system as possible in order to minimize cost and environmental disruption. Location of new facilities was based on field investigation, existing land use, and topographic conditions.

Initiative for plan implementation will rest largely with the policy boards and citizens of the area. The scope of transportation needs throughout the State greatly outweigh the available funding. It is, therefore, necessary that the local areas aggressively pursue funding for desired projects.

Responsibility for the proposed construction must be shared by Kill Devil Hills, Kitty Hawk, Nags Head, and Southern Shores, and the N. C. Division of Highways. With the different governmental agencies involved in providing the elements of the plan, coordination of activities is of prime importance. The plan is formally adopted by both the local governing bodies and the North Carolina Board of Transportation, to serve as a mutual official guide in providing a well coordinated, adequate, and economical major transportation system. In order for the plan to be effective, the Towns and the State must procure in advance or protect by various legal controls the right-of-ways necessary for the improvements which will ultimately be required.

It must be emphasized that the Thoroughfare Plan was based on anticipated growth of the urban area, as provided by Kill Devil Hills, Kitty Hawk, Nags Head, and Southern Shores. Actual growth rates and patterns may differ from those anticipated and it may become necessary to accelerate or retard the development of thoroughfares or to make revisions in the proposed plan. It is desirable to review the plan in detail approximately every ten to fifteen years to adjust the thoroughfare system to the actual rate and type of development.

II THOROUGHFARE PLANNING

Basic Principles

The urban street system typically occupies 25 to 30 percent of the total developed land in the urban area. Since the system is permanent and expensive to build and maintain, much care and foresight are needed in its development. Thoroughfare planning is the process used by public officials to assure the development of the most logical and appropriate street system to meet future travel desires. The major steps involved in the thoroughfare planning process are:

1. *Collection of data concerning existing physical development and travel desires within the area.*
2. *Development of a (computer) model which reflects present travel desires.*
3. *Prediction of future socioeconomic data, and computation of future travel desires using the computer model.*
4. *Evaluation of the adequacy of the existing transportation system in serving present and future travel.*
5. *Formulation of the best thoroughfare plan, on the basis of travel demand, economic benefits, and environmental considerations, to meet future travel desires.*
6. *Development of construction priorities for plan implementation.*
7. *Implementation of the plan.*

Purpose of Planning

There are many benefits to be gained from thoroughfare planning, but the primary objective is to assure that the street system will be progressively developed in such a manner as to adequately serve future travel desires. Thus, the cardinal concept of thoroughfare planning is that provisions be made for street and highway improvements so that as needs arise, feasible opportunities to make improvements exist.

Some of the benefits derived from thoroughfare planning are:

(1) Each street can be designed to perform a specific function. This permits savings in right-of-way and construction costs; and encourages stability in travel and land use patterns. (2) Local officials and citizens are informed as to future improvements. Public facilities can be better located; and damage to property and appearance can be minimized (for example: buildings and plants can be located to permit future street widening). (3) Residents will know which streets will be developed as major thoroughfares and be able to make an informed decision when choosing a home. (4) City officials will know when improvements will be needed and can schedule funds accordingly.

Efficiency

The improvement of the efficiency of existing facilities can be achieved through: (1) improving the **system efficiency**; and (2) improving the **operational efficiency** of thoroughfares.

(1) System Efficiency - A more efficient system can reduce travel distances, time, and cost. Improvements in system efficiency can be achieved through the concept of classification of streets and development of a coordinated major street system.

Classification - There are many different ways in which to classify a street system. The Thoroughfare Plan uses a system which reflects that streets perform two primary functions -they provide **traffic service** and **land service**. **These two functions are basically incompatible**. The conflict is not serious if both traffic and land service demands are low, but when traffic volumes are high, conflicts created by intense land service demands result in intolerable congestion.

The Thoroughfare Plan provides a classification system of streets which permits travel with directness, ease, and safety. Different streets in the system are designed to perform specific functions thus minimizing the traffic and land service conflict. On the Thoroughfare Plan streets are functionally categorized as: local access streets, minor thoroughfares or as major thoroughfares.

"Local Access Streets" provide access to abutting property. They are not intended to carry heavy volumes of traffic and should be located such that only traffic with origins or destinations on the streets would be served. **THEIR FUNCTION IS TO PROVIDE ACCESS**. Local access streets may serve residential, commercial, and/or industrial areas.

"Minor Thoroughfares" are more important streets in the city system. They collect traffic from local access streets and carry it to the major thoroughfare system. They may supplement the major thoroughfare system by aiding minor through movements. A third function which may be performed is that of providing access to abutting property. They should be designed to serve limited areas so that their development as major thoroughfares will be prevented.

"Major Thoroughfares" are the primary traffic arteries of the city. Their function is to move intra-city and inter-city traffic. Although undesirable, the streets which comprise the major thoroughfare system may also serve abutting property; however, **THEIR MAJOR FUNCTION IS TO CARRY TRAFFIC**. They should not be bordered by strip development as it significantly lowers the capacity of the thoroughfare, and each driveway is a danger to traffic flow. Major thoroughfares may range from two lane streets to multi-lane expressways. As a general rule, parking should not be permitted on major thoroughfares.

Figure 1 graphically illustrates the trade-off between the travel service and land service functions of the above types of facilities. Having the proper mix of these different classes improves the efficiency of the street system.

Some of the other ways of classifying street systems include: by funding source; as radial, circumferential or cross-town; by cross-section; and the federal functional classification system of arterials and collectors.

(2) Operational Efficiency - A street's operational efficiency is improved by increasing the capability of the street to carry vehicular traffic and people. In terms of vehicular traffic, a street's **capacity** is defined as "the maximum number of vehicles which can pass a given point on a roadway during a given time period under prevailing roadway and traffic conditions." Capacity is affected by the physical features of the roadway, nature of traffic, and weather. The ability to improve an area's operational efficiency depends on the development patterns established.

Physical ways to improve vehicular capacity include street widening, intersection improvements, improving the vertical and horizontal alignment, and eliminating road-side parking.

Other operational ways to improve street capacity include:

- (1) **Control of access** - A roadway with complete access control can carry over two times the traffic handled by a non-controlled access street.
- (2) **One-way operation** - The capacity of a street can be increased 50% or more, depending upon turning movements and overall street width, by initiating one-way traffic operations. One-way streets can also improve traffic flow by decreasing potential traffic conflicts and simplifying traffic signal coordination.
- (3) **Reversible lanes** - Reversible traffic lanes may be used to increase street capacity in situations where heavy directional flows occur during peak periods.
- (4) **Signal phasing and coordination** - Uncoordinated signals and poor signal phasing restrict traffic flow by creating excessive stop-and-go operation. This is of great importance in the study area.

Altering travel demand is a third way to improve the efficiency of existing streets. Although of limited use in the study area, travel demand can be reduced or altered in the following ways:

- (1) Encourage people to form carpools and vanpools for work and other trips. This reduces the number of vehicles on the roadway while increasing the people carrying capability of the street system.
- (2) Encourage the use of mass transit, bicycles, and pedestrian travel.
- (3) Encourage industries and business to stagger work hours or establish variable work hours for employees. This will reduce travel demand in peak periods and spread peak travel over a longer time period.

Application of Thoroughfare Planning Principles

The above descriptions are of an idealized major thoroughfare system. In actual practice, thoroughfare planning is done for established areas and is constrained by existing land use and street patterns, public attitudes, and expectations of future land use.

BASIC ROLES OF ROADWAYS

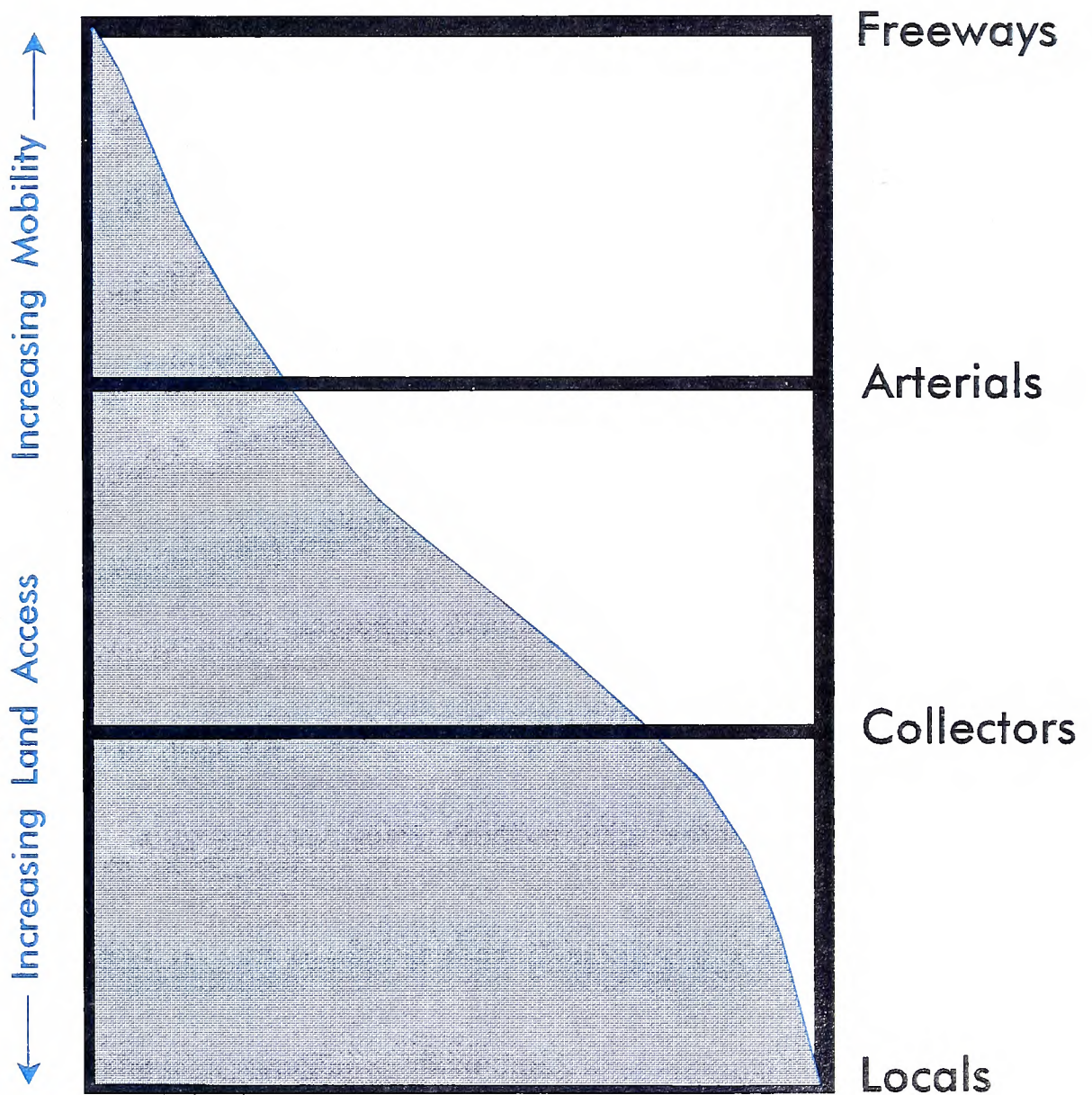


FIGURE 1

III EXISTING AND PROJECTED CONDITIONS

The Planning Area - Past and Present

Dare County is a county of history. The Lost Colony is memorialized in the outdoor theater of the same name on Roanoke Island. The Wright brothers first visited Kill Devil Hills in 1900 and on their fourth trip, in December of 1903, the historic 12 second flight was made. See **Figure 2**, location map.

The Planning Area (**Figure 3**) consists of the urban areas of Kitty Hawk, Kill Devil Hills, Nags Head, Southern Shores, and the unincorporated areas of Duck and Sanderling - all in Dare County.

Principal roads in the Planning Area include US 158, NC 12 and Virginia Dare Trail, which run parallel to the coastline. Access to the area is by means of one of three bridges: the Bonner Bridge which connects Nags Head to Hatteras Island, the Washington Baum Bridge, which ties Nags Head to Roanoke Island (Roanoke Island is then connected to the mainland by means of the William B. Umstead Memorial Bridge), and the Wright Memorial Bridge which comes from the mainland and continues as US 158, providing the boundary between Southern Shores and Kitty Hawk.

Factors Affecting Transportation

In order to formulate an adequate year 2010 thoroughfare plan, reliable forecasts of future travel characteristics must be achieved. The factors of population, vehicle usage trends, economy and land use play a significant role in determining the transportation needs of an area, and must be carefully analyzed. Additional items may include the effects of legal controls such as subdivision regulations and zoning ordinances, availability of public utilities, and physical features of the area. Examination of these factors helps to explain historic travel patterns and lays the ground work for thoroughfare planning.

The first step in the development of the thoroughfare plan is to define the planning period and the planning area. The planning period is typically on the order of 20 years. The base year for the Outer Banks study was 1988, and the year 2010 was chosen to be the endpoint of the study period (22 years). Based on the perception of local planning staff, it was felt that the amount of developable land in the planning area would reach capacity within or prior to a twenty year design period. It was therefore necessary to include two growth scenarios for the planning area. The first scenario was simply labeled "Buildout". Basically, this concept reflects the average growth rate at which development has occurred in the past. The other included more intense development and some renewal at higher densities, labeled "Redevelopment". The planning area is generally chosen to be the limits to which some urbanization is expected to occur during the planning period.

The planning area is then subdivided into zones. **Figure 4** shows the planning area boundary and zones.

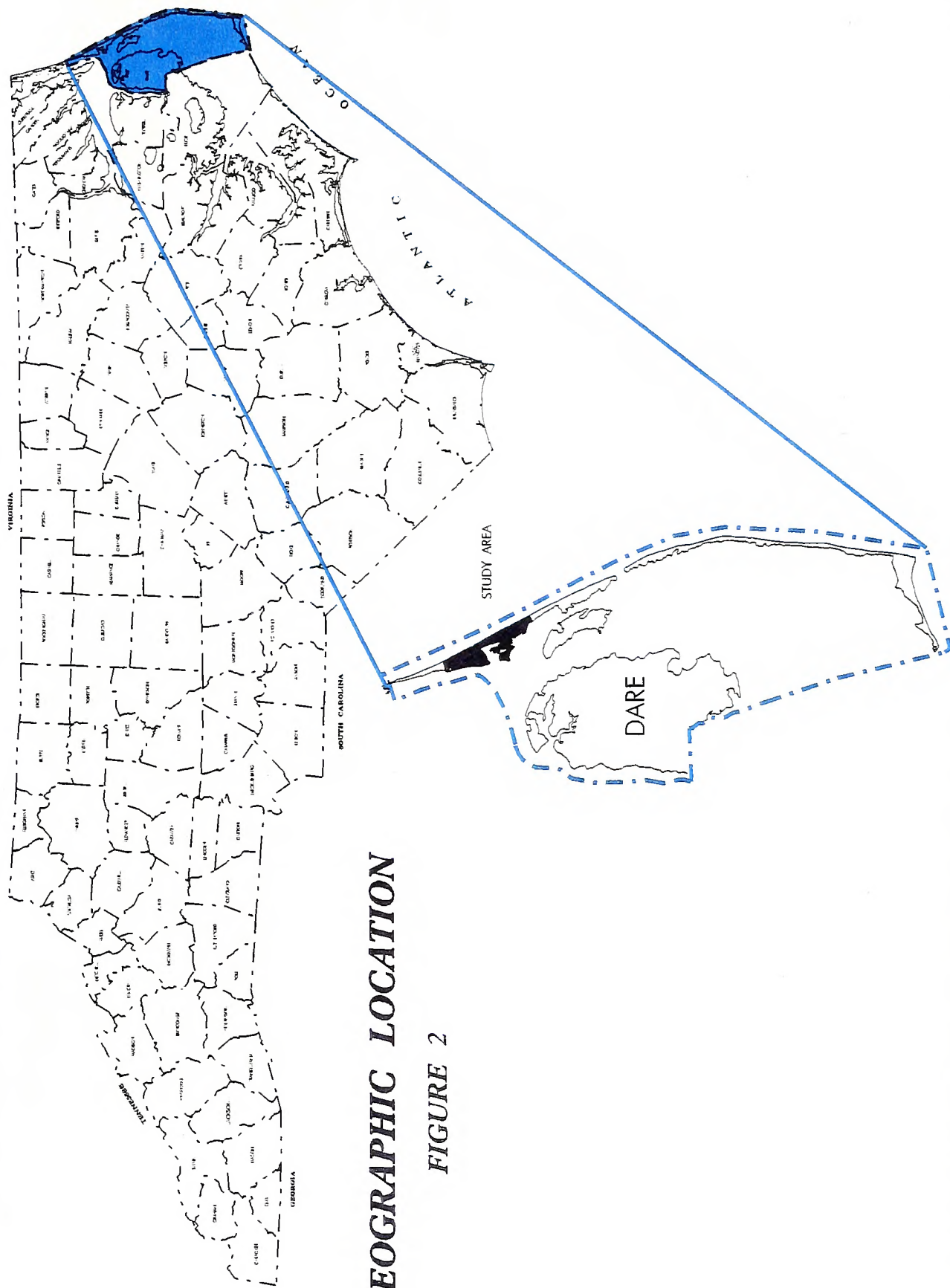
Population

Travel is directly related to population, and the volume of traffic is closely related to the size and distribution of the population which it serves. In the planning area, the population varies considerably with the season. The summer population is much higher than the permanent population. **Table 1** shows population totals and trends for the planning area. For this study, **average summer weekday** population and travel was used. The term "population" throughout this report, shall be interpreted as average summer weekday population.

The population estimate used for development of the thoroughfare plan is that of the planning area. It was necessary to make individual dwelling unit (DU) counts and use an estimated number of persons per DU as provided by local planning staff to calculate the 1988 planning area population. The occupancy rates used for the three housing classifications were as follows: "domicile" (7.3), "hotel/motel" (3.5), and for "campgrounds" (3.8). **Table 2** shows the historical "persons per DU" data for the area.

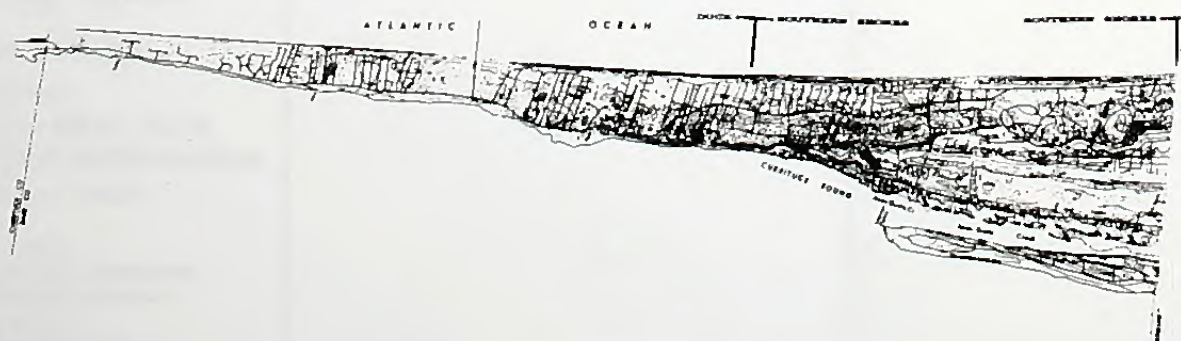
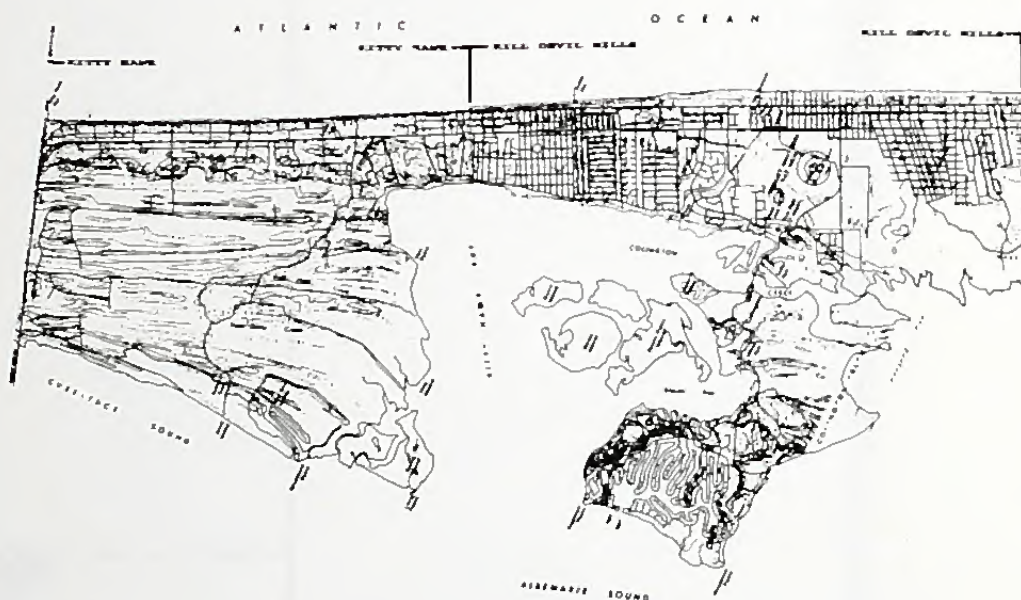
The estimated summer population of the Planning Area for 1988 was 131,818 using this technique. **Appendix A, Table A1** shows dwelling units by zone and classification, and the associated population data and calculations using figures provided by the Town of Nags Head.

Population and employment projections were provided by the Towns of Kitty Hawk, Kill Devil Hills, Nags Head, and Southern Shores. Other studies, such as the electrical company's projections, the Dare County Thoroughfare Plan report, the Long Range Plan for the Community of Southern Shores, and the Carrying Capacity report have also been utilized. Again, the population was calculated based on an average "persons per DU", based on the type of dwelling unit.



GEOGRAPHIC LOCATION

FIGURE 2



PLANNING AREA

TOWNS OF
NAGS HEAD, KILL DEVIL HILLS,
KITTY HAWK AND SOUTHERN SHORES
DARE COUNTY, NORTH CAROLINA

PREPARED BY THE
NORTH CAROLINA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS-STATEWIDE PLANNING BRANCH

IN COOPERATION WITH
U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION



AUGUST 10, 1962

FIGURE 3

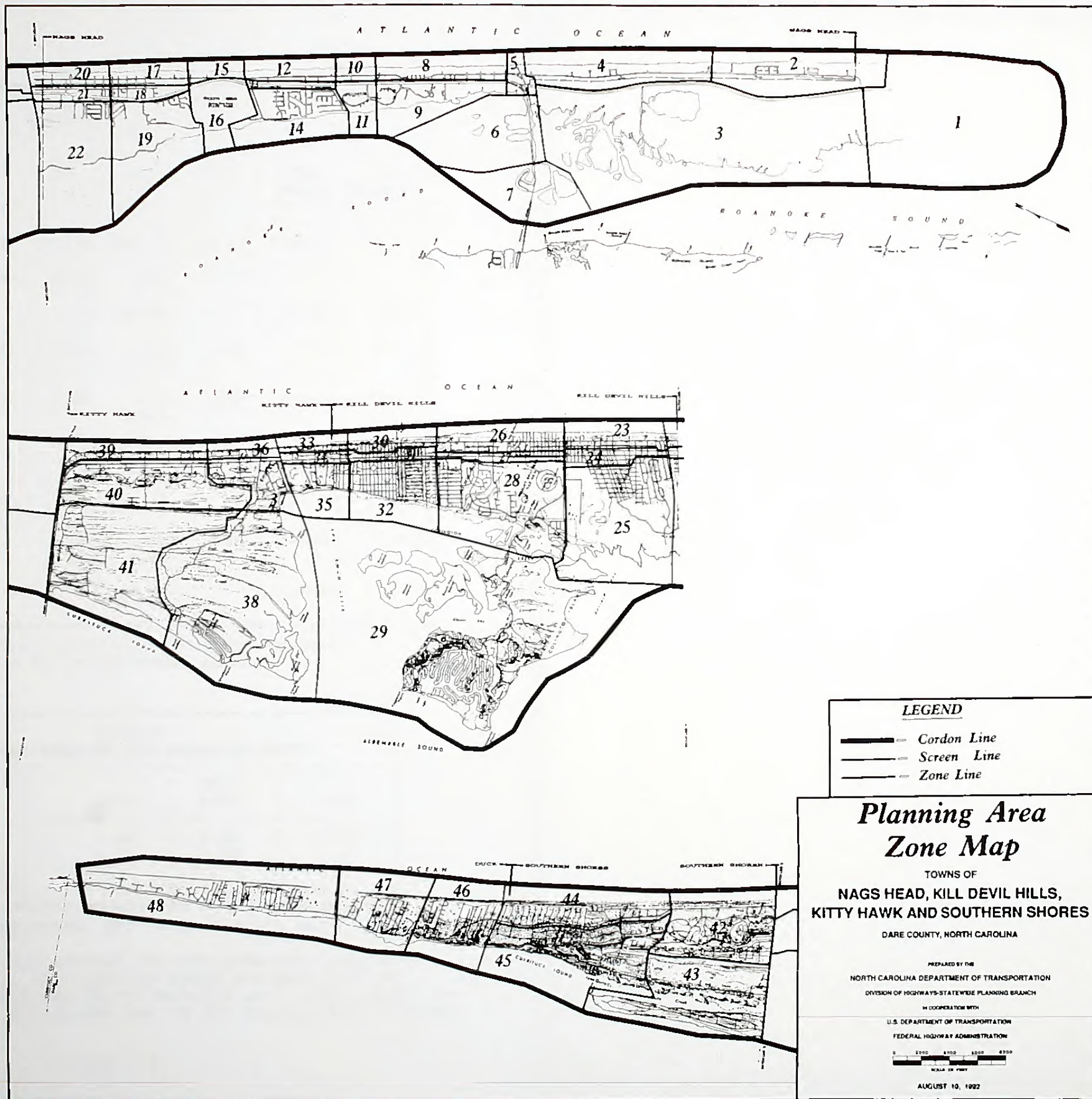


FIGURE 4

**TABLE 1
POPULATION**

LOCATION	YEAR						
	1960	1970	1980	1988	1990	2010 Buildout	2010 Redeveloped
North Carolina	4,556,155	5,082,059	5,881,766	6,487,438	6,628,637	7,776,000	7,776,000
Dare County	5,935	6,995	13,377	21,460	22,746	-	-
Planning Area	-	-	-	131,818*	-	237,426*	254,815*
Nags Head	-	414	1,020	1,811 24,082*	1,838	55,223*	59,510*
Kill Devil Hills	268	357	1,796	3,635 57,063*	4,238	71,370*	76,612*
Kitty Hawk	-	-	-	1,270 14,894*	1,937	42,184*	44,993*
Colington	-	-	-	13,181*	-	28,285*	30,260*
Southern Shores	-	-	395	1,218 9,826*	1,447	20,016*	22,084*
Duck/Sanderling	-	-	-	12,772*	-	20,348*	21,356*

NOTE: Historic populations are from Census data.

* Data is for average summer weekday.

**TABLE 2
AVERAGE PERSONS PER DWELLING UNIT**

	YEAR		
	1960	1970	1980
North Carolina	3.66	3.24	2.78
Dare County	3.17	2.80	2.48
Kill Devil Hills	-	-	2.38
Nags Head	-	-	2.34

permanent population, ref: U.S. Census data

Table 2 indicates the average for the permanent population. As indicated in Table 2, the historic trend has been for the average number of persons per DU to decrease over time. Based on this trend, the estimated persons per DU for the Buildout scenario

was reduced for the design year. The 2010 Buildout population used was 237,426. Since our initial data collection, the planning area has experienced some areas of redevelopment. It is expected that additional redevelopment will continue to occur in select locations of the planning area. Given this, the 2010 Redevelopment population used was 254,815. (The calculations for the year 2010 are shown in Appendix A, Tables A3 and A5.)

TABLE 3 STUDY AVERAGE PERSONS PER DU (summer weekday)			
Dwelling Unit Type	1988	YEAR 2010 BUILDOUT	2010 REDEVELOP
Domicile	7.3	7.2	7.2
Hotel / Motel	3.5	3.4	3.4
Campground	3.8	3.7	3.7

To calculate the rate of growth, the following formula (the simple interest formula) was used:

$$F = B (1+r)^n$$

where F = future year population

B = base year population

r = rate of growth

n = number of years (2010 - 1988 = 22)

Substituting in the known values and solving for "r", the rate of growth over the planning period is + 2.71% per year for 2010 Buildout, and + 3.04% for 2010 Redevelopment. To understand this magnitude of growth, the average population growth rate for North Carolina is approximately 1.0%.

Table 4 shows the growth rate per year, for the State, the planning area, and the local municipalities. **Figure 5** is a graphical representation of the data in Table 4.

TABLE 4
SELECTED GROWTH RATES
(percent per year)

<u>Location</u>	<u>Time Period</u>				
	1960-70	1970-80	1980-90	1988-2010 Bld'out	1988-2010 Redevelop
North Carolina	1.10	1.48	1.20	0.83	0.83
Dare County	1.66	6.70	5.45	-	-
Planning Area	-	-	-	2.89	3.21
Kill Devil Hills	2.91	17.53	8.97	1.66	1.99
Kitty Hawk	-	-	-	4.67	4.90
Nags Head	-	9.44	6.07	3.59	3.98
Southern Shores	-	-	13.86	3.22	3.80
Duck	-	-	-	1.96	2.11

Economy and Employment

The traffic patterns of an area is to a great extent reflected in the economic activity of that area. A field survey was taken of the places of employment and number of employees. These data were classified both by zone and by type of employment. The employment types used were: retail, highway retail, office, services, and industrial. The survey indicated a total of 9,311 persons employed within the planning area. To calculate the employment-to-population (E/P) ratio, only the "permanent" population should be used. To estimate the "permanent" population, only the "Domicile" category was used, and it was acknowledged that this was still in excess of the permanent population. Domicile population was 115,968, resulting in an E/P ratio was 8.0%. This extremely low ratio can be largely attributed to the fact that a large portion of the "domicile" population is in fact tourist population, and thus not employed.

Local staff provided information on the expected allocation of employment within the area and increases in employment activity. For the design year 2010 Buildout, total employment in the planning area was expected to increase to 27,995, yielding E/P ratio (using a "domicile" population of 218,052) of 12.8%. For 2010 Redevelopment, a total employment of 37,260, yielded an E/P ratio (using a "domicile" population of 231,804) of 16.1%.

Table 5 gives a brief summary of the employment information used in the study. The Standard Industrial Classification codes corresponding to each Employment Classification category are: Retail-Wholesale: 50-54, 56, 57, 59; Highway Retail (restaurants, gas stations): 55, 58; Office: 60-67, 91-97; Services: 70-76, 78-89, 99; and Industrial: 1-49. A more detailed description of the employment distribution is given in Appendix A, Table A2 (for the base year, 1988), Table A4 (for the design year, 2010 Buildout), and Table A6 (for the design year, 2010 Redevelopment).

TABLE 5 EMPLOYMENT			
Employment Classification	1988	YEAR	
		2010 Buildout	2010 Redevelop
Retail-Wholesale	2,151	8,060	10,800
Highway Retail	2,702	7,170	9,575
Office	1,046	2,560	3,700
Services	2,672	7,190	10,255
Industrial	740	2,295	2,930
TOTALS	9,311	27,995	37,260

Land Use and Development

The generation of traffic on a particular street is also related to adjacent land uses. Some types of land use generate more traffic than others. For example, a commercial area generates larger traffic volumes than a residential area. The attraction between different land uses varies with development intensity and spatial separation of the uses.

The local staff provided information on anticipated land development and growth areas. This information was then used to allocate employment, population and housing projections on a zonal basis. The resulting travel patterns allow for a determination of the type of facilities required to adequately serve the areas in the future. **Figure 6** displays the anticipated growth patterns in Outer Banks area by highlighting areas of high growth potential.

The impediments to development within the planning area are largely natural. The Outer Banks has the Atlantic Ocean on one side and Roanoke Sound on the other. The Outer Banks is linked to the mainland via two bridges, one at Nags Head, and the other between Kitty Hawk and Southern Shores. Throughout the area there are numerous rivers, creeks, and wetlands. Also restrictive within the planning area is the existence of the Cape Hatteras National Seashore, Jockey's Ridge State Park, Wright Brothers National Memorial, and Nags Head Woods.

Residential development accounts for the majority of the land use in the study area. Commercial growth is mainly centered around US 158, and portions of NC 12 and Virginia Dare Trail.

GROWTH RATES (PERCENT PER YEAR)

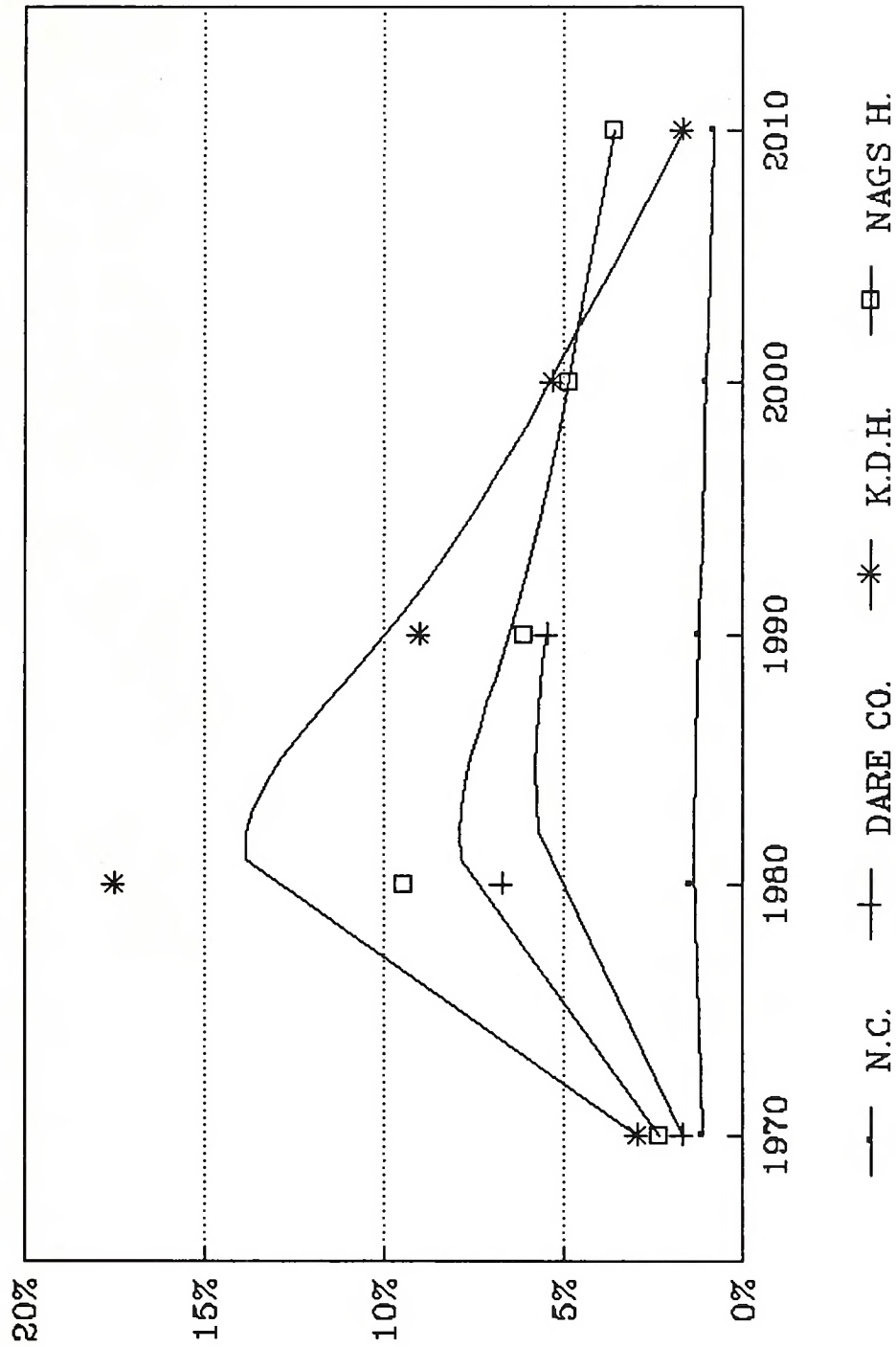


FIGURE 5

STATIONER

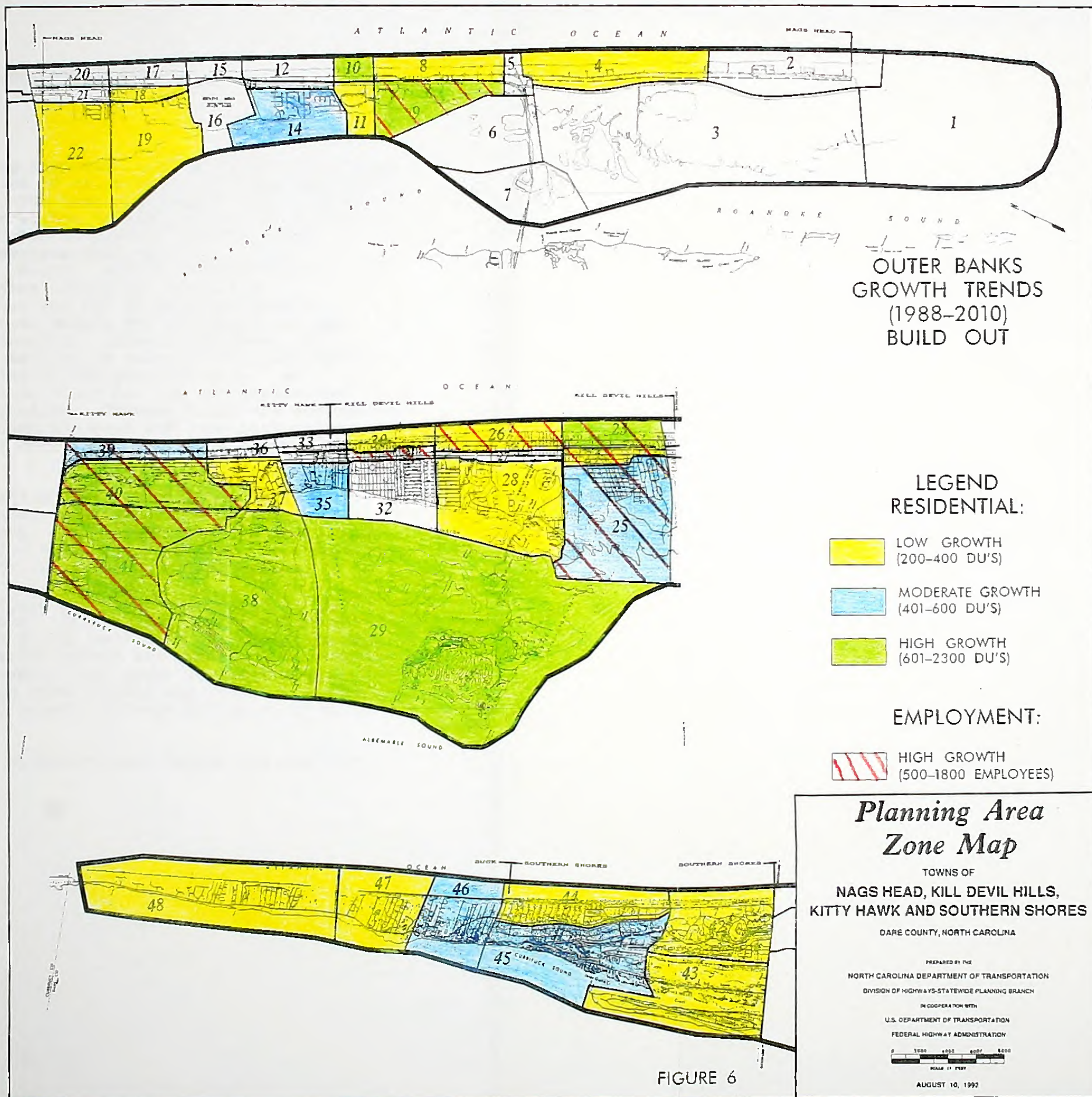


FIGURE 6

IV TRAVEL FORECAST MODELS

General

For thoroughfare planning purposes, a comprehensive knowledge of the origins and destinations of existing traffic and estimated future traffic is essential. The type, intensity, and location of the population and employment within an area largely determine the travel patterns. The method used to predict future travel involves the development of mathematical models relating population and employment to travel. Models are developed to (1) estimate trips produced (origins) and trips attracted (destinations) by traffic zones and (2) to estimate travel patterns between zones. Separate models are developed for the three basic types of trips: internal, internal-external, and through. **Internal trips** are defined as those trips which have both origin and destination inside the planning area. An **internal-external trip** is a trip which has one end inside the planning area and the other outside. **Through trips** are defined as those trips which travel through the area and have both origin and destination outside the study area. Through trips do not stop within the planning area.

The travel forecast models for the Outer Banks area were developed on the basis of travel, employment and population data obtained for the base year 1988. The validity of the models was tested by comparing the traffic volumes computed by the models to traffic volume counts taken on the existing street system.

After travel forecast models have been calibrated so that they adequately duplicate travel, design year travel estimates are produced through the input of design year data on population and employment. The trip distribution models are sensitive to changes in the street system and variation will occur in the travel patterns as alternative future street plans are tested. A more detailed documentation of the travel forecast models is given in Appendix B.

Table 6 gives a summary of travel data trends for the area.

TABLE 6
TRAVEL DATA SUMMARY

Type	1988	2010 Buildout	2010 Redevel.
Average Daily Trips per DU	5.05	5.72	5.72
Internal Trips			
Home Based Work	17,297	41,899	45,841
Other Home Based	53,813	113,126	123,772
Non-Home Based, internal	24,985	54,468	59,594
NHB secondary	6,000	15,500	16,000
Internal <--> External	35,980	82,860	88,565
Through Trips	4,060	7,495	7,917

V ANALYSIS OF THE EXISTING SYSTEM

General

This chapter presents an analysis of the ability of the existing street system to serve the area's travel desires. Emphasis is placed not only on detecting the deficiencies, but on understanding their cause. Travel deficiencies may be localized and the result of substandard highway design, inadequate pavement width, or intersection controls. Alternately, the underlying problem may be caused by a system deficiency such as a need for construction of missing links.

Existing Travel Patterns

An indication of the adequacy of the existing street system is a comparison of traffic volumes versus the ability of the streets to move traffic. In an urban area, a street's ability to move traffic is generally controlled by the spacing of major intersections, access control, width of pavement, and the traffic control devices (such as signals) utilized.

Capacity is the maximum number of vehicles which has a "reasonable expectation" of passing over a given section of a roadway, during a given time period. The relationship of traffic volumes to the capacity of the roadway will determine the **level of service (LOS)**. Six levels of service identify the range of possible conditions. **Figure 7** shows the levels of congestion associated with the various levels of service. Table 7 gives a brief description of each LOS in accordance with 1985 Highway Capacity Manual.

The recommended improvements and overall design of the Thoroughfare Plan were based on achieving a minimum of LOS D on existing facilities, and LOS C on new facilities. LOS D is considered the **"practical capacity"** of a facility, or that at which the public begins to express dissatisfaction. LOS E is the maximum carrying capacity of a facility.

TABLE 7
LEVEL OF SERVICE

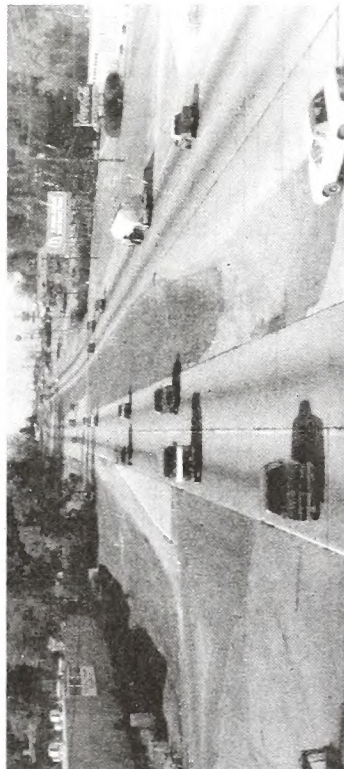
- LOS A** - describes primarily free flow conditions. The motorist experiences a high level of physical and psychological comfort. The effects of minor incidents or breakdowns are easily absorbed. On an urban arterial, LOS A corresponds to a average travel speed of 25 to 35 mph.
- LOS B** - also represents reasonably free flow conditions. The ability to maneuver within the traffic stream is only slightly restricted.
- LOS C** - provides for stable operations, but flows approach the range in which small increases will cause substantial deterioration in service. Freedom to maneuver is noticeably restricted. Minor incidents may still be absorbed, but the local decline in service will be great. Queues may be expected to form behind any significant blockage.
- LOS D** - borders on unstable flow. Small increases in flow can cause substantial deterioration in service. Freedom to maneuver is severely limited, and the driver experiences drastically reduced comfort levels. Minor incidents can be expected to create substantial queuing. On an urban arterial, LOS D corresponds to an average travel speed of 9 to 17 mph.
- LOS E** - The boundary between LOS D and LOS E describes operation at **capacity**. Operations at this level are extremely unstable, because there are virtually no usable gaps in the traffic stream. Any disruption to the traffic stream, such as a vehicle entering from a ramp, or changing lanes, requires the following vehicles to give way to admit the vehicle. This condition establishes a disruption wave which propagates through the upstream traffic flow. At capacity, the traffic stream has no ability to dissipate any disruption. Any incident can be expected to produce a serious breakdown with extensive queuing.
- LOS F** - describes forced or breakdown flow. Such conditions generally exist within queues forming behind breakdown points.



LEVEL OF SERVICE - A



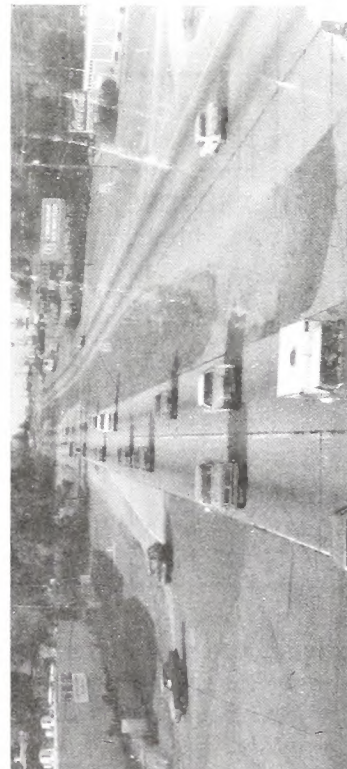
LEVEL OF SERVICE - D



LEVEL OF SERVICE - B



LEVEL OF SERVICE - E



LEVEL OF SERVICE - C



LEVEL OF SERVICE - F

LEVELS OF SERVICE

FIGURE 7

Parking Inventory

Parking is a consideration in any thoroughfare planning study because it directly affects the capacity of individual facilities. On-street parking limits the capacity of a roadway, while a deficiency of parking adds to the congestion of an area by increasing circulation. Since on-street parking reduces the capacity of thoroughfares, parking removal is one of the first procedures applied in attempting to increase the capacity of a facility. Generally, on-street parking is not a problem in the Planning Area.

Effects of 1988 Traffic on the Existing System

High Accident Location - Traffic accident records are of assistance in defining problem areas and often pinpoint a deficiency such as poor design, inadequate signing, ineffective parking, or poor sight distance. Accident patterns developed from analysis of accident data can lead to remedial action reducing the number of accidents.

Both the severity and number of accidents should be considered when investigating accident data. The severity of every accident is measured with a series of weighting factors developed by the Division of Highways of the NCDOT. In terms of these factors, a fatal or incapacitating accident is 47.7 times more severe than one involving only property damage, and an accident resulting in minor injury is 11.8 times more severe than one with only property damage.

TABLE 8
ACCIDENT SUMMARY 5/90-5/93

LOCATION	TOTAL	SEVERITY
1. US 158 / Ocean Acres Road (KDH)	31	12.37
2. US 158 / Third Street (KDH)	27	12.37
3. Croatan Hwy / Ocean (KH)	27	21.70
4. US 158 / Mall Street (NH)	18	12.54
5. Croatan Hwy / Kitty Hawk Road (KH)	14	37.67
6. US 158 / Eighth Street (NH)	12	7.03
7. US 158 / Barn Street (NH)	10	9.11
8. US 158 / Grouse (NH)	10	12.73
9. US 158 / Soundside (NH)	10	10.92

The "Total" column indicates the total number of accidents reported within two hundred (200) feet of the intersection during the indicated time period. The severity listed is the average accident severity for that location.

Capacity Deficiencies - Figure 8 depicts the base year (1988) major street system, and the ADT (Average Daily Traffic). A comparison of the base year ADT to capacities reveals several streets near or over practical capacity (LOS D). These areas are highlighted, and include:

- o NC 12 through Southern Shores
- o US 158 through Kill Devil Hills

There are several locations within the planning area which have high volumes. Traffic volumes in excess of 20,000 vehicles per day (in 1988) occurred on US 158 throughout the Planning Area, while counts as high as 16,000 vpd have occurred on NC 12 north of Kitty Hawk.

System Deficiencies - The geographic nature of the area precludes the "ideal thoroughfare plan" of circumferential and radial routes. System deficiencies are a measure of the extent to which the existing system lacks continuous radials, loops, cross-towns, and bypasses. System deficiencies in the system include:

- Lack of adequate access to Duck and Corolla from the mainland. The current system funnels all traffic through Southern Shores and the Pine Island Audubon Sanctuary. The 1994-2000 TIP (Transportation Improvement Program) includes the Mid-Currituck Bridge project. This bridge would link Coinjock to Corolla and would alleviate a large percentage of traffic travelling through Southern Shores.
- In areas where island width allows, lack of adequate collector system to direct traffic to the thoroughfares.

Another problem arises from the inadequate intersections where travel over the two main bridges changes from east-west to north-south travel. These locations are further complicated by movements from NC 12 and Virginia Dare Trail. Obviously, the number of legs at these intersections and the overall magnitude of traffic would justify the need for a directional flow lane (flyover).

Existing Street System - The street system is characterized by two facilities parallel to the coastline: US 158 and NC 12. These roadways are intersected by cross streets at frequent intervals. The street system is limited by the nature of the area: that it is a coastal community and surrounded by water. Options for the area, therefore, are severely limited.

VI ANALYSIS AND NEEDS

General

This chapter presents an analysis of the ability of the existing street system to serve the area's travel desires as the area continues to grow. The usefulness of transportation planning is in the analysis of different highway configurations for their efficiency in serving the area.

Goals and Objectives

Each area has its own priorities and concerns relating to the transportation system, and related topics. In order to determine the items of importance to the planning area, a "Goals and Objectives" survey was conducted in 1988. The survey included questions about topics such as new roads, improvements to existing roads, adequacy of beach access, sidewalks, landscaping, and intersections. Several questions were intended to force the respondent into choosing only one of several desirable items. A space was provided for comments at the end of the survey.

Many methods were used to distribute the forms. In Kill Devil Hills and Kitty Hawk the forms were distributed to the Planning Board, police department, town staff, and other interested civic groups, as well as were available at Town Hall. In Nags Head the survey was also mailed out in the Town newsletter. Southern Shores sent the survey to permanent residents, as well as having them available at Town Hall. A total of one thousand fifty four useable responses were received, with not all respondents answering every question. Of these, 280 were from Kill Devil Hills, 46 were from Kitty Hawk, 588 were from Nags Head, and 133 from Southern Shores.

The results of the survey indicated that the transportation improvements most desired were improved access (bridges) and provisions for bicycle travel. Improved access to industry and shopping centers ranked last.

Appendix C includes a copy of the survey, as well as the results. The results are tabulated in several sections: Nags Head, Kitty Hawk, Kill Devil Hills, and Southern Shores. A sample of the comments received are also included in Appendix C.

Alternative Plans

The process of developing, testing and evaluating alternate plans involved a number of considerations. These included area goals and objectives (above), identified deficiencies (**Chapter V**), environmental impacts, existing and anticipated land development, and travel service. Travel forecasts provide a basis for

evaluation of alternatives as to ability to serve future travel needs. Aerial photography, topographic mapping, field reconnaissance and discussion with town staff, officials and interested local citizens provided additional basis for identifying and evaluating the alternative alignments.

Some of the major **system** problems which need to be addressed by the thoroughfare plan include: island access; deficiency for serving the US 158 corridor traffic; and determining what to do if erosion breaches Virginia Dare Trail.

Do Nothing Alternative

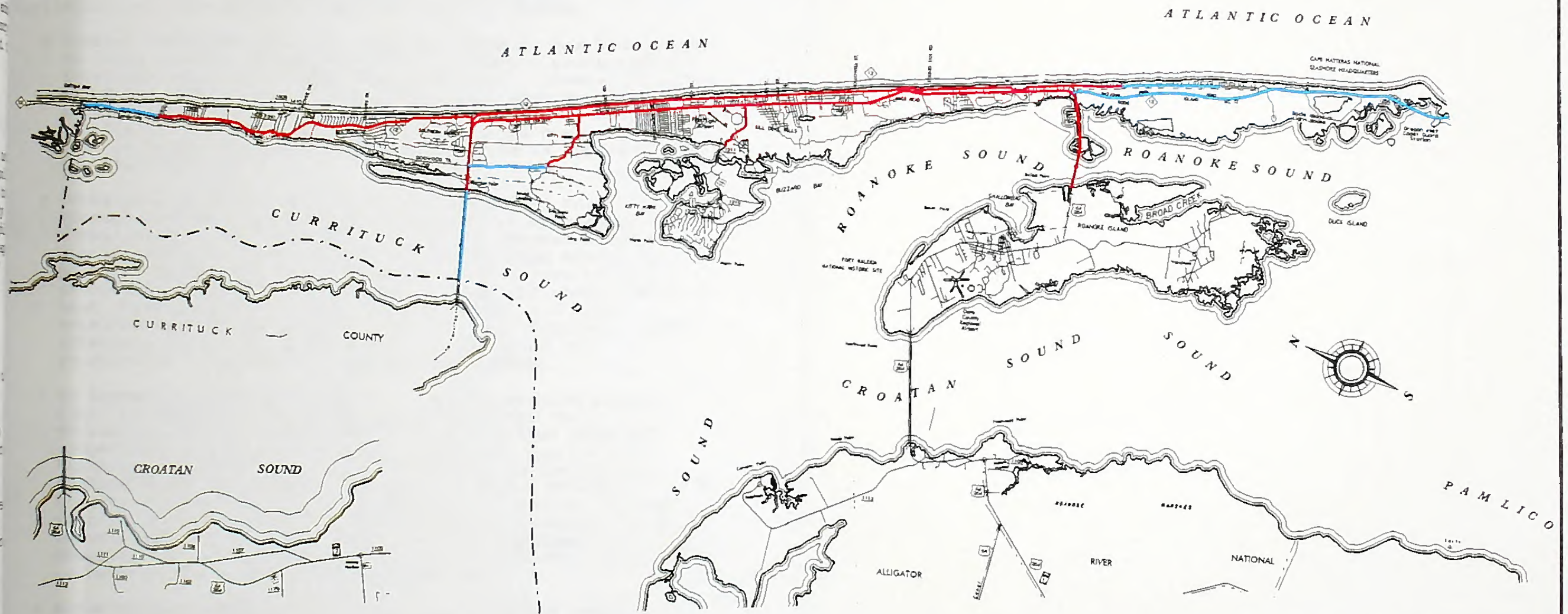
An alternative to any proposed improvement is a "do nothing" option. There can be varying degrees of a "do nothing" policy, meaning no highway improvements to the existing network, to a partial "do nothing" policy, meaning no new highways (only minor improvements to the existing system). **Figure 9** highlights the areas where facilities are at or near the practical capacity under the 2010 Redevelopment scenario. Some of the major advantages of doing nothing are:

- (1) No capital investment.
- (2) No removal of landscaping.
- (3) No additional land acquisition.
- (4) No displacement of people or business as a result of construction.

However, there are a number of disadvantages to a "do nothing" policy which would have significant adverse effects on the urban environment. These include:

- (1) Increased traffic congestion on major facilities resulting in increased air pollution and driver frustration.
- (2) Social, health, and safety standards would deteriorate.
- (3) Increased driving time and road user costs.
- (4) Increased accidents.
- (5) Reduced mobility for emergency vehicles.
- (6) Loss of potential gains in tourist trade due to congestion.

The most dramatic consequence of this option is the poor operations of US 158 corridor. To understand the severity of this problem, an estimated 80% of US 158 will be operating at a Level-of-Service F by the design year of 2010. Effectively, travellers will experience stop-and-go progression on 80% of the 17.8 mile section. Couple this with the inevitable incident along the facility and grid-lock may occur. Whether it is a stalled vehicle or an accident, incidents are the number one cause for highway congestion.



TOWNS OF
NAGS HEAD, KILL DEVIL HILLS,
KITTY HAWK AND SOUTHERN SHORES
DARE COUNTY, NORTH CAROLINA

2010 CAPACITY DEFICIENCIES

— OVER CAPACITY
— NEAR CAPACITY

FIGURE 9

In addition to the do nothing plan, it is desirable to take a more in depth look at the existing street system to determine if **non-construction** projects can enable the existing system to serve future travel. Because of the unique nature of the planning area, most non-construction alternatives have little if any applicability. Non-construction alternatives include:

- o **Control of access** increases capacity where it has been eroded by large volumes of turning traffic at many closely spaced locations, such as a strip commercial zone. Several sections of US 158, NC 12, and Virginia Dare Trail provide examples of such development. As new development occurs, it would be beneficial to attempt to reduce or limit the number of curb cuts to these facilities. This alternative is applicable to the planning area.
- o Prior to a traffic signal being installed on a facility, the through movement is permitted continuously. Upon signal installation, the time permitted for the through movement is significantly reduced to allow for conflicting movements, possibly by as much as fifty percent. This results in a reduced capacity for the facility. It is therefore important that signals only be installed when needed, and with consideration to the overall impact on the facility. Signal locations on US 158 should be highly regulated and coordinated to provide for signal progression.
- o An aggressive **carpool, vanpool, or public transit** program would process the same number of person-trips while decreasing the number of vehicle-trips and thus decrease congestion. Currently the Outer Banks does not have an historical pool of vehicle occupancy data to analyze for evaluation of the possible benefits of such a program. However, as much of the summer traffic is tourist related, a carpool program would have limited impact. Seasonal transit between major areas of interest, such as the beach and local shopping centers could be considered, but would be expected to have little impact on traffic patterns.
- o **Altering work hours** such that the beginning and ending times are staggered, can reduce travel in the peak hour. The resulting peak period would be less congested, but last longer. Therefore, the total traffic carrying ability of an existing street can be increased. While such a program could be beneficial, the impact on projected traffic is expected to be minimal.
- o **Restrictions on growth** would also slow traffic growth, and delay the need for street improvements.

The do nothing concept is not a viable alternative in transportation planning for the planning area. Some of the concepts, such as signal progression and control of access, could be considered to compliment the transportation system.

Regional Proposals

Several alternate plans were considered. The physical realities of the narrow island greatly limits the options. The remainder of the chapter will be divided into sections, with the first section addressing regional proposals.

One factor, above all others, affects the travel patterns throughout the Outer Banks. It is the number and location of access points to the island. A second important issue is the assumptions made as to erosion. Depending upon the decisions made in each of these instances, a different Thoroughfare Plan is required for the area.

Figure 10 shows the regional proposals for the Planning Area. Each project is anticipated to have an area-wide impact, and is discussed below. The following is a list of projects that when implemented in its entirety should provide the most efficient transportation system under prevailing conditions. Locations for cross streets are primarily based on providing system continuity. The idea is to provide spacing of signals on US 158 to allow proper progression by way of platooning vehicles. Realizing Not all of the projects will The implementation of each will be the responsibility of the local municipality. The regional features of this plan include:

Third Bridge to Corolla:

The design year volume of traffic going from the mainland to the Outer Banks is such that the existing bridges are insufficient. A third bridge to Corolla, will divert traffic from the Wright Brothers Memorial Bridge, and have only a negligible effect on the Washington Baum Bridge traffic. Benefits of the third bridge are:

- o traffic at the existing bridge terminal complex at Kitty Hawk/Southern Shores already places a strain on the existing road system;
- o traffic is forced through Southern Shores and Duck on NC 12, which is loaded beyond capacity;
- o for the purposes of evacuation, three exit locations are superior to two; and
- o three access points to the island serves the travel desires better than two.

A third bridge to Corolla is listed in NCDOT's 1996-2002 Transportation Improvement Program (TIP), project R-2576, and is scheduled for design to commence in 1997. The Thoroughfare Plan study acknowledges the need for a third bridge, but does not attempt to further evaluate the exact location.

For the remainder of this report, it was assumed that the third bridge from Corolla to Coinjock is constructed, thus decreasing the traffic on NC 12 from US 158 to Duck. It should be noted that if this is not the case, the resulting traffic on NC 12 and the associated congestion would be increased.

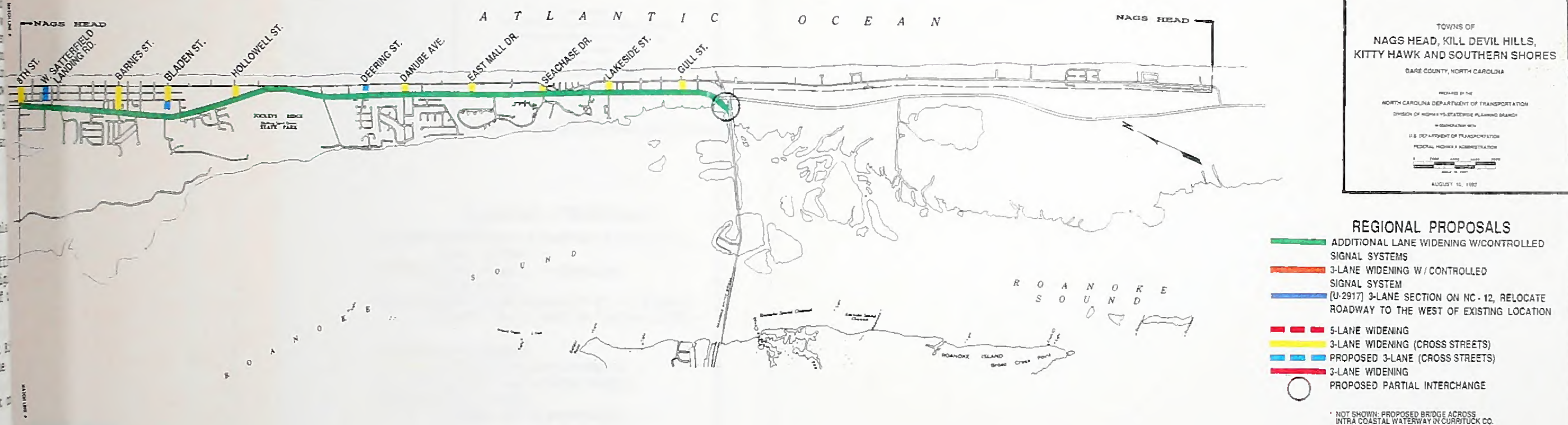
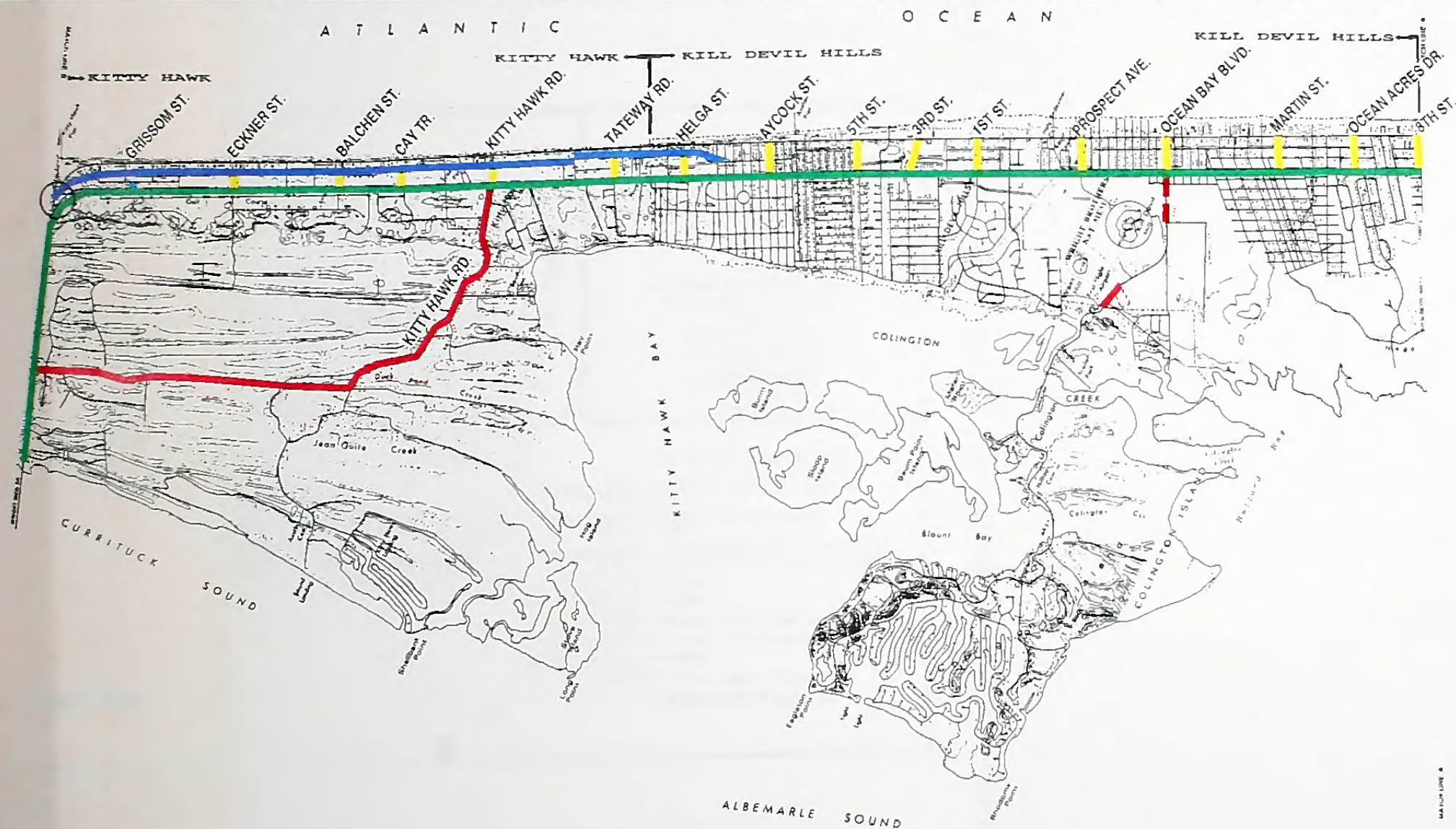


FIGURE 10-A



TOWNS OF
NAGS HEAD, KILL DEVIL HILLS,
KITTY HAWK AND SOUTHERN SHORES
DARE COUNTY, NORTH CAROLINA

PREPARED BY THE
NORTH CAROLINA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS-STATEWIDE PLANNING BRANCH

IN COOPERATION WITH
U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION

0 2000 4000 6000
FEET
SCALE 1" = 1000'

AUGUST 10, 1992

REGIONAL PROPOSALS

- ADDITIONAL LANE WIDENING W/CONTROLLED SIGNAL SYSTEMS
- 3-LANE WIDENING W/CONTROLLED SIGNAL SYSTEM
- [U-2917] 3-LANE SECTION ON NC-12, RELOCATE ROADWAY TO THE WEST OF EXISTING LOCATION
- 5-LANE WIDENING
- 3-LANE WIDENING (CROSS STREETS)
- PROPOSED 3-LANE (CROSS STREETS)
- 3-LANE WIDENING
- PROPOSED PARTIAL INTERCHANGE

* NOT SHOWN: PROPOSED BRIDGE ACROSS INTRA COASTAL WATERWAY IN CURRITUCK CO.

FIGURE 10-B

NC 12 (Virginia Dare Trail):

SECTION 1: From Wilkinson Street to the Wright Memorial Bridge Terminal at Kitty Hawk - Erosion continues to make inroads into the beach along the Outer Banks. In some areas, Virginia Dare Trail is already threatened by overwash. A study was conducted by the North Carolina State University, Center for Transportation Engineering Studies (Executive Summary Options for North Carolina Coastal Highways Vulnerable to Long Term Erosion).

"The report identifies North Carolina highway sections that are susceptible to ocean damage and ranks their relative vulnerability by expected year of damage. After identifying likely vulnerability resulting from long term erosion forces, feasible engineering solutions including shoreline hardening, beach nourishment, and highway relocation or abandonment are matched to appropriate highway sections. Approximate costs, construction year, and other factors are discussed for each critical highway section." (1)

A feasibility study (project U-2917) was published in March, 1994 by the Department of Transportation, Feasibility Studies Unit. The study addressed relocating Virginia Dare Trail to the west of its existing location in Kitty Hawk (KH) and in Kill Devil Hills (KDH). The project limits were from Wilkinson Street in KDH to a new intersection with NC 12 north of Byrd Street in KH. The study focused on providing a 3-lane facility. However, the project was not recommended for programming due to the low annual traffic demands and estimated (right-of-way and construction) costs.

Figure 11 shows the relocation of NC 12 per above report.

It is important to understand that if Summer travel demand continues to increase at its present rate and erosion of NC 12 leads to additional safety and maintenance problems or even closer, then relocation of NC 12 must be considered for programming. Ultimately, the demand for this project depends on the priority needs for the local municipalities. The lobbying efforts by the local governments will determine if the relocation of NC 12 can be programed.

SECTION 2: From Whalebone Junction to Wilkinson Street in Kill Devil Hills - Traffic projections on NC 12 warrant widening this facility to 4-lanes. This would require the acquisition or relocation of many homes, businesses, and parking. Therefore, this is not a feasible alternative. The improvements proposed for US 158 would relieve some through trip traffic on NC 12, although it is expected that NC 12 will operate at a lower level of service.

- (1) Executive Summary Options for North Carolina Coastal Highways Vulnerable to Long Term Erosion, Stone, Overton, and Fisher, August 1991, pg ES-1.

The existing surface on NC 12 includes three feet on both sides for bicycles and pedestrians. It is expected that bicycle and pedestrian traffic will increase. Therefore, it is recommended that this section of NC 12 be widened an additional three feet for a total of six feet on both sides of the roadway and be striped for pedestrian and bicycle use only.

SECTION 3: From the Wright Memorial Bridge Terminal at Kitty Hawk to Currituck County - A major concern for Southern Shores is the amount of through trip traffic that is created by not having an access point from the main-land to Currituck County. It would be difficult to determine the actual number of vehicles that creates this problem.

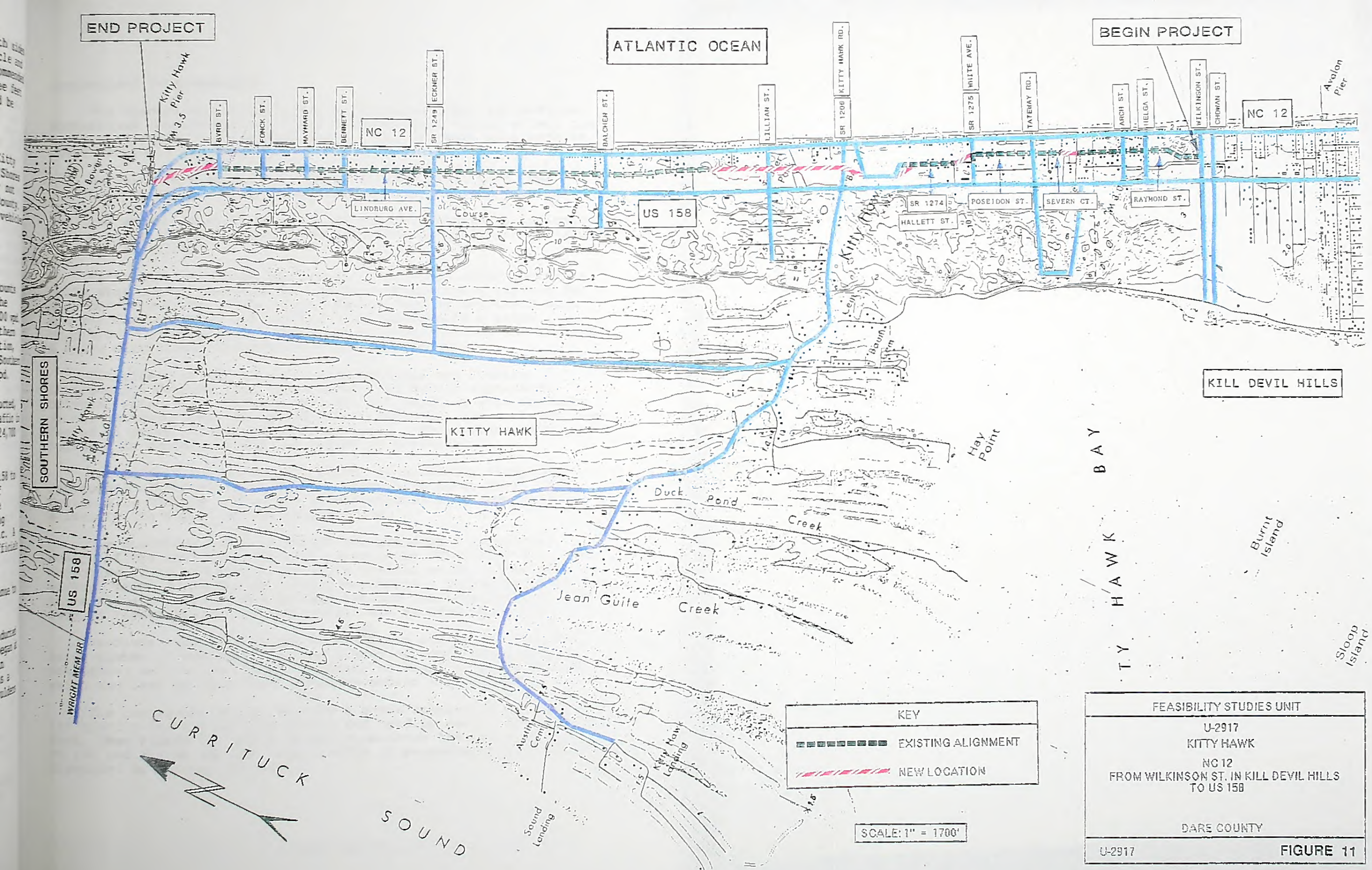
The previous **Figure 9** shows the 1988 (actual) and 2010 (projected) average summer weekday counts for selected locations. Southern Shores shows actual (1988) ground counts ranging from 14,500 to 16,000 vehicles per day (vpd). The actual ground count at the Currituck County line was 4,100 vpd. This means that the total number of through trips in Southern Shores would be less than 4,100 vpd. Using this information, the estimated number of through trip vehicles (2010) in Southern Shores generated by not having a third bridge is 8,000 vpd.

When the third bridge from Corolla to Coinjock is constructed, it is expected that the average summer weekday (2010) traffic in Southern Shores will range from approximately 22,000 to 24,700 vpd.

A 4-lane undivided cross-section is recommended from US 158 to Thirteenth Avenue in Duck. Although, if a lower level of service can be tolerated, a 3-lane cross-section could be implemented. The additional lane would allow left turning vehicles to progress without delay to through trip traffic. A 3-lane widening was supported by Southern Shores local officials on June, 1996.

A 3-lane cross-section is recommended from Thirteenth Avenue to the end of the commercial area in Duck.

Note: In March 1995, a feasibility study (R-3335) was conducted by NCDOT for the widening of NC 12. The project limits began at the intersection of US 158 northward to Ocean Boulevard in Southern Shores. The recommended typical cross-section is a four-lane, 48-foot shoulder section with 3-foot paved shoulders on 90 feet of existing right-of-way.



FEASIBILITY STUDIES UNIT	
U-2917	KITTY HAWK
NC 12	FROM WILKINSON ST. IN KILL DEVIL HILLS
	TO US 158
	DARE COUNTY
U-2917	FIGURE 11

Channelized Intersections:

The channelized intersection at Whalebone Junction and the channelized intersection at Kitty Hawk/Southern Shores affect the traffic patterns on a regional basis. The problems with each of these terminals are twofold.

- 1) Existing development, wetlands, traffic facilities, and limited right of way restrict the design of each terminal.
- 2) The amount of traffic processed at these terminals indicates a necessity for unrestricted east-west as well as north-south traffic flow.

Each location, given the amount of traffic, could warrant a fully directional interchange. An interchange would provide for smoother transition of the higher demand turning movements with minimum disruption to the traffic flow, and create a safer condition. Unfortunately, the area has developed and the cost of a fully directional interchange would be prohibitive.

The effect of the current unconventional and complex intersection designs lead to problems with driver expectancy. Drivers, especially tourists, who are not familiar with the area or the unconventional design of these terminals, are confused when forced to drive through the intersections. This often leads to an increase in congestion and the number of accidents.

The Roadway Design Branch, NCDOT, is currently studying conceptual alternatives at the interchange in Kitty Hawk. This design will include a fly-over or directional flow ramp that would alleviate the signal at the intersection.

The Statewide Planning Branch has prepared a conceptual design of the Whalebone Junction interchange (US 158/NC 12). The objective of this design is to allow traffic on US 158 to flow without being disrupted by turning movements. Currently there are three signals that process traffic through the intersection. As seen on **Figure 12**, two of the signals are located on US 158.

Problems occurring at the intersection are due to: (1) the large number of vehicles; (2) the number of conflicting turning movements; and (3) the close proximity of signals to one another, which can lead to spill-back problems and/or increase delay.

The conceptual design employs a grade separation for the US 158 traffic travelling from Manteo and heading north towards Kitty Hawk. (**See Figure 13**) This design eliminates the two signals on US 158 and allows through trip traffic to progress without being disrupted by turning movements.

US 158:

The last major improvement deals with US 158 Bypass. Since US 158 is the backbone of the transportation system in the planning area, several alternatives were considered for improving the facility. The following is a discussion of the capacity, design characteristics, and some of the alternatives considered during the analysis.

US 158 is critically important in that it is the primary facility that carries traffic north and south on the Outer Banks. This facility provides not only for through trip traffic and local trips, but for emergency evacuation as well. Two of the main constraints faced by planners when rehabilitating US 158 involve the capacity and design characteristics of the facility.

Capacity: (Level of Service)

A capacity analysis was conducted for US 158 within the planning area to achieve a better understanding of its practical limitations under the existing and anticipated traffic conditions. The analysis utilized existing traffic volume count information and the Highway Capacity Manual (HCM) method for evaluating the capacity of an Urban Arterial roadway. In addition, US 158 was separated into sections for the analysis, which takes into account the unique characteristics of the various sections of roadway through the planning area (e.g., varying number of signals).

The analysis indicated that the practical capacity of US 158 (5-lanes) ranges from approximately 35,000 to 42,000 vehicles per day (vpd). The "practical capacity" of a road is based on providing a Level-of-Service "D" (LOS D). This is because LOS D is the point at which the public begins to express dissatisfaction. LOS E is the maximum carrying capacity of a facility.

In 1988, traffic counts were collected at designated locations along US 158. The US 158 section through Kill Devil Hills experienced an Average Summer Weekday Traffic count between 27,000 and 33,000 vehicles (the highest traffic counts in the planning area).

Summer weekend traffic is another consideration. During travel periods where tourist traffic is incoming and outgoing, US 158 experiences a Level-of-Service E to F. This scenario is referred to as the "change over period". Operations at this level are extremely unstable, because there are no usable gaps in the traffic stream. The main factor which causes breakdown of flow is not the number of cars, but the number of conflicting movements. Any disruption to the traffic stream, such as a vehicle entering from a side street, or changing lanes, requires the following vehicles to give way to admit the vehicle. Average summer weekday traffic on US 158 in the planning area is projected to reach the 70,000 mark by the year 2010. Traffic will reach "stand still" if no improvements are employed.

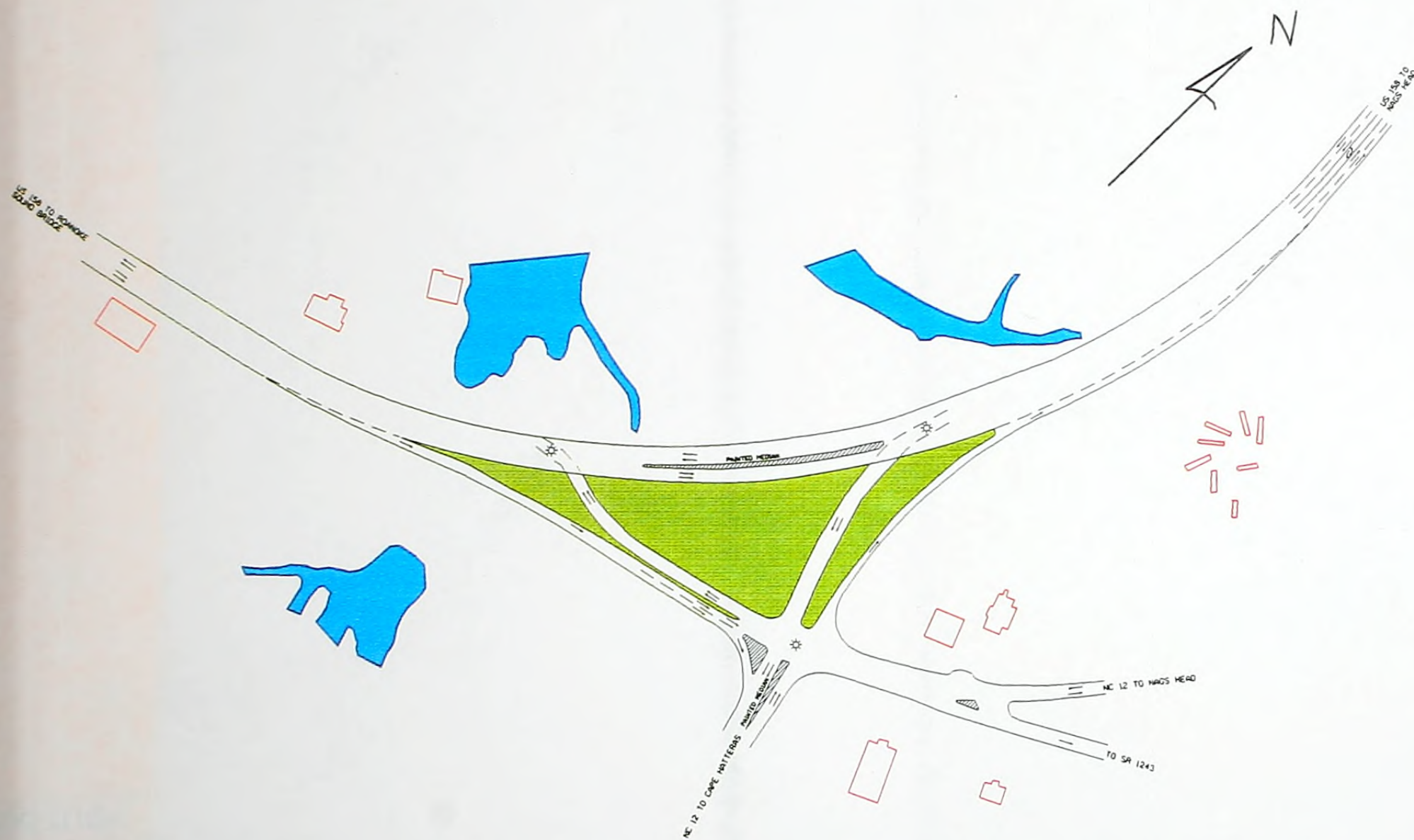


FIGURE 12
EXISTING INTERSECTION
AT WHALEBONE JUNCTION

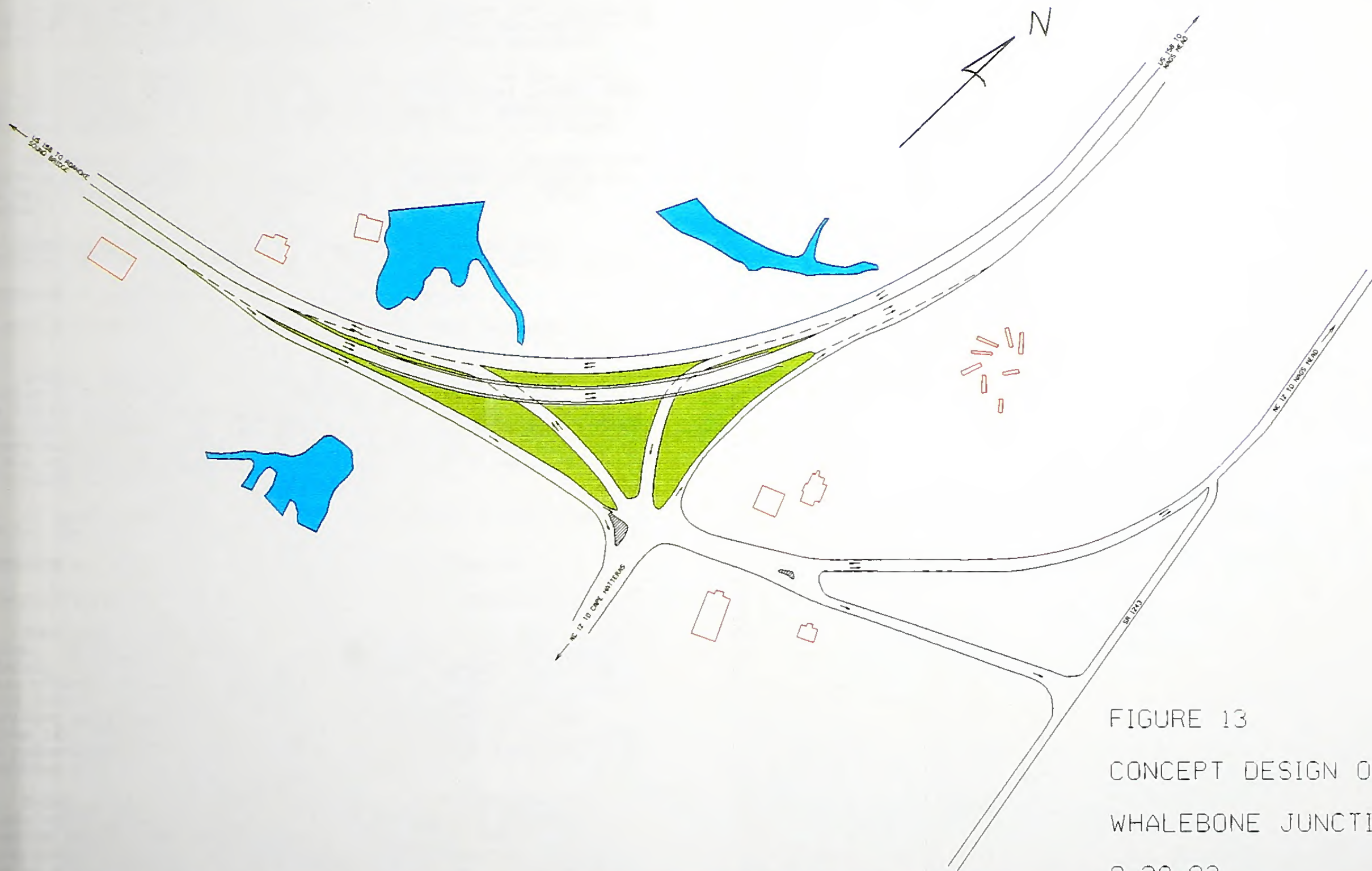


FIGURE 13
CONCEPT DESIGN OF
WHALEBONE JUNCTION
9-28-93

Design Characteristics:

As discussed in Chapter II, one way to improve the efficiency of US 158 is to redesign the facility to allow the maximum flow of traffic, i.e.; with the least amount of conflicting movements.

A limiting factor in redesigning US 158 could be the amount of right-of-way (ROW) available for adding additional lanes. The ROW width for US 158 throughout the Outer Banks is approximately 150 feet, allowing for the addition of two to three lanes total.

The second means for improving the efficiency of US 158 is to maximize the operation of the facility. One method of improving the operation of an urban arterial is to optimize its signal system.

Considering the aforementioned factors, some of the alternatives for alleviating the traffic congestion on US 158 are.

ALTERNATE 1: Seven Lane Section (See Figure 14)

Construction Cost: \$ 28.5 million (based on 1993 construction cost)

Adding one lane in each direction to the current 5-lane section of US 158 is one of the easiest options to employ. While the 7-lane cross section has some benefits, it is not a perfect solution. The practical capacity of a 7-lane section is approximately 56,000 vpd. Compared to the 5-lane capacity of 42,000 vpd and forecasted traffic volumes, this is not a sufficient improvement.

In areas where low volumes are anticipated, a 7-lane cross section could be tolerated.

ALTERNATE 2: Six Lane Divided Section (See Figure 15)

Construction Cost: \$ 36 million (based on 1993 construction cost)

Improving US 158 to a 6-lane divided section is also a viable option. It would increase US 158 capacity to a somewhat higher capacity (56,100-66,000 vpd) than the 7-lane section (56,000 vpd), again without the cost of additional ROW. A 6-lane cross section would control the hazardous left hand turning vehicles by providing median cross-over at predesignated locations. (See **Figure 16**) Furthermore, the median would provide as a safe haven for pedestrian and bicycle crossing at signalized intersections, something a 7-lane section can not provide.

Using controlled signal systems to augment the proposed 6-lane cross section would ultimately increase the operational capacity (approximately 67,000-76,500 vpd) of US 158. Uncoordinated signals and poor signal phasing restrict traffic flow by creating excessive stop-and-go operation. Grouping the signals on US 158 into progression controlled systems would allow

through trip traffic to be processed in platoons. The subsequent gaps in the traffic stream would allow the stored vehicles at the median cross-overs to discharge without disruption to the through trip traffic.

The concept of a median could also prove beneficial to the aesthetics of the community.

ALTERNATE 3: Four Lane Facility on New Location

Construction Cost: \$ 60 million (based on 1993 construction cost)

The lack of available space to build a separate facility makes this option unrealistic.

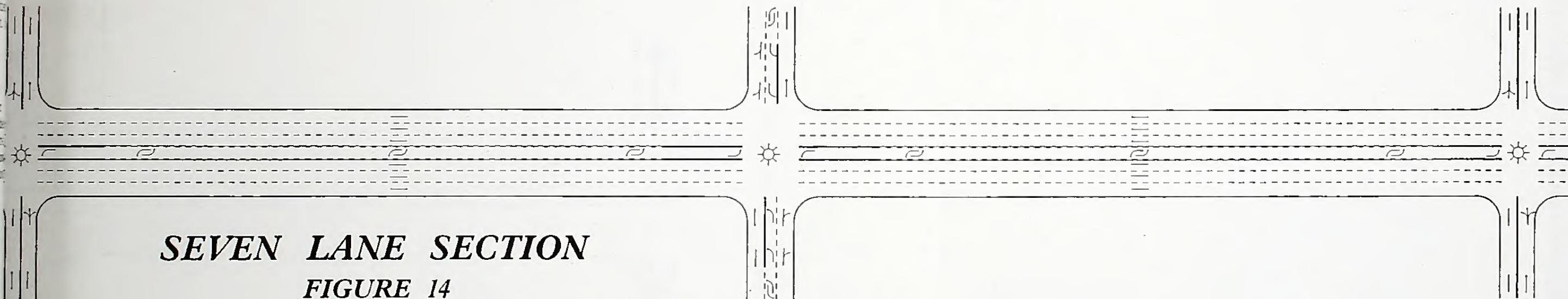
Other options, such as a continuous grade separated facility were considered in the preliminary planning stages of the study. Unfortunately, design and geographical constraints eliminated them from the number of feasible options. A grade separation at the interchange in Kitty Hawk/Southern Shores and at Whalebone Junction is still recommended, as noted on **page VI-17**.

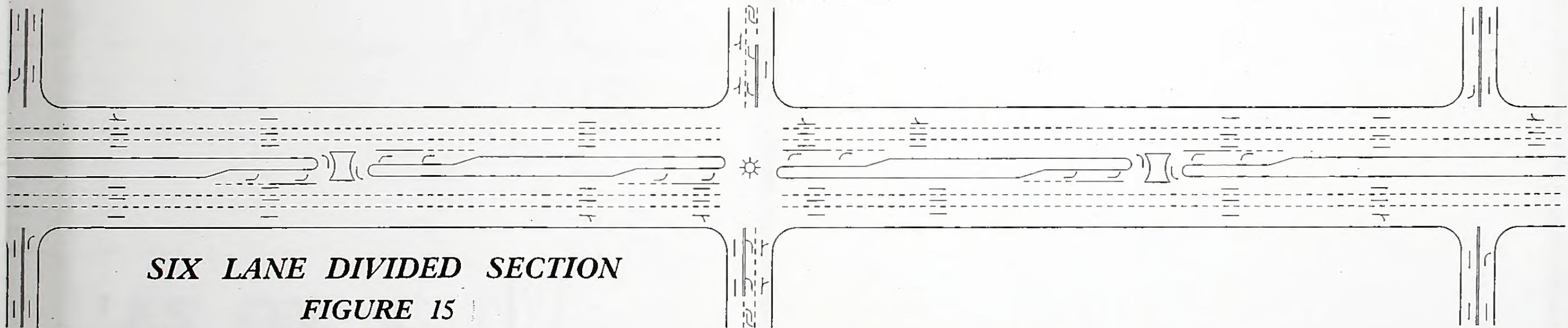
Given the alternatives for alleviating the congestion problems with US 158, and acknowledging the constraints for implementing these options, a combination of a 6-lane divided and a 7-lane cross section is preferred for the the planning area. In either case, a progression controlled signal system is recommended for US 158 from Dogwood Trails in Southern Shores to Whalebone Junction.

Ultimately, the local area would be responsible for determining the cross section that would best suit their community.

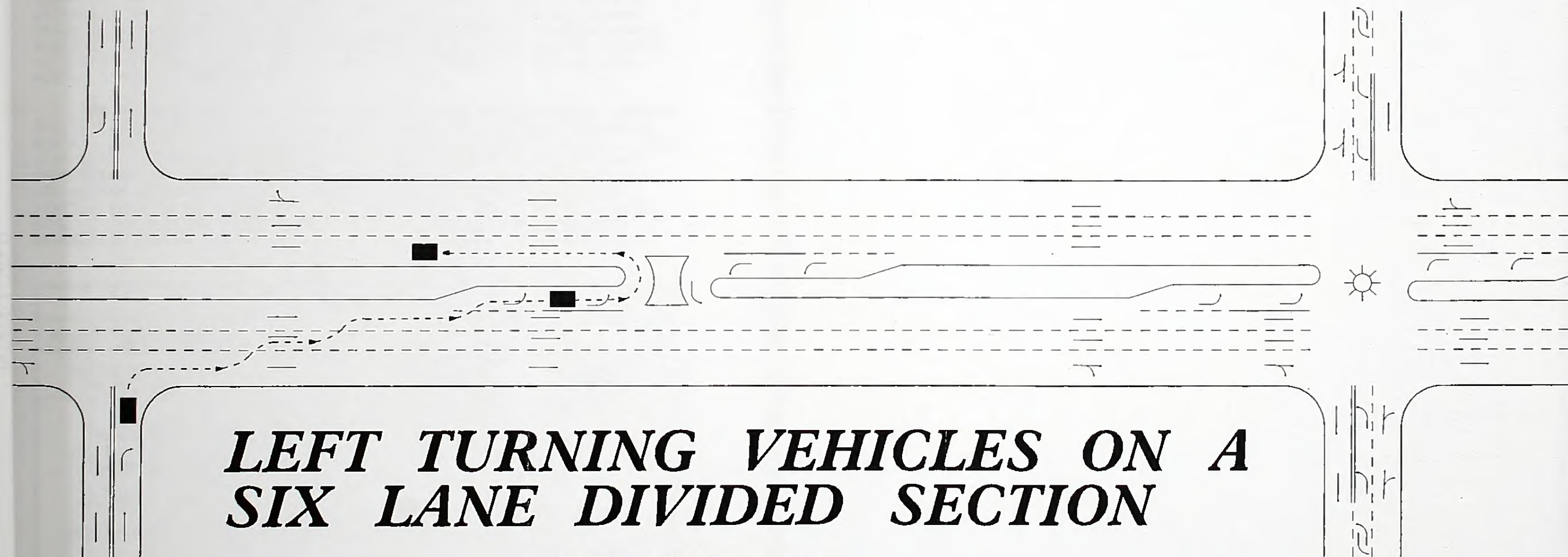
SEVEN LANE SECTION

FIGURE 14





SIX LANE DIVIDED SECTION
FIGURE 15



***LEFT TURNING VEHICLES ON A
SIX LANE DIVIDED SECTION***

FIGURE 16

Signal Plan:

Providing six through lanes for traffic will not resolve the future capacity problems with US 158. It is therefore necessary to enhance the operational efficiency of the facility to counterbalance the increase in traffic. The existing signal system must be improved.

As discussed in Chapter II (Thoroughfare Planning), the efficiency of US 158 is dramatically affected by the number and operation of its signals. When signals are relatively close in proximity to one another and are uncoordinated, this increases delay by creating excessive stop and go operation. Establishing 6-signal systems for US 158 will enable large platoons of traffic to travel through each system without the need to stop at each signal within a system.

The signal system is comprised of 6 individual signal systems for the planning area. The following tables list each existing signal location and a proposal to have it removed or remain. The list also proposes additional locations acceptable for signals. (see Figure 17, SIGNAL SYSTEM) The proposed signal systems will not effect signals on other routes. Furthermore, pedestrian traffic will be provided at all signal locations. As an integral part of the system, each signal should be equipped to protect pedestrian travel by allowing a protected phase for pedestrians.

Table 9		
Signal System #1:		
Existing signals on US 158	Proposed to remain	Proposed to remove
NC 12 (Whalebone Jct.)	X	
Seachase Drive	X	
E. Mall Drive	X	
Additional locations acceptable for signals		
* Gull Street		
* Lake Side Drive		

* Approximate location to provide consistent spacing between signals. This will improve platooning as well as meet driver expectations.

Table 10		
Signal System #2:		
Existing signals on US 158	Proposed to remain	Proposed to remove
E. Mall Drive	X	
Deering Street	X	
Pedestrian crossing	X	
Barnes Street	X	
W. Satterfield Landing Rd.	X	
Eighth Street	X	
Ocean Acres Drive (south)		X
Ocean Acres Drive (north)	X	
Martin Street	X	
Ocean Bay Boulevard	X	
Additional locations acceptable for signals		
* Danube Avenue		
* Hollowell Street		
* Villa Dunes Drive		

* Approximate location to provide consistent spacing between signals. This will improve platooning as well as meet driver expectations.

Table 11

Signal System #3:

Existing signals on US 158	Proposed to remain	Proposed to remove
Ocean Bay Boulevard	X	
Prospect Avenue	X	
First Street	X	
Third Street	X	
Fifth Street	X	
Kitty Hawk Road	X	
Additional locations acceptable for signals		
* Aycock Street		
* Helga Street		
* Tateway Road		

Signal System #4:

Existing signals on US 158	Proposed to remain	Proposed to remove
Kitty Hawk Road	X	
Eckner Street	X	
NC 12	X	
Additional locations acceptable for signals		
* Cay Street		
* Belchen Street		
* Grissom Street		

* Approximate location to provide consistent spacing between signals. This will improve platooning as well as meet driver expectations.

Table 12		
Signal System #5:		
Existing signals on US 158	Proposed to remain	Proposed to remove
NC 12	X	
Juniper Trail	X	
N. Market Place	X	
Dogwood Trail (Kitty Hawk)	X	
No additional locations acceptable for signals		
Signal System #6:		
Existing signals on US 158	Proposed to remain	Proposed to remove
NC 12	X	
Chicahawk Trail	X	
Dogwood Trail (Southern Shores)	X	
Hillcrest Drive	X	
Thirteenth Avenue	X	
No additional locations acceptable for signals		

* Approximate location to provide consistent spacing between signals. This will improve platooning as well as meet driver expectations.

Cross Streets: A Cross Street is defined in this report as any street that connects US 158 and Virginia Dare Trail, and intersects US 158 at a signal. To assist in controlling turning movements at the signal locations on US 158, 3-lane cross sections are recommended for all cross streets. (See FIGURE 15) The extra lane allows left turning vehicles to be stored at the signal. Providing 3-lanes at the cross streets will also benefit drivers who are unfamiliar with the signal system and the area. Tourist who are travelling on NC 12 (Virginia Dare Trail) will recognize a 3-lane cross section and realize that all turning movements can be made at the signal on US 158.

It is recommended that all Cross-Streets connecting US 158 and NC 12 be a continuous 3-lane cross section.

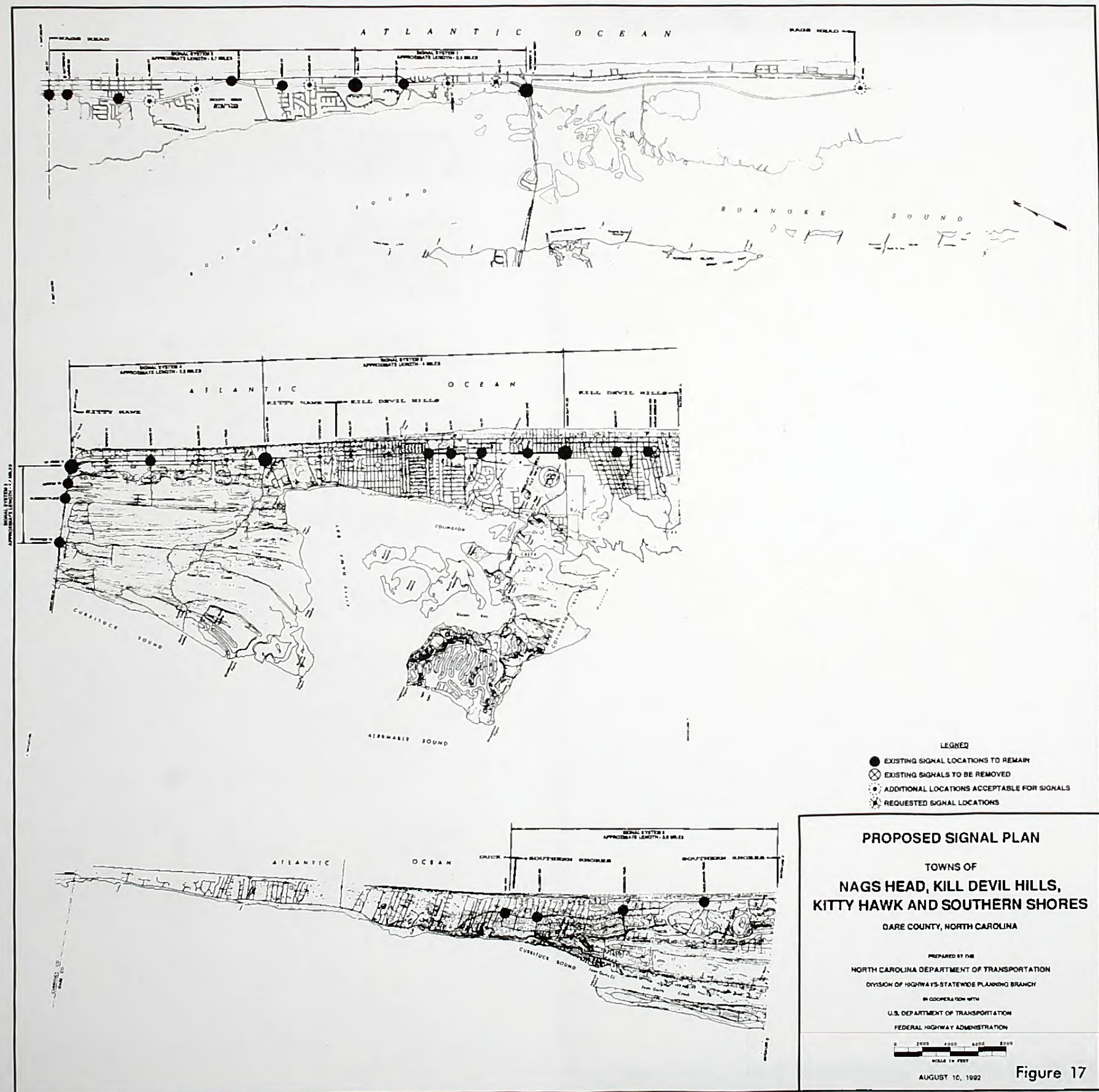


Figure 17

BICYCLE SYSTEM

This section of the report is dedicated solely to addressing the bicycle needs of the Outer Banks Planning Area. Ultimately, a multi-jurisdictional bicycle plan would benefit recreational as well as transportation needs for the Outer Banks.

To date, the only designated bicycle route in the Outer Banks Planning Area consists of US 158. This route is designated as an "off-season route" on the North Carolina Coastal Bicycle Map (ref: Around Pamlico Sound). The following lists several proposals for providing a comprehensive bicycle route system for the Outer Banks Planning Area. (See Figure 18)

1) The amount of traffic on US 158 will increase substantially within the design period. In addition, the current recommendation for US 158 changes the cross-section of the facility and could include the implementation of a median. Based on this, it would be undesirable to continue with the current designation of US 158 as a bicycle route. As an alternative, it is recommended that NC 12 from Currituck County to Whalebone Junction be widened an additional 3 feet on both sides (paved shoulders) to accommodate the additional bicycle and pedestrian traffic.

2) To provide additional bicycle access to residential areas within the planning area, several bicycle improvements are recommended. They are:

- Woods Road & Kitty Hawk Road (highlighted in yellow): From US 158 to NC 12. It is recommended that these two facilities be widened an additional 2 feet of paved shoulders for bicycles.
- US 158: From Wright Memorial Bridge to the intersection of US 158/NC 12 (highlighted in green). To provide proper access to the bridge without impeding the operation of US 158, it is recommended that a separate bike path be constructed on the north side of US 158. A separate 8' path (paved) would allow bicyclist to access the key attraction points along this corridor including; Martins Point; Kitty Hawk Elementary School; and the Market Place Shopping Center. The path would be shared by bicyclist travelling in both directions.

A 4' paved shoulder improvement (highlighted in yellow) is recommended on the south shoulder of US 158 from the Wright Memorial Bridge to Woods Road. This would allow east-bound bicycle traffic to access the Promenade or continue to the intersection of Woods Road/US 158. At this signalized intersection a pedestrian and bicycle crosswalk is recommended.

- SR 1243 (highlighted in green): From NC 12 at Whalebone Junction to NC 12 (south). It is recommended that a separate 8' bike path (paved) be constructed on the east side of SR 1243. The path would be shared by bicyclist travelling in both directions. The separate path could be constructed adjacent to this facility given the amount of existing right of way.

A field investigation was conducted to determine the impacts of a separate facility. The existing utility poles appear to be set back far enough to allow for the construction of the bike path. Garbage containers would require relocation.

3) The last bicycle recommendation would require coordination between several agencies as well as local property owners. A north-south bicycle route connecting Kitty Hawk Road and Old Nags Head Road would provide a continuous bicycle route on the west side of US 158. The route (highlighted in yellow) would utilize Moor Shore Road, Tateway Road, Bay Drive, Canal Drive, First Street, Veterans Drive, Sixth Avenue, Clark Street, and Old Nags Head Road. A cooperative effort would be required to purchase or dedicate right of way, build, and maintain the bicycle route.

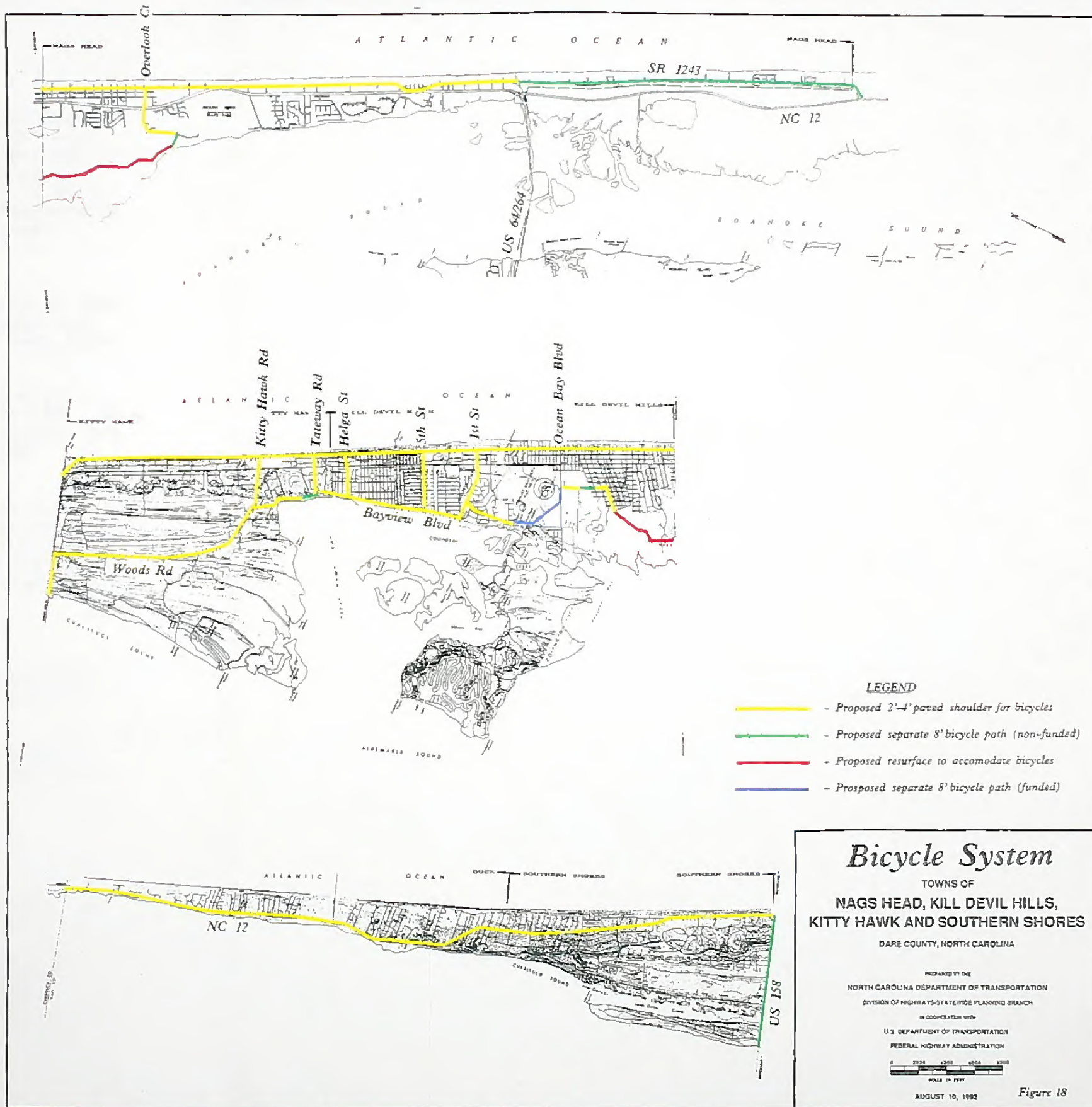
Due to the environmental sensitivity of the area, an agreement between the Nature Conservancy and the local communities would have to be reached prior to any improvements to Old Nags Head Road.

Due to the shared, or multi-modal, use of these facilities, it is recommended that the sub-standard facilities be widened to a standard 24 foot cross-section with additional 2 feet of paved shoulders for bicycles (for a total of 28 feet). This widening will enhance safety and improve the functional design of the facility.

A bicycle path connection would be required in locations where facilities do not exist (highlighted in green). It is recommended that a separate 8' bike path (paved) be constructed for the shared use by bicyclist travelling in both directions. Barricade poles should be placed at the entrances of the bike path to restrict vehicular traffic.

When considering the widening of these facilities, it is recommended that the Office of Bicycle and Pedestrian Transportation (NCDOT) be consulted. They can help provide the most appropriate cross-section for the widening. They may also provide assistance in identifying the need for bicycle improvements based on present and future bicycle traffic. The Towns should contact the coordinator of this branch for further consideration and assistance.

Bicycle and Pedestrian Program
NC Department of Transportation
P. O. Box 25201
Raleigh, NC 27611



VII. RECOMMENDATIONS

The recommended Thoroughfare Plan resulting from this study is shown in **Figure 19**. This plan provides for a system of major and minor thoroughfares, which, when progressively implemented, should accommodate the expected travel desires and anticipated land development within the planning period. **Appendix E** of this report includes a street by street description of the thoroughfare system with the associated volumes under several conditions.

Adoption of the Plan

A series of public meetings and hearings will be held on the Thoroughfare Plan. Throughout the study process, the local and NCDOT staff worked together. To date two meetings have been held on this plan:

August 3, 1993: The Kill Devil Hills Planning Board conducted a Public Hearing to discuss the proposed improvements to US 158. As a result, the Board endorsed the recommendations for additional lane widening of US 158 through K.D.H., including signal system coordination.

September 13, 1993: A Public Hearing was conducted by the Kill Devil Hills Board of Commissioners to discuss the proposed improvements to US 158. The Board agreed by consensus to support the proposed widening of US 158. In addition, the Board was most interested in seeing emphasis placed on bike paths and pedestrian walkway improvements.

Recommendations

The main consideration is US 158, which runs from Nags Head to Southern Shores. Other major considerations include island access (bridges) and NC 12.

The following sections describe the thoroughfare system in detail. A street by street inventory which includes existing and recommended cross sections, are shown in Appendix E.

KILL DEVIL HILLS

Major Thoroughfares:

US 158: From Nags Head City limits to Kitty Hawk City limits. This facility will exceed its present capacity within the design year and should be widened to include additional lanes. Additional recommendations include signal system coordination. The local planning officials will be responsible for determining the appropriate cross section that will best suit the local community. It is recommended that the proposed cross section be one of the following. 1) 7-lanes; 2) 6-lane divided; 3) A combination of the two.

NC 12: From Nags head City limits to Kitty Hawk City limits. A feasibility study (project U-2917) was published in March, 1994 by the Department of Transportation, Feasibility Studies Unit. The study addressed relocating Virginia Dare Trail to the west of its existing location in Kitty Hawk (KH) and in Kill Devil Hills (KDH). The project limits were from Wilkinson Street in KDH to a new intersection with NC 12 north of Byrd Street in KH. The study focused on providing a 3-lane facility. However, the project was not recommended for programming due to the low annual traffic demands and estimated (right-of-way and construction) costs. See **Figure 11**

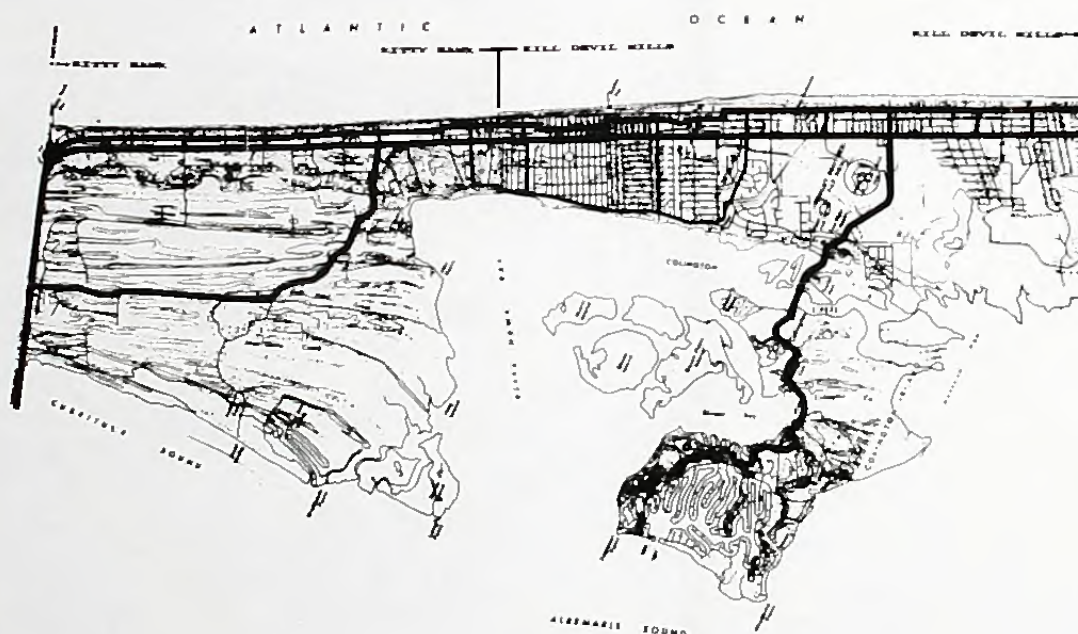
As noted, it is important to understand that if Summer travel demand continues to increase at its present rate and erosion of NC 12 leads to additional safety and maintenance problems or even closer, then relocation of NC 12 must be considered for programming. Ultimately, the demand for this project depends on the priority needs for the local municipalities. The lobbying efforts by the local governments will determine if the relocation of NC 12 can be programed.

It is recommended that this section include four feet of additional pavement (striped) on both sides for the use of bicyclist and pedestrians. The remaining section of NC 12, from Nags Head to Wilkinson Street is recommended to be widened an additional three feet (for a total of six feet) on both sides of the roadway and be striped for pedestrian and bicycle use only.

In addition, turn lanes at the signalized intersections with three lanes connecting from US 158 (signalized intersection) are recommended to control the conflicting turning movements.

Ocean Bay Boulevard (Colington): From US 158 to Colington. Ocean Bay Boulevard, west of US 158 carries approximately 9000 vehicles on an average summer weekday and is further complicated by a high school in the immediate vicinity. This facility will exceed its present capacity by the design year and should be widened to 5-lanes from US 158 to Veterans Drive. It is currently a 3-lane section from Veterans Drive to the K.D.H. Public Works Complex. Three lanes are recommended from the Complex to Kill Devil Hills City limits at Colington.

Ocean Bay Boulevard from K.D.H. City limits to the end of the facility on Colington Island warrants widening to 3-lanes (Average Summer Weekday projections range from 16,000 vpd - 23,000 vpd). Unfortunately, limited right-of-way and adjacent wetland areas restrict additional lane widening. The additional traffic will have to be tolerated if residential growth of Colington continues as forecasted. There are some potentially hazardous sharp curves located along this section of Ocean Bay Boulevard. A roadway alignment investigation should be conducted by the Traffic Engineering Branch (NCDOT) for possibly correcting this potentially dangerous roadway section.



LEGEND:

EXISTING PROPOSED

MAJOR THOROUGHFARE

SENIOR THOROUGHFARE

PARTIAL INTERCHANGE

REVISIONS:

NO.	DATE	BY	REVISION
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

ADOPTED BY:

TOWN OF NAGS HEAD _____
 TOWN OF KITTY HAWK _____
 TOWN OF KILL DEVIL HILLS _____
 TOWN OF SOUTHERN SHORES _____
 STATEWIDE PLANNING BRANCH _____
 N.C. DEPARTMENT OF TRANSPORTATION _____
 PUBLIC HEARING DATE: _____

PROPOSED THOROUGHFARE PLAN FOR THE

TOWNS OF
 NAGS HEAD, KILL DEVIL HILLS,
 KITTY HAWK AND SOUTHERN SHORES

DARE COUNTY, NORTH CAROLINA
 FEBRUARY, 1994

PREPARED BY THE
 NORTH CAROLINA DEPARTMENT OF TRANSPORTATION
 DIVISION OF HIGHWAYS-STATEWIDE PLANNING BRANCH

IN COOPERATION WITH
 U.S. DEPARTMENT OF TRANSPORTATION
 FEDERAL HIGHWAY ADMINISTRATION



DAY 1992

Minor Thoroughfares:

Ocean Bay Boulevard: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section.

Ocean Acres Drive: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12.

Martin Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to Blue Jay Avenue to accommodate left turning vehicles at the signal.

Prospect Avenue: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12.

First Street: (SR 1172, Cross Street) From NC 12 to Canal Drive. This facility should be widened to a 3-lane cross-section from NC 12 to First Flight Lane, plus 2' paved shoulders for bicyclists and pedestrians. The remaining section, from First Flight Lane to Canal Street should be improved to 2-lane, 24'-wide standards with 2' paved shoulders for bicycles and pedestrians.

Canal Street: From First Street to Bay Drive. This facility should be improved to 2-lane, 24'- wide standards with 2' paved shoulders for bicyclists and pedestrians.

Bay Drive: From Canal Street to Kitty Hawk City limits. This facility should be improved to 2-lane, 24'- wide standards with 2' paved shoulders for bicyclists and pedestrians.

Third Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to Seminole Street. The middle lane will provide the storage for left turning vehicles at the intersection of US 158.

Fifth Street: (Cross Street) From NC 12 to Bay Drive. This facility should be widened to a 3-lane cross-section from NC 12 to Orville Court, including 2' paved shoulders for bicyclists and pedestrians. The remaining section, from Orville Court to Bay Drive should be improved to 2-lane, 24'- wide standards with 2' paved shoulders for bicyclists and pedestrians.

Aycock Street: (Cross Street) From NC 12 to US 158. This facility should be widened to a 3-lane cross-section from US 158 to NC 12. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to 500' West of US 158. The middle lane will provide the storage for left turning vehicles at the intersection of US 158.

Helga Street: (Cross Street) From NC 12 to Bay Drive. This facility should be widened to a 3-lane cross-section from NC 12 to Susan Drive, plus 2' paved shoulders for bicyclists and pedestrians. The remaining section, from Susan Drive to Bay Drive should be improved to 2-lane, 24'- wide standards with 2' paved shoulders for bicyclists and pedestrians.

KITTY HAWK

Major Thoroughfares:

US 158: From Kill Devil Hills City limits to the Wright Memorial Bridge. This facility will exceed its present capacity within the design year and should be widened to include additional lanes. Additional recommendations include signal system coordination. The local planning officials will be responsible for determining the appropriate cross section that will best suit the local community. It is recommended that the proposed cross section be one of the following. 1) 7-lanes; 2) 6-lane divided; 3) A combination of the two.

NC 12: From Kill Devil Hills City limits to Southern Shores City limits. Refer to Kill Devil Hills discussion of NC 12. (See Figure 11)

It is recommended that this section include four feet of additional pavement (striped) on both sides for the use of bicyclists and pedestrians.

In addition, turn lanes at the signalized intersections with three lanes connecting from US 158 are recommended to control the conflicting turning movements.

Main Street/Woods Road: From US 158 to Kitty Hawk Road. A three lane cross-section is recommended for this facility with the center lane providing for left turns. This corridor provides for the transportation needs of local residents. Additional improvements include 2' paved shoulders for bicyclists and pedestrians. The Average Summer Weekday projection (2010) for Woods Road at US 158 is 11,000 vpd.

Kitty Hawk Road: (Cross Street) From NC 12 to Woods Road. This facility should be widened to a 3-lane cross-section from US 158 to NC 12, including 2'- paved shoulders for bicyclists and pedestrians. A 5-lane cross-section is recommended from US 158 to Woodard Road. Three lanes are recommended from Woodard Road to Woods Road. In addition, 2'- paved shoulders for bicyclists and pedestrians are recommended for this section. The Average Summer Weekday projection (2010) for Kitty Hawk Road at US 158 is 25,000 vpd.

Minor Thoroughfares:

Tateway Road: (Cross Street) From NC 12 to Velway Street/Bay Drive. This facility should be widened to a 3-lane cross-section from NC 12 to Rabbit Hollow Drive, plus 2' paved shoulders for bicyclists and pedestrians. The remaining section, from Rabbit Hollow Drive to Velway Street should be improved to 2-lane, 24'-wide standards with 2' paved shoulders for bicyclists and pedestrians.

Velway Street: From Tateway Road to Kill Devil Hills City Limits. This facility should be improved to 2-lane, 24'-wide standards with 2' paved shoulders for bicyclists and pedestrians.

Cay Trail: (Cross Street) From US 158 to NC 12. This facility is a proposed Cross Street connecting US 158 to NC 12. Its cross-section should be built to 3-lane standards. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to 500' West of US 158. The middle lane will provide the storage for left turning vehicles at the intersection of US 158.

Balchen Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12. Presently, the intersection of US 158 and Balchen Street is offset. Realignment of Balchen Street at the intersection is recommended to correct the offset. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to Johnston Lane.

Eckner Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to Johnston Lane. The middle lane will provide the storage for left turning vehicles at the intersection of US 158.

Grissom Street: (Cross Street) From US 158 to NC 12. This facility is a proposed Cross Street connecting US 158 to NC 12. Its cross-section should be built to 3-lane standards. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to Putter Lane.

NAGS HEAD

Major Thoroughfares:

US 158: From Whalebone Junction to Kill Devil Hills City limits. This facility will exceed its present capacity within the design year and should be widened to include additional lanes. Additional recommendations include signal system coordination. The local planning officials will be responsible for determining

the appropriate cross section that will best suit the local community. It is recommended that the proposed cross section be one of the following. 1) 7-lanes; 2) 6-lane divided; 3) A combination of the two.

NC 12: From Whalebone Junction (US 64) to Kill Devil Hills City limits. It is recommended that this section include three feet of additional pavement (striped) on both sides for the use of bicyclists and pedestrians.

In addition, turn lanes at the signalized intersections with three lanes connecting from US 158 are recommended to control the conflicting turning movements.

SR 1243: From NC 12 (Whalebone Junction) to NC 12 (South). No improvements necessary.

Minor Thoroughfares:

Gull Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12. It is also recommended that access to the Soundings Shopping Center be allowed at the signalized intersection of Gull Street/US 158.

Lakeside Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to 500' West of US 158. The middle lane will provide the storage for left turning vehicles at the intersection of US 158.

Seachase Drive: (Cross Street) From US 158 to NC 12. No additional improvements are necessary for Seachase Drive between US 158 and NC 12. It is recommended that this facility be widened to a 3-lane cross-section from US 158 to Baymeadow Drive to provide the proper storage length for left turning vehicles at the signal.

Mall Drive: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12.

Danube Avenue: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12.

Deering Street: (Cross Street) From US 158 to NC 12. This facility is a proposed Cross Street connecting US 158 to NC 12. Its cross-section should be built to 3-lane standards. The cross street locations are based on providing appropriate signal spacing which will allow for platooning on US 158. Since the cross streets are designated as minor thoroughfares, it is the responsibility of the local community to determine when to implement each proposal.

Hollowell Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12.

Villa Dunes Drive: (Cross Street) From US 158 to NC 12. This facility is a proposed Cross Street connecting US 158 to Bladen Street. Its cross-section should be built to 3-lane standards with 2' paved shoulders for bicyclists and pedestrians. The cross street locations are based on providing appropriate signal spacing which will allow for platooning on US 158. Since the cross streets are designated as minor thoroughfares, it is the responsibility of the local community to determine when to implement each proposal.

Bladen Street: From Wrightsville Avenue to NC 12. This facility connects the proposed Villa Dunes Drive to NC 12. It is recommended that this section of Bladen Street be improved to a 3-lane cross-section with 2' paved shoulders for bicyclists and pedestrians.

Barnes Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12. It is also recommended that this facility be widened to a 3-lane cross-section from US 158 to Meekins Avenue.

West Satterfield Landing Road: (Cross Street) From US 158 to NC 12. This facility is a proposed Cross Street connecting US 158 to NC 12. Its cross-section should be built to 3-lane standards. The cross street locations are based on providing appropriate signal spacing which will allow for platooning on US 158. Since the cross streets are designated as minor thoroughfares, it is the responsibility of the local community to determine when to implement each proposal.

Eighth Street: (Cross Street) From US 158 to NC 12. This facility should be widened to a 3-lane cross-section from US 158 to NC 12.

Note: The Signal Plan is an *ideal* system of controlling access and traffic flow on US 158 and NC 12. Understandably, existing development may limit the effectiveness of the system if cross street locations are eliminated. If alterations are made to the plan which would limit the number and/or spacing between signals, then traffic operations and progression of traffic will become less effective. Three of the eleven Cross Street connectors in Nags Head have existing development in the areas of proposed right-of-way. They are: Deering Street, Villa Dunes Drive, and West Satterfield Landing Road.

SOUTHERN SHORES

Major Thoroughfares:

NC 12: From US 158 to the North City limits of Southern Shores. This facility will exceed its present capacity within the design year and should be widened from 2-lanes to a 3-lane cross-section. Additional recommendations include signal system coordination. The Average Summer Weekday projection for NC 12 through Southern Shores ranges from 21,600 vpd to 26,000 vpd.

It is also recommended that this section include three feet of additional pavement (striped) on both sides for the use of bicyclists and pedestrians.

Minor Thoroughfares:

Chicahawk Trail: From Trinite Trail to NC 12. No improvements necessary.

Trinite Trail: From Juniper Trail to Chicahawk Trail. No improvements necessary.

Juniper Trail: From US 158 to Trinite Trail. No improvements necessary.

UNINCORPORATED AREAS - DUCK/SANDERLING

Major Thoroughfares:

NC 12: From the North City limits of Southern Shores to Currituck County. This facility will exceed its present capacity within the design year and should be widened from 2-lanes to a 3-lane cross-section from Southern Shores to the end of the commercial area in Duck.

It is also recommended that this section include three feet of additional pavement (striped) on both sides for the use of bicyclists and pedestrians.

Minor Thoroughfares:

None

Traffic Operations

Traffic signal progression

As a part of this study, locations for future signals should be considered to provide for future progression patterns. It should be stressed that contrary to popular belief, traffic signals are not always the solution to traffic problems, and in fact, typically worsen the problem if utilized incorrectly.

Facilities should be designed such that the integrity of the road is not compromised with excessive curb cuts.

Summary of Recommendations

This report contains numerous recommendations for the Outer Banks transportation system. The following is a brief review of these recommendations, with additional notation as to where in the report a more detailed discussion can be found.

Cross-sections - Each facility on the Thoroughfare Plan is discussed in detail earlier in this chapter. A summary of the recommended cross-sections for each facility is in Appendix E. The minimum desirable cross-section is twenty-four feet with paved shoulders or curb and gutter. The desirable lane width is twelve feet.

New Facilities - Few new facilities are needed within the planning area. They provide for continuity of travel, corridor spacing, and/or a more direct travel path.

Systems Improvements - Often system improvements can provide additional capacity or improved traffic conditions, with a minimum of capital outlay. Recommended system improvements include:

- A continuing program to assure proper timing and phasing of all traffic signals. Proper signal progression can have significant positive impact on a corridor, this is especially true for the US 158 corridor.
- Protection of access control is one of the two areas where a significant contribution can be made. Specific corridors, such as US 158, NC 12, and Virginia Dare Trail as well as all other major thoroughfares should be protected from unrestricted access.
- A single unprotected left turning car can expend the capacity equivalent of five through vehicles. The cross-section recommendations presume that left turns at key intersections are provided for, as indicated earlier in this report.

VIII ADMINISTRATIVE CONTROLS AND IMPLEMENTATION TOOLS

Spiraling right-of-way cost has greatly reduced the ability of the State to fund highway improvements. Initiative for plan implementation will rest largely with the local policy boards involved in the Plan's development (Kill Devil Hills, Kitty Hawk, Nags Head, Southern Shores Town Councils, and Dare County Commissioners).

Unless implementation is an integral part of the transportation planning process, the effort and expense associated with developing a plan is lost. To neglect the implementation process is a three-fold loss the loss of the costs associated with developing a plan, the opportunity cost of capital expenditures, and more importantly the loss of the benefits from an improved transportation system.

Administrative controls and implementation tools which can aid in the implementation process are generally available to municipalities through Federal and State Legislation. These will be discussed in this chapter. They include: Subdivision Regulations; Zoning Ordinances, Official Maps, Urban Renewal, Capital Improvements Programs and construction. Generally, two issues play a major role in the implementation process -available finances and citizen involvement. Effective use of the tools listed above minimize the effects of limited finances. It is through good planning that maximum use is made of every available dollar and that citizen involvement and approval of the transportation plan is obtained.

State and Municipal Adoption of the Thoroughfare Plan

Chapter 136, Article 3A, Section 136-66.2 of the General Statutes of North Carolina provides that after development of a thoroughfare plan, the plan may be adopted by the governing body of the municipality and the Board of Transportation to serve as the basis for future street and highway improvements. The General Statutes also require that, as part of the plan, the governing body of the municipality and Department of Transportation shall reach agreement on responsibilities for existing and proposed streets and highways included in the plan. Facilities which are designated a State responsibility will be constructed and maintained by the Division of Highways. Facilities which are designated a municipal responsibility will be constructed and maintained by the municipality.

After mutual plan adoption, the Department of Transportation will initiate negotiations leading to determination of which existing and proposed thoroughfares will be a Departmental responsibility and which will be a municipal responsibility. Chapter 136, Article 3A, Section 136-66.1 of the Federal Statutes

provides guidance in the delineation of responsibilities. This statute stipulates that the Department of Transportation shall be responsible for those facilities which serve volumes of through traffic and traffic from outside the area to major business, industrial, governmental, and institutional destinations located inside the municipality. The municipality is responsible for those facilities which serve primarily internal travel.

Thoroughfare plan adoption enables other planning tools such as the subdivision ordinance, zoning ordinance, official street map, and capital improvement program to be used to assist in plan implementation and thus minimize public cost, environmental cost, and land use disruption.

Subdivision Control

Subdivision regulations are locally adopted laws governing the process of converting land into building sites. A subdivision ordinance requires that subdividers submit to the Town Planning Commission a plot of the proposed subdivision. Certain standards must be met by the developer before he can be issued a building permit to construct his development. Through this process, it is possible to reserve or protect the necessary rights-of-way for projected streets which are a part of the thoroughfare plan and to require street construction in accordance with the plan.

Subdivision regulations may provide such benefits as requiring portions of major streets to be constructed in accordance with the Thoroughfare Plan, or requiring subdividers to provide for the dedication and/or reservation of rights-of-way in advance of construction. These practices reduce the overall cost of the Plan by having some costs borne by developers. Recommended Subdivision Ordinances are included in **Appendix D**.

Zoning

Zoning is probably the single most commonly used legal device available for implementing a community's land-use plan. A zoning ordinance can be beneficial to thoroughfare planning in that planned locations of various land uses and planned densities of dwellings can be realized. This provides a degree of stability on which to make future traffic projections and to plan streets and highways.

Other benefits of a good zoning ordinance are: (1) the establishment of standards of development which will aid traffic operations on major thoroughfares, and (2) the minimization of strip commercial development which creates traffic friction and increases the traffic accident potential. The zoning ordinance should be structured to control strip development along the major traffic-carrying thoroughfares.

The zoning ordinance does not regulate the design of streets, utilities, the dedication of parks, school sites, and related matters. These are typically controlled by subdivision regulations. The zoning ordinance should be carefully coordinated with subdivision regulations and other control measures.

Official Map

The roadway corridor official map (or official map) is a document, adopted by the legislative body of the community, that pinpoints and preserves the location of proposed streets against encroachment. In effect, the official map serves notice on developers that the State or municipality intends to acquire certain specific property. The official map serves as a positive influence for sound development by reserving sites for public improvements in anticipation of actual need.

The NCDOT limits its use of official maps to large scale, fully access controlled facilities planned for developing areas outside of municipal jurisdictions. For projects within municipal jurisdictions, official maps should be prepared and adopted by the local government. Municipalities may adopt official maps that extend beyond its extraterritorial jurisdiction with approval from the Board of County Commissioners.

It should be recognized that an official map places severe but temporary restrictions on private property. These restrictions are in the form of a prohibition, for up to three years, on the issuance of building permits or the approval of subdivisions on property lying within an official map alignment. The three year reservation period begins with the request for development approval. This authority should be used carefully and only in cases where less restrictive powers are found to be ineffective.

Requests for NCDOT to prepare and adopt an official map should be directed to the manager of the Program and Policy Branch. For cities contemplating the adoption of an official map, there are two ways in which the city may proceed. The first is to consider the official map statute as a stand-alone authority and use it as the basis for local adoption of an official map. Alternatively, the second approach is to adopt a local ordinance modeled after the statute, but modified to fit local circumstances and clarify the statute. Regardless of the approach taken, several procedural steps will need to be considered, such as establishing procedures for consideration of variance petitions.

Once the project has been selected and the alignment determined, maps must be prepared that are suitable for filing with the county Register of Deeds Office. The map should show the proposed alignment in sufficient detail to identify the functional design and the preliminary right-of-way boundaries. Since the purpose of the map is to show the effect on properties along the

project path, the existing property boundaries should be identified. Within one year of the adoption of an official map, work must begin on an environmental study or preliminary engineering.

It is important to recognize the risks inherent in the adoption of an official map prior to completing the environmental studies. Projects to be funded using any federal funds require the unbiased evaluation of alternative alignments. This means that other alternatives will be studied and compared to the protected alignment. ¹

The above information serves as an introduction to official maps, and in no way provides the information necessary to begin development of an official map. The Program and Policy Branch of the North Carolina Department of Transportation is responsible for facilitating the adoption of Official Street Maps.

Urban Renewal

Urban renewal plays a minor role in the transportation planning implementation process. However, under the right circumstances, renewal programs can make significant contributions. Provisions of the New Housing Act of 1974 (as amended) call for the conservation of good areas, rehabilitation of declining areas, and clearance of slum areas. In the course of renewal, it is important to coordinate with the Thoroughfare Plan to see if additional set-backs or rights-of-way are needed.

Continued use of urban renewal programs to improve the transportation system is encouraged. Every effort should be made to ensure that community development and transportation plans are compatible.

¹ "Guidelines for Municipalities Considering Adoption of Roadway Corridor Official Maps", prepared by NCDOT Program and Policy Branch.

Cities considering Official Street Map projects should contact this Branch for their "Guidelines for Municipalities Considering Adoption of Roadway Corridor Official Maps" at:

Programming and Policy Branch
NC Department of Transportation
P.O. Box 25201
Raleigh, North Carolina 27611

Capital Improvements Program

One of the tools which makes it easier to build a planned thoroughfare system is a capital improvements program. This is a long range plan for the spending of money on street improvements, acquisition of rights-of-way, and other capital improvements within the bounds of projected revenues. Municipal funds should be available for construction of street improvements which are a municipal responsibility, right-of-way cost sharing on facilities designated as a Division of Highways responsibility, and advance purchase of right of way where such action is required.

Historically cities and towns have depended, to a great degree, on Federal or State funding to solve their transportation problems. Chapter 136-Article 3A of the Road and Highway Laws of North Carolina outlines the responsibilities of governmental bodies regarding highway improvements. North Carolina Highway Bill 1211, passed in 1988, limits the role of municipalities to specific limits in right-of-way cost sharing. Set-back regulations, right-of-way dedications and reservations play a major role in the ultimate cost of many facilities.

IX. IMPLEMENTATION

The Thoroughfare Plan is a representation of the existing highway system by functional use (major thoroughfare, minor thoroughfare, or local street) plus any new facilities which are needed. The planning methodology enables identification of deficiencies in the existing system, allowing compilation of a list of needed improvements. The list is divided into four priority groups indicating the degree of importance associated with each proposal from a technical perspective.

Priority Groups

Table 15 depicts the recommended priority groups as evaluated from a technical perspective. Priority Group 1 and 2 improvements are centered around alleviating current traffic deficiencies and strengthening the existing highway system. Strong enforcement of subdivision and zoning ordinances will enhance the implementation of these projects. Priority Group 3 projects are needed in the future to provide for anticipated growth. As many of these projects will not be implemented in the near future, continued protection of the corridors is critical. Strong enforcement of local ordinances, particularly set-back and right-of-way reservation requirements, will enhance the implementation of these projects by ensuring there will be a corridor to locate the facility. Priority Group 4 includes the remaining improvements listed in the Street Tabulation (Appendix E). Project types found in Priority Group 4 are not included in Table 15, they include widening to eleven or twelve foot lanes, minor projects for facility continuity and spacing, and projects which have merit but are not warranted within the planning period.

Physical Environmental Factors

Air quality studies have reported that the internal combustion engine used in motor vehicles causes about three-fourths of the carbon monoxide, half the hydrocarbons, and nearly half the nitrogen oxides in the atmosphere. The design of the thoroughfare system can have a crucial effect on the amount of pollutants added to the atmosphere. Auto emissions of two major pollutants causing air quality concerns in the United States today, carbon monoxide and hydrocarbons, are reduced wherever traffic is permitted to flow smoothly, and where congestion and "stop and go" driving conditions are reduced. This reduction of pollutants is created by the more efficient burning of fuel under free flow conditions. The layout of the major street system will also have an effect on air quality. A street system that provides easy and direct movement between all sections will reduce travel time and distances, thereby reducing pollutant emissions.

Water quality is a prime asset of all cities and every effort should be made during highway construction for the prevention of pollution. Guidelines have been issued outlining procedures for maintaining water quality and the reduction of possible soil erosion that takes place during and following highway construction. Due to the coastal nature of the area, it is reasonable to anticipate that wetlands will be of concern for almost any project.

Staging a Highway Project

Many considerations go into determining the actual staging of a highway project. Political, social, environmental, and economic considerations all have their influence. The adoption of the Thoroughfare Plan by each Town and the State does not include the priorities. Priorities are reviewed and set by the towns and county yearly. **Table 15** should be used to assist the decision makers - the Kill Devil Hills, Kitty Hawk, Nags Head and Southern Shores Town Councils, Dare County Commissioners, and the North Carolina Board of Transportation. **Table 15** provides an alphabetical listing, by priority grouping, of the recommended highway projects, and the estimated cost in 1992 dollars of each project. Appendix E contains the Thoroughfare Plan Street Tabulation and recommendations.

Explanation of Items in Table 15

Priority Group - There are four Priority Groups, as explained earlier in this chapter. Priority Group 4 projects not listed in **Table 15**. Priority Group 4 includes all improvements listed in Appendix E, and not included in Priorities 1 through 3.

Length - The length of the project, in miles.

Existing X-Section - This is a shortened term for "Existing Cross-Section". Additional information on cross-sections is available in Appendix E.

Proposed X-Section - This is a shortened term for "Proposed Cross-Section". For additional information, see Appendix E.

Prop. Access Control - "Proposed Access Control". "Full" denotes access with public streets only, with no private property driveways. "Partial" indicates access at a limited number of points, which can be implemented through local subdivision ordinances by requiring private lots to be served by the development's internal street network. Facilities with full access control can have a combination of interchanges and at-grade intersections. "None" in the access control column indicates all lesser forms of access control, including no control at all.

Subdivisions on major thoroughfares should be strongly encouraged to provide internal street access for all lots. This results in higher capacity for the thoroughfare, and increased safety overall due to fewer turning conflicts. In commercial areas, lack of access control results in undesirable strip development, resulting in numerous traffic signals, congestion and delays.

Const. Cost (X \$1,000) - The estimated construction cost, in thousands of dollars, of the project (in 1992 dollars). Note that right-of-way costs are in addition to construction costs, and can be 50% or more of construction costs.

Probable Impacts - Probable impacts include both economic development and environmental impact.

Economic Development - "A project can be successful by stimulating economic development in the immediate area of the project or by increasing the level of service of accessibility to an adjacent area. The probability of a project's success in achieving this objective is affected by the overall potential for economic growth in the urban area. . . . The estimate of probabilities is a subjective evaluation by the analyst based on his knowledge of the proposed project, urban development characteristics, and land development potential."

Table 13 may be used as a guideline for interpreting the "Probable Impact" values in **Table 15**.

Environmental Impact - "Environmental factors usually considered in highway project evaluation can be separated into three major categories--(1) physical environmental considerations, (2) social and/or cultural environmental considerations, and (3) economic environmental considerations. Factors included in each of these categories are given in **Table 14**."

PROBABILITY ESTIMATION GUIDE - TABLE 13	
Subjective Evaluation	Impact Probability
Excellent - very substantial	0.90
Very good - substantial	0.60
Fair - some	0.40
Poor - none	0.10

"It is desirable that the evaluation of environmental impacts be condensed to a single measure or measures easily understood by the decision maker. In the benefits matrix model the approach is to estimate the probability of a positive or negative impact for each of the environmental factors. The summations of both positive and negative probabilities are then measures of the relative environmental impact of a project.

"Information from public involvement and environmental analysis elements of urban transportation studies provide a basis for estimation of probabilities for projects. If no urban transportation studies have been made, it would be necessary to rely on other planning studies and reports, or field surveys."

Table 13 may be used as a basis for assigning probability estimates to the positive and negative environmental impacts. These values may be used as a guideline for interpreting **Table 15**.

ENVIRONMENTAL CONSIDERATIONS - TABLE 14		
Physical Environment	Social and/or Cultural Environment	Economic Environment
Air quality Water resources Soils and geology Wildlife Vegetation	Housing Neighborhoods Noise Education facilities Churches Parks and recreation Public health and safety National defense Aesthetics	Business Employment Economic Development Public Utilities Transportation costs Capital costs Operation and maintenance costs

TABLE 15

		CONSTRUCTION PRIORITIES AND COST ESTIMATES					ENVIRON. IMPACT	ECONOMIC DEVELOPMENT
HIGHWAY SECTION	CROSS SECTION	RECOMMENDED LENGTH (miles)	COST ESTIMATES (\$1,000)	CONSTRUCTION ROW TOTAL BENEFITS	POTENTIAL PROB.			
FIRST PRIORITY:								
1. US 158 (R-3419)	C/E	14.60	31,000	0	31,000	264,000	+ .40	
NC 12 Nags Head to NC 12 Kitty Hawk							- .20	
2. US 158	C/E	1.50	2,100	0	2,100	15,000	+ .40	
WCL Kitty Hawk to NC 12							- .20	
							+ .20	
SECOND PRIORITY:								
3. NC 12 (U-2917)	I	6.14	5,100	21,600	26,700	188,810	+ .30	
US 158 Kitty Hawk to Wilkinson St. KDH							- .20	
4. NC 12	I	14.32	11,500	0	11,500	39,000	+ .20	
Duck to US 158							- .00	
THIRD PRIORITY:								
5. Woods Road	I	2.06	1,500	0	1,500	2,000	+ .30	
US 158 to Kitty Hawk Road							- .00	
6. Kitty Hawk Road	I/D	1.64	1,200	500	1,700	8,000	+ .30	
US 158 to Woods Rd							- .00	

Implementation Scenarios

The construction of the third bridge to Corolla will reduce the travel burden on other major facilities, such as the Wright Memorial Bridge and NC 12 through Southern Shores and Duck. The timing of improvements to Virginia Dare Trail is critical as it may be breached, causing a diversion of traffic onto US 158. The importance can not be over emphasized for the towns to provide a united effort to protect major corridors from strip development and unregulated curb cuts. This is best accomplished through subdivision regulations. The facility which most critically needs this type of protection is US 158.

All proposed facilities on new location should be protected and implemented where possible with subdivision and zoning ordinances. These are extremely valuable tools, especially where the proposed facility traverses prime developable land.

Summary

The goal in preparing this report was to help decision makers focus on, and take advantage of opportunities to exercise the available tools to develop the transportation system for the Outer Banks Planning Area.

No single control or tool discussed is adequate to serve the transportation implementation function in and of itself. Since transportation serves development, development decisions should consider the impact that they will have on the transportation system. Lead planning time and land acquisition often requires many years before construction can begin. The existence of a Thoroughfare Plan, and use of the tools discussed in this report will allow a timely response to development.

APPENDIX A

APPENDIX A

PLANNING AREA EMPLOYMENT AND HOUSING

TABLE A-1 DWELLING UNIT DATA - 1988

ZONE	DWELLING UNITS				SPECIAL VEHICLES			TOTDU
	EXC	AAV	AVE	BA	TAXI	TRUCK	COMM AUTO	
1	2	0	120	0	0	19	0	122
2	473	0	2	0	0	0	0	475
3	0	0	0	0	0	0	0	0
4	444	170	0	0	0	1	3	614
5	75	145	0	0	0	5	2	220
6	7	0	0	0	0	1	2	7
7	37	10	0	0	0	1	2	47
8	150	470	0	0	0	12	11	620
9	43	0	30	0	0	3	1	73
10	40	0	0	0	0	0	0	40
11	0	0	0	0	0	3	0	0
12	117	126	0	0	0	3	1	243
13	20	0	0	0	0	1	0	20
14	200	0	0	0	0	1	0	200
15	80	0	0	0	0	19	0	80
16	9	0	0	0	0	3	0	9
17	222	136	0	0	0	2	0	358
18	139	0	0	0	0	2	0	139
19	316	0	0	0	0	2	1	316
20	63	219	0	0	0	30	2	282
21	84	0	0	0	0	10	0	84
22	87	0	0	0	0	47	0	87
23	258	967	0	0	0	15	10	1225
24	265	40	0	0	0	75	11	305
25	482	0	0	0	0	45	0	482
26	650	410	0	0	0	2	0	1060
27	52	0	0	0	0	10	22	52
28	2047	0	106	0	0	12	0	2153
29	1375	0	443	0	0	18	0	1818
30	360	180	0	0	0	6	9	540
31	159	0	0	0	0	27	33	159
32	1596	0	0	0	0	1	0	1596
33	210	28	150	0	0	0	0	388
34	42	0	0	0	0	24	2	42
35	179	6	0	0	0	0	0	185
36	131	105	120	0	0	1	0	356
37	264	0	0	0	0	32	0	264
38	366	20	0	0	0	112	16	386
39	389	99	0	0	0	0	0	488
40	151	0	0	0	0	8	7	151
41	44	0	0	0	0	6	0	44
42	339	0	0	0	0	1	8	339

TABLE A-1 DWELLING UNIT DATA - 1988

ZONE	DWELLING UNITS				SPECIAL VEHICLES			TOTDU
	EXC	AAV	AVE	BA	TAXI	TRUCK	COMM AUTO	
43	155	0	0	0	0	2	0	155
44	386	0	0	0	0	4	0	386
45	466	0	0	0	0	0	0	466
46	297	0	260	0	0	24	29	557
47	612	0	0	0	0	9	1	612
48	676	61	0	0	0	7	1	737
49	0	0	0	0	0	0	0	0
50	0	0	0	0	0	0	0	0
51	0	0	0	0	0	0	0	0
52	0	0	0	0	0	0	0	0
53	0	0	0	0	0	0	0	0
54	0	0	0	0	0	0	0	0
55	0	0	0	0	0	0	0	0
56	0	0	0	0	0	0	0	0
57	0	0	0	0	0	0	0	0
58	0	0	0	0	0	0	0	0
59	0	0	0	0	0	0	0	0
	14559	3192	1231	0	0	606	174	18982

TABLE A-2

POPULATION - 1988

ZONE	DWELLING UNITS				TOTPOP
	EXC	AAV	AVE	BA	
1	7	0	456	0	463
2	1750	0	8	0	1758
3	0	0	0	0	0
4	1643	476	0	0	2119
5	278	406	0	0	684
6	26	0	0	0	26
7	137	28	0	0	165
8	555	1316	0	0	1871
9	159	0	114	0	273
10	148	0	0	0	148
11	0	0	0	0	0
12	433	353	0	0	786
13	74	0	0	0	74
14	740	0	0	0	740
15	296	0	0	0	296
16	33	0	0	0	33
17	821	381	0	0	1202
18	514	0	0	0	514
19	1169	0	0	0	1169
20	233	613	0	0	846
21	311	0	0	0	311
22	322	0	0	0	322
23	955	2708	0	0	3663
24	981	112	0	0	1093
25	1783	0	0	0	1783
26	2405	1148	0	0	3553
27	192	0	0	0	192
28	7574	0	403	0	7977
29	5088	0	1683	0	6771
30	1332	504	0	0	1836
31	588	0	0	0	588
32	5905	0	0	0	5905
33	777	78	570	0	1425
34	155	0	0	0	155
35	662	17	0	0	679
36	485	294	456	0	1235
37	977	0	0	0	977
38	1354	56	0	0	1410
39	1439	277	0	0	1716
40	559	0	0	0	559
41	163	0	0	0	163
42	1254	0	0	0	1254

TABLE A-2

POPULATION - 1988

ZONE	DWELLING UNITS				TOTPOP
	EXC	AAV	AVE	BA	
43	574	0	0	0	574
44	1428	0	0	0	1428
45	1724	0	0	0	1724
46	1099	0	988	0	2087
47	2264	0	0	0	2264
48	2501	171	0	0	2672
49	0	0	0	0	0
50	0	0	0	0	0
51	0	0	0	0	0
52	0	0	0	0	0
53	0	0	0	0	0
54	0	0	0	0	0
55	0	0	0	0	0
56	0	0	0	0	0
57	0	0	0	0	0
58	0	0	0	0	0
59	0	0	0	0	0
	53867	8938	4678	0	67483
OCCUPANCY RATES					
	3.70	2.80	3.80	.00	

TABLE A-3 EMPLOYMENT DATA 1988

ZONE	X1	X2	X3	X4	X5	X6	X7	X8	X9
1	0	0	0	150	0	0	0	160	56
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
4	0	0	0	34	0	0	0	92	0
5	9	37	123	25	0	0	0	23	0
6	0	10	88	5	2	0	0	0	0
7	0	18	67	2	0	0	0	0	0
8	0	28	176	127	39	0	0	108	0
9	0	10	138	62	4	0	0	0	0
10	0	0	0	0	0	0	0	23	0
11	0	285	35	30	10	0	0	0	0
12	75	67	36	27	19	0	0	14	0
13	0	0	20	2	0	0	0	0	0
14	4	0	0	4	0	0	0	16	0
15	0	175	22	0	8	0	0	20	0
16	0	0	0	4	0	0	0	200	0
17	4	22	31	92	26	0	0	147	0
18	10	104	5	31	15	0	0	0	0
19	0	0	0	133	0	0	0	0	0
20	0	79	62	123	0	0	0	30	0
21	14	187	220	126	77	0	0	0	0
22	36	0	0	4	39	0	0	10	0
23	0	138	412	414	12	0	0	63	0
24	71	91	755	194	195	0	0	0	0
25	0	0	0	0	43	0	0	0	0
26	0	7	102	170	0	0	0	229	0
27	0	25	20	113	5	0	0	0	0
28	24	4	0	16	8	0	0	150	10
29	82	5	4	10	0	0	0	0	20
30	0	3	64	61	24	0	0	4	0
31	87	54	45	108	102	0	0	0	0
32	3	0	0	0	0	0	0	3	0
33	0	0	5	28	0	0	0	2	0
34	15	22	77	51	51	0	0	0	0
35	0	0	0	0	0	0	0	0	0
36	0	36	0	8	0	0	0	41	0
37	65	164	66	62	4	0	0	0	0
38	206	0	0	18	15	0	0	0	0
39	0	29	18	15	0	0	0	12	0
40	22	32	5	37	105	0	0	0	0
41	5	28	0	9	0	0	0	0	0
42	4	122	30	54	69	0	0	40	0

TABLE A-3 EMPLOYMENT DATA 1988

ZONE	X1	X2	X3	X4	X5	X6	X7	X8	X9
43	0	0	0	36	11	0	0	0	0
44	0	0	0	0	0	0	0	0	36
45	0	0	0	0	0	0	0	75	0
46	0	250	17	160	90	0	0	0	0
47	4	116	59	42	61	0	0	0	0
48	0	3	0	85	12	0	0	33	0
49	0	0	0	0	0	0	0	0	0
50	0	0	0	0	0	0	0	0	0
51	0	0	0	0	0	0	0	0	0
52	0	0	0	0	0	0	0	0	0
53	0	0	0	0	0	0	0	0	0
54	0	0	0	0	0	0	0	0	0
55	0	0	0	0	0	0	0	0	0
56	0	0	0	0	0	0	0	0	0
57	0	0	0	0	0	0	0	0	0
58	0	0	0	0	0	0	0	0	0
59	0	0	0	0	0	0	0	0	0
	740	2151	2702	2672	1046	0	0	1495	122

X1 - INDUSTRY
 X2 - RETAIL
 X3 - HIGHWAY RETAIL
 X4 - SERVICE
 X5 - OFFICE
 X6 - DWELLING UNITS
 X7 - TOURIST DAY PARKING
 X8 - CHARTER BOATS
 X9 - BOATS AT MARINA

TABLE A-4 DWELLING UNIT DATA - 2010 "BUILDOUT" SCENARIO

ZONE	DWELLING UNITS				SPECIAL VEHICLES			TOTDU
	EXC	AAV	AVE	BA	TAXI	TRUCK	COMM AUTO	
1	10	0	120	0	0	21	4	130
2	600	0	0	0	0	3	10	600
3	0	0	0	0	0	0	0	0
4	625	200	0	0	0	7	5	825
5	80	220	0	0	0	19	6	300
6	25	0	0	0	0	5	3	25
7	55	0	0	0	0	3	3	55
8	405	525	0	0	0	35	18	930
9	1250	0	0	0	0	35	11	1250
10	180	500	0	0	0	27	8	680
11	350	0	0	0	0	12	3	350
12	225	140	0	0	0	18	6	365
13	30	0	0	0	0	30	9	30
14	615	0	0	0	0	3	1	615
15	130	0	0	0	0	28	3	130
16	10	0	0	0	0	6	1	10
17	320	170	0	0	0	13	3	490
18	350	0	0	0	4	30	9	350
19	625	0	0	0	0	8	3	625
20	180	200	0	0	0	40	5	380
21	210	0	0	0	0	25	5	210
22	410	0	0	0	0	53	2	410
23	850	1150	0	0	0	68	25	2000
24	500	50	0	0	0	160	36	550
25	1000	75	0	0	0	155	32	1075
26	750	600	0	0	0	54	16	1350
27	150	0	0	0	6	81	42	150
28	2450	0	0	0	0	34	7	2450
29	3800	0	250	0	0	43	7	4050
30	440	300	0	0	0	25	15	740
31	210	75	0	0	0	85	50	285
32	2500	0	0	0	0	5	2	2500
33	380	100	100	0	0	19	6	580
34	140	0	0	0	0	54	11	140
35	750	0	0	0	0	3	1	750
36	250	155	100	0	0	27	8	505
37	620	0	0	0	0	56	7	620
38	1100	0	0	0	0	129	21	1100
39	600	400	0	0	0	37	11	1000
40	880	0	0	0	0	51	20	880
41	800	100	0	0	0	113	31	900
42	650	0	0	0	0	23	25	650

TABLE A-4 DWELLING UNIT DATA - 2010 "BUILDOUT" SCENARIO

ZONE	DWELLING UNITS				SPECIAL VEHICLES			TOTDU
	EXC	AAV	AVE	BA	TAXI	TRUCK	COMM AUTO	
43	480	0	0	0	0	5	7	480
44	750	0	0	0	0	6	1	750
45	900	0	0	0	0	0	0	900
46	720	0	260	0	0	51	37	980
47	930	0	0	0	0	33	13	930
48	1000	90	0	0	0	26	7	1090
49	0	0	0	0	0	0	0	0
50	0	0	0	0	0	0	0	0
51	0	0	0	0	0	0	0	0
52	0	0	0	0	0	0	0	0
53	0	0	0	0	0	0	0	0
54	0	0	0	0	0	0	0	0
55	0	0	0	0	0	0	0	0
56	0	0	0	0	0	0	0	0
57	0	0	0	0	0	0	0	0
58	0	0	0	0	0	0	0	0
59	0	0	0	0	0	0	0	0
	30285	5050	830	0	10	1764	556	36165

TABLE A-5

POPULATION - 2010 "BUILDOUT" SCENARIO

ZONE	DWELLING UNITS				TOTPOP
	EXC	AAV	AVE	BA	
1	36	0	444	0	480
2	2160	0	0	0	2160
3	0	0	0	0	0
4	2250	560	0	0	2810
5	288	616	0	0	904
6	90	0	0	0	90
7	198	0	0	0	198
8	1458	1470	0	0	2928
9	4500	0	0	0	4500
10	648	1400	0	0	2048
11	1260	0	0	0	1260
12	810	392	0	0	1202
13	108	0	0	0	108
14	2214	0	0	0	2214
15	468	0	0	0	468
16	36	0	0	0	36
17	1152	476	0	0	1628
18	1260	0	0	0	1260
19	2250	0	0	0	2250
20	648	560	0	0	1208
21	756	0	0	0	756
22	1476	0	0	0	1476
23	3060	3220	0	0	6280
24	1800	140	0	0	1940
25	3600	210	0	0	3810
26	2700	1680	0	0	4380
27	540	0	0	0	540
28	8820	0	0	0	8820
29	13680	0	925	0	14605
30	1584	840	0	0	2424
31	756	210	0	0	966
32	9000	0	0	0	9000
33	1368	280	370	0	2018
34	504	0	0	0	504
35	2700	0	0	0	2700
36	900	434	370	0	1704
37	2232	0	0	0	2232
38	3960	0	0	0	3960
39	2160	1120	0	0	3280
40	3168	0	0	0	3168
41	2880	280	0	0	3160
42	2340	0	0	0	2340

TABLE A-5

POPULATION - 2010 "BUILDOUT" SCENARIO

ZONE	DWELLING UNITS				TOTPOP
	EXC	AAV	AVE	BA	
43	1728	0	0	0	1728
44	2700	0	0	0	2700
45	3240	0	0	0	3240
46	2592	0	962	0	3554
47	3348	0	0	0	3348
48	3600	252	0	0	3852
49	0	0	0	0	0
50	0	0	0	0	0
51	0	0	0	0	0
52	0	0	0	0	0
53	0	0	0	0	0
54	0	0	0	0	0
55	0	0	0	0	0
56	0	0	0	0	0
57	0	0	0	0	0
58	0	0	0	0	0
59	0	0	0	0	0
	109026	14140	3071	0	126237
OCCUPANCY RATES					
	3.60	2.80	3.70	.00	

TABLE A-6 EMPLOYMENT DATA 2010 "BUILDOUT" SCENARIO

ZONE	X1	X2	X3	X4	X5	X6	X7	X8	X9
1	0	0	0	175	0	0	0	180	70
2	0	0	0	40	0	0	0	15	0
3	0	0	0	0	0	0	0	0	0
4	0	25	40	65	0	0	0	130	0
5	15	70	210	100	15	0	0	25	0
6	0	30	110	25	5	0	0	0	10
7	0	40	85	0	0	0	0	0	10
8	0	210	240	210	65	0	0	125	0
9	0	270	300	120	20	0	0	0	10
10	0	40	55	320	0	0	0	35	0
11	0	330	90	60	30	0	0	0	0
12	100	120	85	115	45	0	0	40	0
13	30	180	170	70	30	0	0	0	0
14	20	0	0	15	0	0	0	25	15
15	0	220	80	30	25	0	0	40	0
16	0	20	0	30	0	0	0	240	0
17	15	60	75	140	40	0	0	160	0
18	50	250	125	100	75	0	0	0	0
19	0	15	15	210	0	0	0	0	0
20	0	110	130	180	0	0	0	40	0
21	35	240	280	200	105	0	0	0	0
22	70	0	0	40	60	0	0	10	0
23	0	300	600	850	50	0	0	90	0
24	350	450	1250	350	250	0	0	0	0
25	680	300	250	300	250	0	0	0	0
26	0	200	400	450	40	0	0	250	0
27	15	600	350	200	100	0	0	0	0
28	55	120	50	100	80	0	0	170	15
29	150	200	50	60	40	0	0	0	40
30	20	80	150	160	60	0	0	25	0
31	110	400	280	350	150	0	0	0	0
32	20	0	0	50	0	0	0	3	0
33	0	80	100	150	0	0	0	20	0
34	40	150	200	150	140	0	0	0	0
35	0	0	0	35	0	0	0	0	0
36	0	160	100	180	0	0	0	60	0
37	100	300	150	150	30	0	0	0	0
38	300	0	0	150	50	0	0	0	10
39	0	150	220	250	0	0	0	50	0
40	50	250	200	200	170	0	0	0	0
41	30	900	200	500	50	0	0	0	0
42	15	220	110	120	150	0	0	40	0

TABLE A-6 EMPLOYMENT DATA 2010 "BUILDOUT" SCENARIO

ZONE	X1	X2	X3	X4	X5	X6	X7	X8	X9
43	0	80	40	230	40	0	0	0	10
44	0	0	0	0	25	0	0	20	40
45	0	0	0	0	0	0	0	75	10
46	0	350	150	280	150	0	0	0	0
47	25	250	100	100	150	0	0	15	10
48	0	60	110	200	20	0	0	50	0
49	0	0	0	0	0	0	0	0	0
50	0	0	0	0	0	0	0	0	0
51	0	0	0	0	0	0	0	0	0
52	0	0	0	0	0	0	0	0	0
53	0	0	0	0	0	0	0	0	0
54	0	0	0	0	0	0	0	0	0
55	0	0	0	0	0	0	0	0	0
56	0	0	0	0	0	0	0	0	0
57	0	0	0	0	0	0	0	0	0
58	0	0	0	0	0	0	0	0	0
59	0	0	0	0	0	0	0	0	0
	2295	7830	7150	7810	2510	0	0	1933	250

X1 - INDUSTRY
 X2 - RETAIL
 X3 - HIGHWAY RETAIL
 X4 - SERVICE
 X5 - OFFICE
 X6 - DWELLING UNITS
 X7 - TOURIST DAY PARKING
 X8 - CHARTER BOATS
 X9 - BOATS AT MARINA

TABLE A-7 DWELLING UNIT DATA - 2010 "REDEVELOPMENT" SCENARIO

ZONE	DWELLING UNITS				SPECIAL VEHICLES			TOTDU
	EXC	AAV	AVE	BA	TAXI	TRUCK	COMM AUTO	
1	15	0	150	0	0	22	10	165
2	650	0	0	0	0	6	12	650
3	0	0	0	0	0	0	0	0
4	650	250	0	0	0	14	7	900
5	80	250	0	0	0	23	8	330
6	30	0	0	0	0	6	4	30
7	60	0	0	0	0	4	3	60
8	410	550	0	0	0	51	23	960
9	1425	75	0	0	0	47	14	1500
10	200	520	0	0	0	46	14	720
11	350	0	0	0	0	27	9	350
12	245	150	0	0	0	19	7	395
13	40	0	0	0	0	46	14	40
14	650	0	0	0	0	9	3	650
15	150	50	0	0	0	32	4	200
16	15	0	0	0	0	6	2	15
17	330	210	0	0	0	18	5	540
18	325	0	0	0	8	46	14	325
19	700	0	0	0	0	14	5	700
20	150	260	0	0	0	47	8	410
21	160	0	0	0	0	36	8	160
22	460	0	0	0	0	55	5	460
23	900	1250	0	0	0	89	32	2150
24	500	50	0	0	0	150	42	550
25	1150	100	0	0	0	160	43	1250
26	850	650	0	0	0	78	23	1500
27	150	0	0	0	10	104	50	150
28	2600	0	0	0	0	45	10	2600
29	4100	0	200	0	0	70	16	4300
30	500	400	0	0	0	41	20	900
31	210	50	0	0	0	91	52	260
32	2600	0	0	0	0	7	2	2600
33	400	150	80	0	0	29	9	630
34	130	0	0	0	0	66	15	130
35	750	0	0	0	0	7	2	750
36	250	250	50	0	0	41	12	550
37	650	0	0	0	0	71	12	650
38	1200	0	0	0	0	120	26	1200
39	600	450	0	0	0	47	14	1050
40	880	0	0	0	0	74	27	880
41	850	150	0	0	0	121	33	1000
42	700	50	0	0	0	65	30	750

TABLE A-7 DWELLING UNIT DATA - 2010 "REDEVELOPMENT" SCENARIO

ZONE	DWELLING UNITS				SPECIAL VEHICLES			TOTDU
	EXC	AAV	AVE	BA	TAXI	TRUCK	COMM AUTO	
43	550	0	0	0	0	47	13	550
44	820	50	0	0	0	12	3	870
45	950	0	0	0	0	0	0	950
46	760	50	120	0	0	78	50	930
47	1000	0	0	0	6	75	20	1000
48	1050	150	0	0	0	37	10	1200
49	0	0	0	0	0	0	0	0
50	0	0	0	0	0	0	0	0
51	0	0	0	0	0	0	0	0
52	0	0	0	0	0	0	0	0
53	0	0	0	0	0	0	0	0
54	0	0	0	0	0	0	0	0
55	0	0	0	0	0	0	0	0
56	0	0	0	0	0	0	0	0
57	0	0	0	0	0	0	0	0
58	0	0	0	0	0	0	0	0
59	0	0	0	0	0	0	0	0
	32195	6115	600	0	24	2299	745	38910

TABLE A-8

POPULATION - 2010 "REDEVELOPMENT" SCENARIO

ZONE	DWELLING UNITS				TOTPOP
	EXC	AAV	AVE	BA	
1	54	0	555	0	609
2	2340	0	0	0	2340
3	0	0	0	0	0
4	2340	700	0	0	3040
5	288	700	0	0	988
6	108	0	0	0	108
7	216	0	0	0	216
8	1476	1540	0	0	3016
9	5130	210	0	0	5340
10	720	1456	0	0	2176
11	1260	0	0	0	1260
12	882	420	0	0	1302
13	144	0	0	0	144
14	2340	0	0	0	2340
15	540	140	0	0	680
16	54	0	0	0	54
17	1188	588	0	0	1776
18	1170	0	0	0	1170
19	2520	0	0	0	2520
20	540	728	0	0	1268
21	576	0	0	0	576
22	1656	0	0	0	1656
23	3240	3500	0	0	6740
24	1800	140	0	0	1940
25	4140	280	0	0	4420
26	3060	1820	0	0	4880
27	540	0	0	0	540
28	9360	0	0	0	9360
29	14760	0	740	0	15500
30	1800	1120	0	0	2920
31	756	140	0	0	896
32	9360	0	0	0	9360
33	1440	420	296	0	2156
34	468	0	0	0	468
35	2700	0	0	0	2700
36	900	700	185	0	1785
37	2340	0	0	0	2340
38	4320	0	0	0	4320
39	2160	1260	0	0	3420
40	3168	0	0	0	3168
41	3060	420	0	0	3480
42	2520	140	0	0	2660

TABLE A-8 POPULATION - 2010 "REDEVELOPMENT" SCENARIO

ZONE	DWELLING UNITS				TOTPOP
	EXC	AAV	AVE	BA	
43	1980	0	0	0	1980
44	2952	140	0	0	3092
45	3420	0	0	0	3420
46	2736	140	444	0	3320
47	3600	0	0	0	3600
48	3780	420	0	0	4200
49	0	0	0	0	0
50	0	0	0	0	0
51	0	0	0	0	0
52	0	0	0	0	0
53	0	0	0	0	0
54	0	0	0	0	0
55	0	0	0	0	0
56	0	0	0	0	0
57	0	0	0	0	0
58	0	0	0	0	0
59	0	0	0	0	0
	115902	17122	2220	0	135244
OCCUPANCY RATES					
	3.60	2.80	3.70	.00	

TABLE A-9 EMPLOYMENT DATA 2010 "REDEVELOPMENT" SCENARIO

ZONE	X1	X2	X3	X4	X5	X6	X7	X8	X9
1	0	0	0	195	0	0	0	200	70
2	0	0	20	60	0	0	0	25	0
3	0	0	0	0	0	0	0	0	0
4	0	60	80	95	0	0	0	150	0
5	20	100	235	100	30	0	0	25	0
6	0	35	125	35	0	0	0	0	10
7	0	50	90	0	0	0	0	0	10
8	0	260	300	320	90	0	0	140	0
9	0	300	380	150	70	0	0	0	25
10	0	135	155	400	20	0	0	35	0
11	0	400	160	90	140	0	0	0	0
12	105	130	100	130	65	0	0	50	0
13	50	250	240	130	70	0	0	0	0
14	25	30	30	50	0	0	0	25	15
15	0	240	95	40	40	0	0	55	0
16	0	25	0	35	0	0	0	260	0
17	20	75	95	160	60	0	0	180	0
18	65	300	175	190	150	0	0	0	0
19	0	40	40	240	0	0	0	0	3
20	0	150	170	220	0	0	0	40	0
21	40	260	330	260	135	0	0	0	0
22	85	35	30	70	90	0	0	15	0
23	0	400	675	950	100	0	0	110	0
24	425	550	1300	425	250	0	0	0	0
25	800	400	325	400	375	0	0	0	0
26	0	300	500	600	60	0	0	260	0
27	40	650	500	270	150	0	0	0	0
28	60	170	75	150	120	0	0	200	15
29	225	400	110	90	80	0	0	0	50
30	30	150	250	200	80	0	0	35	0
31	130	500	340	220	200	0	0	0	0
32	30	20	0	60	0	0	0	10	5
33	0	110	130	240	0	0	0	20	0
34	55	200	250	200	160	0	0	0	0
35	0	60	0	50	0	0	0	0	0
36	0	250	200	240	0	0	0	80	0
37	130	350	200	225	60	0	0	0	0
38	360	100	40	175	75	0	0	0	20
39	0	250	250	300	0	0	0	85	0
40	75	400	300	250	200	0	0	0	0
41	40	930	240	525	65	0	0	0	5
42	30	350	140	300	200	0	0	65	0

TABLE A-9 EMPLOYMENT DATA 2010 "REDEVELOPMENT" SCENARIO

ZONE	X1	X2	X3	X4	X5	X6	X7	X8	X9
43	0	200	150	330	55	0	0	0	15
44	0	20	25	30	50	0	0	40	40
45	0	0	0	0	0	0	0	75	10
46	0	450	300	400	200	0	0	10	0
47	100	375	200	250	200	0	0	25	12
48	0	120	200	275	60	0	0	80	0
49	0	0	0	0	0	0	0	0	0
50	0	0	0	0	0	0	0	0	0
51	0	0	0	0	0	0	0	0	0
52	0	0	0	0	0	0	0	0	0
53	0	0	0	0	0	0	0	0	0
54	0	0	0	0	0	0	0	0	0
55	0	0	0	0	0	0	0	0	0
56	0	0	0	0	0	0	0	0	0
57	0	0	0	0	0	0	0	0	0
58	0	0	0	0	0	0	0	0	0
59	0	0	0	0	0	0	0	0	0
	2940	10580	9550	10125	3700	0	0	2295	305

X1 - INDUSTRY
 X2 - RETAIL
 X3 - HIGHWAY RETAIL
 X4 - SERVICE
 X5 - OFFICE
 X6 - DWELLING UNITS
 X7 - TOURIST DAY PARKING
 X8 - CHARTER BOATS
 X9 - BOATS AT MARINA

APPENDIX B

APPENDIX B

TRAVEL FORECAST MODELS

1988 Travel

A synthetic method was used to estimate 1988 internal trip patterns. This method consisted of the following general steps:

1. Determination of zonal trip productions based on trip generation rates.
2. Determination of trip attraction factors by using a multiple regression analysis procedure.
3. Trip distribution by a three purpose gravity model using trip length frequency curves.
4. Traffic assignments to the existing network and accuracy checks on the procedures.

Internal Trip Productions

Average daily trip productions were estimated on a zonal basis in four categories: (1) trips produced by dwelling units, (2) trips produced by trucks, (3) trips produced by commercial passenger vehicles, and (4) trips produced by taxis.

Trips produced by dwelling units were computed by multiplying the appropriate trip generation rate by the number of dwelling units in each zone. For each category of dwelling unit (domicile, hotel/motel, and campground) a different generation rate was used.

The trip generation rate for trucks was obtained from home interviews conducted in conjunction with a study in Wilson N.C. in the mid-1960's. The truck generation rate was estimated to be 6.7 trips per vehicle per day. Commercially owned passenger vehicles were also estimated to make 6.7 trips per vehicle, based on the assumption that the trip generation rates would be similar to that for trucks. The trip generation rate for taxis was estimated to be 40 trips per vehicle per day.

Since the total trips generated by these rates contained both internal-to-external and internal-internal trips, it was necessary to reduce the total trip production in order to not double count the internal-external trips originating within the planning area.

It was estimated that 95% of the internally generated trips remained internal, leaving 5% for the internal <--> external movement. The total internal trip table was therefore reduced by five percent.

TABLE B-1 1988 TRIPS BY HOUSING TYPE			
Housing Category	1988 Trip Generation Rate	Number of DU's	Trips
Domicile	5.5	14,759	81,175
Hotel / Motel	4.0	3,192	12,768
Campground	2.5	1,321	3,303
TOTAL		19,272	97,246

Internal-internal (zonal trip) productions were then divided into the three purposes: home based work trips (HBW), other home based trips (OHB), and non-home based trips (NHB). The percentages used were: HBW - 18%, OHB - 56%, and NHB - 26%. A low HBW percentage was chosen as the tourist traffic does not contribute to this purpose.

The remaining component of internal trips are "secondary" trips; that is NHB trips produced by vehicles garaged outside, but having both trip ends, within the planning area. It was initially assumed that 45% of all external-internal trips made by vehicles garaged outside also make a "secondary" or NHB trip inside the cordon.

$$(147,544) \times (0.05) = 7,377 \text{ Internal } \leftrightarrow \text{ External trips}$$

$$35,980 - 7,377 = 28,603 \text{ External } \rightarrow \text{ Internal trips}$$

$$\text{NHB2} = (28,603) \times 0.45 = 12,872$$

due to the nature of the area, this number was reduced to 6,000.

where: 147,544 = total internally generated trips

0.05 = 5% of the total internally generated trips are Internal $\rightarrow$ External trips.

35,980 = External $\leftrightarrow$ Internal trips from Station Analysis

The figure arrived at by this method was then added to the previously calculated NHB trips produced by internally garaged vehicles. The total NHB trips were then redistributed to the traffic zones based on the trip attraction factor for NHB trips. Trip generation rates for both the base and future years are shown in Table B-2.

Trip Attractions

Trip attractions for the various zones in the study area were considered to be related to employment characteristics. Home-based work trip attraction factors were taken to be the total employment within the zone. Factors for other-home based and non-home based trips were based on employment stratification for each zone. These employment figures were used as the independent variable in a multiple regression analysis. Employment by zone for the base year is shown in Appendix A, **Table A-3**.

The equation for the regression analysis is as follows:

$$Y = + 0.70X_1 + 1.30X_2 + 5.80X_3 + 0.2805X_4 + 0.70X_5 + 4.00X_8 + 0.21X_9 + 0.50X_{10}$$

where Y = attraction factor for each zone

X₁ = Industry (SIC codes 1-49)

X₂ = Retail (SIC codes 50-54, 56, 57, 59)

X₃ = Highway retail (SIC codes 55, 58)

X₄ = Services (SIC codes 70-76, 78-89, 99)

X₅ = Office (SIC codes 60-67, 91-97)

X₈ = tourist parking

X₉ = boat slips

X₁₀ = housing

The widespread development of fast food restaurants, convenience stores, and banks with drive-in windows have caused these traditionally Retail (X₂) establishments to operate at much higher traffic generation rates than their predecessors. Consequently, these categories were grouped together as Highway Related Retail (X₃).

The zonal attraction factors thus derived were adjusted so that the total productions equaled the total attractions. This adjustment was done by multiplying each zonal attraction factor by the ratio of total productions to total unadjusted attractions for each trip category.

Trip Distribution

The gravity model trip distribution program was used to distribute internal trips. Input to this program included (1) zone-to-zone travel times obtained from a "traffic paths" computer program utilizing the existing 1988 street network, (2) individual zonal trip productions and attractions, and (3) trip length frequency curves.

1988 External - Internal and Through Trips

Traffic counts were taken at 4 Stations located around the cordon. These locations represented the only four vehicle access points into the study area: the three bridges and NC 12 to the

north. These counts were hand adjusted and distributed between the stations such that the resulting count totals at each station was consistent with the ground counts. The resulting trip data is shown in **Table B-4**.

Accuracy Checks

The models ability to simulate travel patterns in the area was checked by a comparison of assigned traffic to actual traffic counts taken by the North Carolina Department of Transportation. For the Outer Banks study, four screenlines (A through D) were identified for traffic count comparisons. Screenlines are shown on the Zone Map, **Figure 4**.

Screenline A runs east-west across NC 12 at the northern City Limits of Southern Shores. There was one traffic count along the screenline on NC 12. The computer count was 82% of the ground count. Screenline B evaluated the east-west traffic flow on US 158 and NC 12 near the City Limits of Kitty Hawk/Kill Devil Hills (KDH). The total computer count of 33,962 vpd was 105% of the total ground count of 32,300 vpd. Screenline C was an east-west line with two traffic counts on US 158 and NC 12 at the City Limits of KDH/Nags Head. The total computer count of 46,970 vpd was 102% of the total actual count of 46,000 vpd. Screenline D was again an east-west line, which ran across NC 12 and SR 1243 south of the US 158/US 264 intersection. This screenline was low, with the computer count only 73% of the ground count. Future traffic counts were hand adjusted to lessen the effect of this.

Design Year Travel (year 2010)

2010 Internal Trips

These travel patterns were estimated by projecting the socio-economic data to the year 2010, and then utilizing the 1988 internal travel development procedures to estimate the 2010 travel. The towns of Kill Devil Hills, Kitty Hawk, Nags Head, and Southern Shores provided projections of housing by zone for the year 2010 buildout scenario. For 2010 redevelopment, the 2010 buildout was factored up and reviewed by the local areas.

Trip generation rates were determined for the year 2010 by increasing the 1988 rates by an amount determined using the methodology explained below.

$$\text{Increase in GR} = (\text{CF} \times \text{Avg GR}) - \text{avg GR}$$

CF = Composite factor

GR = Generation Rate

$$\text{Average 1988 GR} = \text{Trips} / \text{DU} = 5.05$$

$$CF = \frac{\text{Vehicle Occupancy 1988}}{\text{Vehicle Occupancy 2010}} \times \frac{\text{Persons per DU 2010}}{\text{Persons per DU 1988}} \times 0.95$$

Veh.Occup. 1988 = 1.5
Veh.Occup. 2010 = 1.2

Persons/DU 1988 = 3.7
Persons/DU 2010 = 3.6

$$CF = \frac{1.5}{1.2} \times \frac{3.6}{3.7} \times 0.95 = 1.1$$

$$\begin{aligned} \text{Increase in GR} &= (CF \times \text{avg. GR}) - \text{avg. GR} \\ &= (1.1 \times 5.05) - 5.05 = 0.5 \end{aligned}$$

The trip generation rates for the year 2010 were then derived by adding 0.5 to the generation rates used for the base year 1988. The trip generation rates for the study are shown in Table B-2 below. The same generation rates were used for 2010 Buildout and 2010 Redevelopment.

The year 2010 trip generation rates were then applied to the DU's projected for each zone. The 2010 trip generation rates for trucks and commercially owned passenger cars was assumed to remain at 6.7 trips per vehicle. It was assumed that the ratio of these vehicles to employment would remain constant in each zone throughout the planning period. Based on the projected employment in the area, it was estimated that the number of commercially owned passenger cars would grow from 174 in 1988 to 707 in 2010 Redevelopment scenario, and trucks from 606 to 2119.

TABLE - B2
DWELLING UNIT TRIP GENERATION RATES

Housing Category	Trip Generation Rates		
	1988	2010 Bldout	2010 Redev.
Domicile	5.5	6.0	6.0
Hotel / Motel	4.0	4.5	4.5
Campground	2.5	3.0	3.0

As no taxis currently operate in the planning area, taxi projections were made based on input from local staff.

The breakdown of internal trips by purpose and percentage of non-home based trips generated externally are shown in Table B-3.

TABLE B-3 TRAVEL DATA SUMMARY			
Type	1988	2010 Bldout	2010 Red
Average Daily Trips per DU	5.05	5.72	5.72
Internal Trips			
Home Based Work	17,297	41,899	45,841
Other Home Based	53,813	113,126	123,772
Non-Home Based, internal	24,985	54,468	59,594
NHB secondary	6,000	15,500	16,000
Internal <-> External	35,980	82,860	88,565
Through Trips	4,060	7,495	7,617
TOTAL DAILY TRIPS	134,015	300,358	326,155

* differences are due to rounding

Trip attraction factors for OHB and NHB purposes were determined by using the 1988 regression equation with projected 2010 zonal employment and dwelling unit data. Trip attraction factors for HBW trips were taken as the total projected zonal employment for the design year. The distribution of 2010 buildout employment, as shown in Appendix A, **Table A-6**, was based on expected land use development as determined by the staffs of the local governing areas. The distribution of 2010 Redevelopment employment, as shown in Appendix A, **Table A-9**, was based on expected land use development and redevelopment. Design year internal trips were again distributed by the gravity model.

TABLE B-4
CORDON STATION TRAVEL

COMPUTER STATION AND LOCATION	BASE YEAR - 1988			2010 BUILDOUT			% Grwth Ftr
	Total ADT	Thru Trip End	Ext-Int Trips	Total ADT	Thru Trip End	Ext-Int Trips	
59 SR 1200 Dare Co. line	4,100	560	3,540	*12,000	1,440	10,560	5.0
56 NC 12 near Bonner Br	7,500	3,040	4,460	12,650	4,550	8,100	2.4
57 US 64/264 Nags Head	18,500	2,500	16,000	37,800	4,750	33,050	3.3
58 US 158 Southern Shore	14,000	2,020	11,980	*35,400	4,250	31,150	4.3

% Growth is the growth rate per year over the planning period, as determined using the formula:

$$\text{Future ADT} = \text{Present ADT} (1 + \text{growth rate})^{\text{number of years}}$$

* Table B-4 reflects a worst case scenario. It does not assume a third bridge to Corolla. As discussed in Chapter IV, if a third bridge is built, the total ADT at Station 59 should be reduced to approximately 8,000 ADT for the 2010 Buildout Scenario. In addition, the projection for Station 58 should be reduced to 31,400 ADT.

COMPTON
STREET
MAY
MOUNTAIN

SR 111
Date Rec
AC
Near
US 111
US 111
US 111
US 111

US 111
US 111
US 111
US 111
US 111
US 111
US 111
US 111

APPENDIX C



APPENDIX C

SURVEY INFORMATION

GOALS AND OBJECTIVE SURVEY:

General

A copy of the Goals and Objectives Survey, entitled "ROADS" is included in Appendix C. It was distributed to the local public, elected officials, and staff. Each returned survey was assigned a survey number. Those surveys distributed in Kill Devil Hills were assigned a number beginning with "KDH", and then numbered sequentially. The same tabulation was applied to the surveys received from Kitty Hawk (KH), Nags Head (NH), and Southern Shores (SS). The remainder of the surveys were assigned a number beginning with "OT", and then numbered sequentially. The survey is followed by a compilation of the results of the survey.

Next is a partial listing of the comments received from each Town as a part of the survey. Each comment is preceded by the unique identifying number assigned to the returned survey. Most surveys did not include a comment.

Because of the large number of returned comments, only ten comments from each town are included in this report. The remaining comments have been submitted to the planning department of each town. Contact your local representative for additional information.

ROADS

Thank you for taking the time to complete this questionnaire. Each town has its own priorities and goals for transportation planning. The goals are to a large extent a series of trade-offs. This survey will help to determine your preference, and will be used in transportation planning.

I. Which type of facilities are needed most ? Please rank the following, 1 through 6, in order of need (1 being most needed):

- _____ Bicycle travel facilities
- _____ Improved access to industry and shopping centers
- _____ Improved access to residential areas
- _____ New sidewalks
- _____ More beach access (parking)
- _____ Better beach access (bridges)

II. Please indicate how desirable each of the following items is by circling the appropriate number on the scale.

	Not Desirable		Neutral		Most Desirable
1. Bicycle travel facilities	1	2	3	4	5
2. Taxi service	1	2	3	4	5
3. Bus transit	1	2	3	4	5
4. Sidewalks	1	2	3	4	5
5. Transportation for elderly and handicapped	1	2	3	4	5
6. Preservation of historic building	1	2	3	4	5
7. Trees and shrubs along streets	1	2	3	4	5
8. Carpool / vanpool program	1	2	3	4	5
9. Improve industry and shopping access	1	2	3	4	5
10. Improve intersections	1	2	3	4	5

III. Please answer the following questions:

a) Which of the following would you support in order to promote carpooling / vanpooling ? (Check as many as apply)

- _____ restricted parking
- _____ high parking rates
- _____ lack of new roads

OVER PLEASE

- b) Building a new highway has many impacts (better traffic service, yet taking land, homes and businesses). However, not building a highway may have negative impacts because traffic will increase on roads through developments. Therefore, traffic congestion, gas consumption, and air and noise pollution, near existing homes would increase. Should environmental factors be (check one):

_____ more important than other factors (cost, taking
of homes, convenience of travel, etc)?
_____ equal to other factors?
_____ only moderately important?

- c) It is often difficult to decide what damages a neighborhood. For example; is it more damaging to widen an existing roadway, or to leave the road congested? The heavy traffic can be bad for a neighborhood because of the noise and air pollution, hazards to children, and other problems. Should widening be considered for residential streets with heavy traffic?

_____yes _____no

- d) How often do you use the Wright Memorial Bridge?

_____ daily
_____ at least once a week
_____ less than once a week

- e) Would you be in favor of charging bridge tolls as a means of improving access to the Outer Banks from the Currituck County mainland?

_____yes _____no

- f) The amount of money available to build highways is limited. Typically, the more a street costs, the better it will function. A more expensive project (a new road, or major widening project) will be better and more convenient than less expensive alternatives (such as minor widening, using narrow lanes, or intersection improvements). For the same total cost, a series of less expensive improvements may provide equal convenience to one large, "first class" improvement. Would you prefer to see...

_____ fewer improvements, but "first class" ?
_____ more, but less expensive improvements ?

- g) Are you a permanent resident of....

_____ Dare County _____ Currituck County
_____ Other N.C. County _____ Other

- h) Any comments?

Return to: Any place this form is being distributed
or G. Ferguson, Planning Department, Nags Head Town Hall
or D. Diaz, Kill Devil Hills Town Hall
or D. Monroe, Planning Department, Kitty Hawk Town Hall
or C. Hackney, Southern Shores Town Hall
or D. Hutchings / Statewide Planning / PO Box 25201 / Raleigh N.C. 27611

OUTER BANKS ROAD SURVEY TABULATION

	T	NH	KDH	KH	SS
1) Which type of facilities are most needed (1 being most needed)					
Bicycle facilities	3.0	3.2	2.7	3.5	3.0
Bridges	3.0	3.0	3.1	2.3	3.4
Beach parking	3.4	3.2	3.8	2.8	3.5
Residential access	3.9	4.0	4.2	4.0	3.3
Sidewalks	4.1	3.9	4.5	4.9	4.1
Commercial access	4.2	4.2	4.3	4.1	4.1
2) Indicate the desirability of each item (5 is most desirable)					
Bicycle	3.8	3.7	3.9	3.9	3.9
Historic building preservation	3.8	4.0	3.6	4.1	3.4
Improve intersections	3.7	3.8	3.6	4.0	3.8
Handicap/elderly transportation	3.6	3.6	3.6	4.0	3.7
Trees and shrubs	3.6	3.5	3.7	3.6	3.5
Sidewalks	3.0	3.2	2.5	2.7	2.9
Bus transit	2.9	3.0	2.6	3.6	3.0
Commercial access	2.8	2.9	2.6	3.2	2.9
Taxi service	2.6	2.7	2.5	2.8	2.6
Carpool/Vanpool	2.5	2.5	2.4	2.9	2.5
3a) Which would you support to promote carpooling/vanpooling?					
Restricted parking	49%	39%	67%	81%	60%
Lack of new roads	17%	13%	25%	19%	25%
High parking rates	15%	12%	24%	19%	17%
3b) Compared to other factors, should environmental factors be...					
equally important	49%	51%	44%	48%	51%
more important	41%	38%	49%	41%	38%
moderately important	10%	11%	8%	11%	8%
3c) Should widening be considered for residential streets with heavy traffic?					
yes	54%	53%	50%	73%	57%
no	46%	47%	50%	27%	45%
3d) How often do you use the Wright Memorial Bridge?					
less than once a week	69%	81%	59%	41%	48%
at least once a week	26%	18%	35%	54%	35%
daily	5%	2%	6%	4%	18%

	T	NH	KDH	KH	SS
3e) Would you favor bridge tolls for improving access from Currituck Co.?					
yes	53%	54%	54%	58%	48%
no	46%	46%	46%	42%	52%
3f) As funding is limited, would you prefer...					
fewer but first class improvements	69%	66%	75%	67%	72%
more but less expensive imprvmnts.	31%	34%	25%	33%	28%
3g) Are you a permanent resident of...					
Dare County	39%	30%	43%	89%	50%
Currituck County	2%	2%	1%	0%	16%
Other NC County	8%	13%	4%	4%	0%
Other	50%	57%	52%	7%	38%
Range of number of people responding					
high	1054	588	280	46	133
low	916	543	184	31	113

Note: Of the 493 people listing "Other" as the place of residence, only 49% were in favor of tolls to improve bridge access.

SURVEY RESPONSES
KILL DEVIL HILLS

KDH01 We are very concerned about the safety of pedestrians trying to cross NC 12 (Duck Road) to get to the beach. Vehicles travel 40-60 mph with not stop signs, stop lights or anything to slow them down while traveling through Southern Shores. Two or three crossover bridges might well save lives. Do keep in mind that "going to the beach" is the very reason people are in Southern Shores during their vacations and relaxation times.

KDH04 Duck Road shoulder improvements needed for safety of joggers and bicycles - also better speed enforcement needed during summer season.

KDH05 Another access to the northern beaches on the Outer Banks should be the #1 priority of the Planning Board not only for traffic congestion but more importantly for the safety of residents and tourists in case of a needed evacuation. Two access bridges is not enough for the area and population covered, especially with the growth taking place.

KDH07 At present we do not permanently reside in North Carolina. We do plan, however, to reside or build a summer residence on our land in the future.

KDH08 I don't feel that we should alter our environment and life style for the three month pulse of tourists that come to the Outer Banks only in the summer time. We should strive to protect the Outer Banks not continue to pave and develop what is left.

KDH10 Add another bridge to access northern areas above Duck.

KDH13 Take a lesson from Carmel, CA. Stop working so hard to make Outer Banks into Myrtle Beach. There are benefits to be gained from exclusiveness.

KDH15 We need more bridges and complete four lane to Virginia. It will be criminal to experience a loss of lives in the event of a major storm in order to motivate the powers that be to approve these projects.

KDH23 Avoid much change - no new highways. Keep Southern Shores quaint. Traffic congestion will in itself limit and discourage excessive travel.

KDH24 It is important to me that the area not turn into a Fort Lauderdale, Daytona Beach, or even Virginia Beach. One consideration is, by limiting access, you would also limit or at least slow growth. My feeling is that any transportation improvements should serve to improve the quality of life in the area and not be a means to accelerate growth to repeat the cycle later.

NAGS HEAD

NH12 If lights are put at important intersections for pedestrian crossings. Only if property owners are paying much less.

NH14 More beach access and foot paths to beach.

NH30 Would prefer to see less rapid growth and development. Divert traffic away from residential areas.

NH31 The beach road is too narrow, poorly marked, and very dangerous. It should be improved on a top priority basis.

NH38 In my opinion priority should be given to the bridge systems connecting The Banks to the mainland.

NH39 Tolls with a lower rate available for locals than tourists-something like a 1 year sticker. Please keep bikes off roads and bridges, they are a hazard to everyone on the road-if you give a bike lane make it extra wide so they can stay clear of traffic.

NH42 Ambulance: Have emergency ambulance service cost free to the people of the Outer Banks.

NH53 Do not think carpooling/vanpooling will be a viable solution. People will not pool.

NH61 High parking rates for non-residents.

NH73 I think that after the last 2 storms we have more important things to worry about-beach erosion and ocean front building. We may have to use tax dollars to build up our beaches. This would be a problem for some tax payers to accept. But the fact is our economy is based on tourism and its the beach that brings people here. We also need to look at our sewage problem-how much more building can we do without having a sewage treatment plant? I am also tired of people driving on our dunes and driving too fast on the beach. Some people enter anywhere they want. A few people spoil it for others. I would suggest having fewer beach accesses, having gates at non-access areas (but open for ped.), making beach permits, especially non-residents, more difficult to obtain. Maybe they already are, I've never applied. We really need more supervision on the beaches. Perhaps the NC Beach Buggy group would be willing to patrol the area. I do believe that certain areas should be closed to traffic at all times and at high tide for all areas. The fines for breaking the laws should be very high. I think that there should be a toll on the Wright Memorial Bridge-people living or working here should have a pass. There should be a toll on the Mann's Harbor Bridge also. Could we stop people who are obviously drunk from coming to Dare County? We do need several municipal pools at least one being indoors for year round swimming. There are many times that the beaches are closed and pools would be a welcome thing.

NH74 I think this is a seasonal problem (carpooling). I don't feel that year round growth supports this.

SOUTHERN SHORES

SS01 I am a 66 year old male, retired disabled military veteran (50%), and a permanent resident of Dare Co. The Outer Banks development, with the possible exception of Southern Shores, has been controlled by myopic elected officials influenced primarily by high roller speculators and fast buck developers. Recently much progress has been made with respect to recognizing that we are at risk. In my judgement, we should have a county type government, to avoid costly duplication of services and facilities. We should have taxes re-oriented to insure the tourists pay their fair share, when they use our facilities. Primary concerns are roads and bridges for emergency evacuation. AFTER they are in place, secondary concerns should be hospital facilities, educational facilities and much much further down the line should be listed such items as bicycles, bushes, transit etc. services. QUESTION: HOW MANY WHEEL CHAIRS HAVE YOU SEEN ON THE BEACH?

SS03 Each town has there own priorities which are not coordinated at the county level.

SS24 I do not consider the taking of homes in the same category as cost, convenience, etc.

SS34 Have a county wide lottery to raise the money for improvements rather than tax money.

SS42 Toll charging of persons employed on the Outer Banks should be limited to an annual fee.

SS43 Don't charge the local people that use the bridge daily.

SS44 No toll charge for local residents.

SS47 Cost of street building and improvement should be from current taxes and new roads from those who will benefit-not from a bond!

SS48 My husband and I own a house (not a rental) in Southern Shores. We have been enriched by staying in SS since 1966, and we are sad to see what was once a quiet retreat, becoming another Virginia Beach. We strongly believe that protection of the environment - protection of the very special qualities provided in the SS community should be fought for.

SS51 Stop excessive growth in population, government, employees, and underground septics. We have too many Real Estate developers and sales offices, and tourists, all of whom take array the good and leave problems behind. Who needs more restaurants, crowded roads and beaches?

OTHER

OT02 We need a new road! Needs to be higher, wider, straighter, and stronger. NOW!

OT04 I would like to see a bicycle and pedestrian lane clearly marked with painted pictures of bicycles on the shoulder. Also traffic signs indicating pedestrian and bike traffic. At present walking or biking is hazardous and the road to Collington is inadequate to support the daily flow of traffic.

OT07 First priority is bridges - Collington bridge is a disaster waiting to happen - collapse is imminent. Wright Memorial needs the same number of lanes as 158 to the north and to the south. These are safety concerns, not amenities. We are talking about lives, not just alleviating a nuisance saving.

APPENDIX D

APPENDIX D

RECOMMENDED SUBDIVISION ORDINANCES

DEFINITIONS:

I. Streets and Roads:

A. Urban Streets

1. Major Thoroughfares - Major thoroughfares consist of Interstate, other freeway, expressway, or parkway roads, and major streets that provide for the expeditious movement of high volumes of traffic within and through urban areas.
2. Minor Thoroughfares - Minor thoroughfares perform the function of collecting traffic from local access streets and carrying it to the major thoroughfare system. Minor thoroughfares may be used to supplement the major thoroughfare system by facilitating minor through-traffic movements and may also serve abutting property.
3. Local Street - A local street is any street not on a higher order urban system and serves primarily to provide direct access to abutting land.

B. Specific Type Urban Streets

1. Freeway, expressway, or parkway - (Major Thoroughfare)
Divided multilane road-ways designed to carry large volumes of traffic at high speeds. A freeway provides for continuous flow of vehicles with no direct access to abutting property and with access to selected crossroads only by way of interchanges. An expressway is a facility with full or partial control of access and generally with grade separations at major inter-sections. A parkway is a for non-commercial traffic, with full or partial control of access.
2. Major Thoroughfare - A street which provides for higher volume and longer distance travel. The primary function of a major thoroughfare is to serve travel.
3. Minor Thoroughfare - A local street which serves as a connector street between local residential streets and the major thoroughfare system. Serves both travel and property access functions.
4. Local Residential Street - Cul-de-sacs, loop streets less than 2,500 feet in length, or streets less than one mile in length that do not connect thoroughfares, or serve major traffic generators, and do not collect traffic from more than 100 dwelling units.

5. Cul-de-sac - (Local Street) A short street having only one end open to traffic and the other end being permanently terminated and a vehicular turn-around provided.
6. Frontage Road - (Local Street) A road that is parallel to a partial or full access controlled facility and provides access to adjacent land.

II. Property

- A. Building Setback Line - A line parallel to the street in front of which no structure shall be erected.
- B. Easement - A grant by the property owner for use by the public, a corporation, or person(s), of a strip of land for a specific purpose.
- C. Lot - A portion of a subdivision, or any other parcel of land, which is intended as a unit for transfer of ownership or for development or both. The word "lot" includes the words "plat" and "parcel".

III. Subdivision

- A. Subdivider - Any person, firm, corporation or official agent thereof, who subdivides or develops any land deemed to be a subdivision.
- B. Subdivision - All divisions of a tract or parcel of land into two or more lots, building sites, or other divisions for the purpose, immediate or future, of sale or building development and all divisions of land involving the dedication of a new street or change in existing streets; provided, however, that the following shall not be included within this definition nor subject to these regulations: (1) the combination or recombination of portions of previously platted lots where the total number of lots is not increased and the resultant lots are equal to or exceed the standards contained herein; (2) the division of land into parcels greater than ten acres where no street right-of-way dedication is involved, (3) widening or opening of streets; (4) the division of a tract in single ownership whose entire area is no greater than two acres into not more than three lots, where no street right-of-way dedication is involved and where the resultant lots are equal to or exceed the standards contained herein.
- C. Dedication - A gift, by the owner, of his property to another party without any consideration being given for the transfer. The dedication is made by written instrument and is completed with an acceptance.
- D. Reservation - Reservation of land does not involve any transfer of property rights. It constitutes an obligation to keep property free from development for a stated period of time.

DESIGN STANDARDS

I. Streets and Roads

The design of all roads within _____* shall be in accordance with the accepted policies of the North Carolina Department of Transportation, Division of Highways, as taken or modified from the American Association of State Highway Officials' (AASHTO) manuals.

The provision of street rights-of-way shall conform and meet the recommendations of the Thoroughfare Plan, as adopted by _____*.

The proposed street layout shall be coordinated with the existing street system of the surrounding area. Normally the proposed streets should be the extension of existing streets if possible.

- A. Right-of-way Widths - Right-of-way (ROW) widths shall not be less than the following and shall apply except in those cases where ROW requirements have been specifically set out the Thoroughfare Plan.

1. Rural	Min. ROW
a. Principle Arterial	
Freeways	350 ft.
Other	200 ft.
b. Minor Arterial	100 ft.
c. Major Collector	100 ft.
d. Minor Collector	70 ft.
e. Local Road	60 ft.**
2. Urban	
a. Major Thoroughfare other than Freeway and Expressway	80 ft.
b. Minor Thoroughfare	70 ft.
c. Local Street	60 ft.**
d. Cul-de-sac	Variable***

* The governing body, such as "Kill Devil Hills", "Kitty Hawk", "Nags Head", "Southern Shores", or "Dare County"

The subdivider will only be required to dedicate a maximum of 100 feet of right-of-way. In cases where over 100 feet of right-of-way is desired, the subdivider will be required only to reserve the amount in excess of 100 feet. On all cases in which right-of-way is sought for a fully controlled access facility, the subdivider will only be required to make a reservation. It is strongly recommended that subdivisions

provide access to properties from internal streets, and that direct property access to major thoroughfares, principle and minor arterials, and major collectors be avoided. Direct property access to minor thoroughfares is also undesirable.

A partial width right-of-way, not less than sixty feet in width, may be dedicated when adjoining undeveloped property that is owned or controlled by the subdivider; provided that the width of a partial dedication be such as to permit the installation of such facilities as may be necessary to serve abutting lots. When the said adjoining property is subdivided, the remainder of the full required right-of-way shall be dedicated.

- B. Street Widths - Widths for street and road classifications other than local shall be as recommended by the Thoroughfare Plan. Width of local roads and streets shall be as follows:

1. Local Residential

Curb and Gutter section: 26 feet, face to face of curb
Shoulder section: 20 feet to edge of pavement, 4 foot
shoulders

2. Residential Collector

Curb and Gutter section: 37 feet, face to face of curb
Shoulder section: 22 feet to edge of pavement, 6 foot
shoulders

- C. Geometric Characteristics - The standards outlined below shall apply to all subdivision streets proposed for addition to the State Highway System or Municipal Street System. In cases where a subdivision is sought adjacent to a proposed thoroughfare corridor, the requirements of dedication and reservation discussed under Right-of-Way shall apply.

** The desirable minimum right-of-way (ROW) is 60 ft. If curb and gutter is provided, 50 feet of ROW is minimum acceptable on local residential streets.

*** The ROW dimension will depend on radius used for vehicular turnaround. Distance from edge of pavement of turnaround to ROW should not be less than distance from edge of pavement to ROW on street approaching turnaround.

1. Design Speed - The design speed for a roadway should be a minimum of 5 mph greater than the posted speed limit. The design speeds for subdivision type streets shall be:

TABLE D-1			
DESIGN SPEEDS			
Facility Type	Desirable	Design Speed	
		Level	Minimum Rolling
RURAL			
Minor Collector Roads	60	50	40
Local roads including Residential Collectors and Local Residential	50	50*	40*
URBAN			
Major Thoroughfares other than Freeway or Expressway	60	50	50
Minor Thoroughfares	60	50	40
Local Streets	40	40**	30**

2. Maximum and Minimum Grades

a. The maximum grades in percent shall be:

TABLE D-2		
MAXIMUM VERTICAL GRADE		
Design Speed	Terrain	
	Level	Rolling
60	4	5
50	5	6
40	6	7
30		9

b. Minimum grade should not be less than 0.5% .

* Based on projected annual average daily traffic of 400-750. In cases where road will serve a limited area and small number of dwelling units, minimum design speeds can be reduced further.

** Based on projected annual average daily traffic of 50-250.

- c. Grades for 100 feet each way from intersections (measured from edge of pavement) should not exceed 5%.
 - d. For streets and roads with projected annual average daily traffic less than 250, short grades less than 500 feet long, may be 150% of the value in the above table.
3. Minimum Sight Distance - In the interest of public safety, no less than the minimum sight distance applicable shall be provided. Vertical curves that connect each change in grade shall be provided and calculated using the following parameters:

TABLE D-3				
SIGHT DISTANCE				
Design Speed	30	40	50	60
Stopping Sight Distance				
Minimum (ft.)	200	275	400	525
Desirable Minimum (ft.)	200	325	475	650
Minimum K* Value for:				
Crest curve	30	80	160	310
Sag curve	40	70	110	160

(General practice calls for vertical curves to be multiples of 50 feet. Calculated lengths shall be rounded up in each case.)

Sight distance provided for stopped vehicles at intersections should be in accordance with "A Policy on Geometric Design of Highways and Streets, 1984".

4. The "Superelevation Table" below shows the maximum degree of curve and related maximum superelevation for design speeds. The maximum rate of roadway superelevation (e) for rural roads with no curb and gutter of 0.08. The maximum rate of superelevation for urban streets with curb and gutter is 0.08, with 0.04 being desirable in the Central Business District.

* K is a coefficient by which the algebraic difference in grade may be multiplied to determine the length in feet of the vertical curve which will provide the desired sight distance.

TABLE D-4			
SUPERELEVATION TABLE			
Design Speed	Maximum e*	Minimum Radius ft.	Max. Deg. of Curve.
30	0.04	302	19 00'
40	0.04	573	10 00'
50	0.04	955	6 00'
60	0.04	1,528	3 45'
30	0.06	273	21 00'
40	0.06	509	11 15'
50	0.06	849	6 45'
60	0.06	1,380	4 15'
30	0.08	252	22 45'
40	0.08	468	12 15'
50	0.08	764	7 30'
60	0.08	1,206	4 45'

e* = rate of roadway superelevation, foot per foot

D. Intersections

1. Streets shall be laid out so as to intersect as nearly as possible at right angles, and no street should intersect any other street at an angle less than sixty-five (65) degrees.
2. Property lines at intersections should be set so that the distance from the edge of pavement, of the street turnout, to the property line will be at least as great as the distance from the edge of pavement to the property line along the intersecting streets. This property line can be established as a radius or as a sight triangle. Greater offsets from the edge of pavement to the property lines will be required, if necessary, to provide sight distance for the stopped vehicle on the side street.
3. Off-set intersections are to be avoided. Intersections which cannot be aligned should be separated by a minimum length of 200 feet between survey centerlines.

E. Cul-de-sacs

Cul-de-sacs shall not be more than seven hundred (700) feet in length. the distance from the edge of pavement on the vehicular turnaround to the right-of-way line should not be less than the distance from the edge of pavement to right-of-

way line on the street approaching the turnaround. Cul-de-sacs should not be used to avoid connection with an existing street or to avoid the extension of an important street.

F. Permits For Connection To State Roads

An approved permit is required for connection to any existing state system road. This permit is required prior to any construction on the street or road. The application is available at the office of the District Engineer of the Division of Highways.

G. Offsets To Utility Poles

Poles for overhead utilities should be located clear of roadway shoulders, preferably a minimum of at least 30 feet from the edge of pavement. On streets with curb and gutter, utility poles shall be set back a minimum distance of 6 feet from the face of curb.

H. Wheel Chair Ramps

All street curbs being constructed or reconstructed for maintenance purposes, traffic operations, repairs, correction of utilities, or altered for any reason, shall provide wheelchair ramps for the physically handicapped at intersections where both curb and gutter and sidewalks are provided and at other major points of pedestrian flow.

I. Horizontal Width on Bridge Deck

1. The clear roadway widths for new and reconstructed bridges serving 2 lane, 2 way traffic should be as follows:

- a. Shoulder section approach

- i. Under 800 ADT design year

- Minimum 28 feet width face to face of parapets of rails or pavement width plus 10 feet, whichever is greater.

- ii. 800 - 2000 ADT design year

- Minimum 34 feet width face to face of parapets of rails or pavement width plus 12 feet, whichever is greater.

- iii. Over 2000 ADT design year

- Minimum width of 40 feet, desirable width of 44 feet width face to face of parapets or rails.

b. Curbs and gutter approach

i. Under 800 ADT design year

Minimum 24 feet face to face of curbs.

ii. Over 800 ADT design year

Width of approach pavement measured face to face of curbs.

Where curb and gutter sections are used on roadway approaches, curbs on bridges shall match the curbs on approaches in height, in width of face to face of curbs, and in crown drop. The distance from face of curb to face of parapet or rail shall be 1'6" minimum, or greater if sidewalks are required.

2. The clear roadway widths for new and reconstructed bridges having 4 or more lanes serving undivided two-way traffic should be as follows:

- a. Shoulder section approach - Width of approach pavement plus width of usable shoulders on the approach left and right. (Shoulder width 8' minimum, 10' desirable.)
- b. Curb and gutter approach - Width of approach pavement measured face to face of curbs.

APPENDIX E



APPENDIX E

STREET TABULATION

The Street Tabulation consists of the facilities listed in in Chapter VII, Recommendations. They are listed in the same order as they appear in Chapter VII with base year and future year traffic, and the recommended cross section for each street.

Definitions

Practical Capacity: Volume of traffic a roadway can at Level of Service D.

2010 ADT: Average summer weekday traffic (2010) on existing system + 3rd bridge to Corrola

ADQ Adequate

N/A Not applicable

ROW Right of Way

Street - Section Reference	LENGTH (mi.)	EXISTING RDWAY (Ft)	EXIST. ROW	PRACT. CAPACITY	1988 ADT	2010 ADT	RECOMMENDED CROSS-SECTION
US 64:							
Roanoke Sound Bridge - US 158	1.04	48	N/A	54,000	18,500	37,800	ADQ
US 158:							
WCL KITTY HAWK -							
1.5m E of WCL K.H.	1.50	55	180	33,500	20,000	33,800	72-E ADQ
1.5m E of WCL K.H. -							
NCL K.D.H.	3.63	60	150	28,000	22,000	45,200	72-E ADQ
NCL K.D.H. - NCL NAGS HEAD	4.69	60	150	28,000	30,000	67,500	72-E ADQ
NCL NAGS HEAD - US 64	6.74	60	150	28,000	18,300	48,000	72-E ADQ
NC 12:							
CURRITUCK CO - 3.7m S of							
COUNTY LINE	3.70	26	60	12,500	4,100	8,000	ADQ
3.7m S of CURR. CO. -							
DUCK	1.29	20	60	10,500	8,500	15,900	36-I ADQ
DUCK - WCL SOUTHERN SHORES	0.92	20	60	10,500	16,000	21,600	36-I ADQ
WCL of SOUTH. SH. - 3.74m							
S of WCL SOUTH. SHORES	3.74	20	60	10,500	14,500	26,200	36-I ADQ
3.74m S of WCL S.S. - US 158	0.19	24	50	12,000	16,000	23,100	36-I ADQ
US 158 - .16m S of US 158	0.16	30	60	12,000	4,000	11,200	TIP: U-2917
.16m S of US 158 - NCL K.D.H.	3.63	20	60	10,500	5,100	21,600	TIP: U-2917
NCL K.D.H. - NCL NAGS HEAD	4.68	20	60	10,500	13,000	25,400	ADQ
NCL NAGS HEAD - US 64	6.60	24	60	12,000	13,200	19,200	ADQ

Street - Section Reference	LENGTH (mi.)	EXISTING RDWAY (Ft)	EXIST. ROW	PRACT. CAPACITY	1988 ADT	2010 ADT	RECOMMENDED CROSS-SECTION
KILL DEVIL HILLS							
OCEAN BAY BOULEVARD: SR 1217							
NC 12 - US 158	0.20	24	--	12,000	6,500	-----	36-I 60
US 158 - .3mi WEST OF US 158	0.30	48	--	20,000	7,000	23,500	60-D 90
.3mi WEST OF US 158 - .88mi							
WEST OF US 158	0.58	36	--	16,000	7,000	23,500	ADQ ADQ
.88mi WEST OF US 158 - K.D.H							
CITY LIMITS	0.70	24	--	12,000	5,000	18,000	36-I 60
OCEAN ACRES DRIVE:							
NC 12 - US 158	0.20	20	--	10,500	4,650	-----	36-I 60
MARTIN STREET:							
NC 12 - US 158	0.20	20	--	10,500	---	-----	36-I 60
PROSPECT AVENUE:							
NC 12 - US 158	0.20	20	--	10,500	---	-----	36-I 60
FIRST STREET:							
NC 12 - FIRST FLIGHT LANE	0.48	20	--	10,500	2,700	-----	36-I 60
FIRST FLIGHT LANE - CANAL STREET	0.24	20	--	10,500	2,700	-----	24-L 60
CANAL STREET:							
FIRST STREET - BAY DRIVE	0.27	18	--	8,000	---	-----	24-L 60
BAY DRIVE:							
CANAL STREET - SPORTSMAN DRIVE	0.85	18	--	8,000	---	-----	24-L 60
SPORTSMAN DRIVE - HAYMAN BOULEVARD	0.30	18(dirt)	--	8,000	---	-----	24-L 60
HAYMAN BOULEVARD - KILL DEVIL HILLS							
CITY LIMITS	0.80	18	--	8,000	---	-----	24-L 60

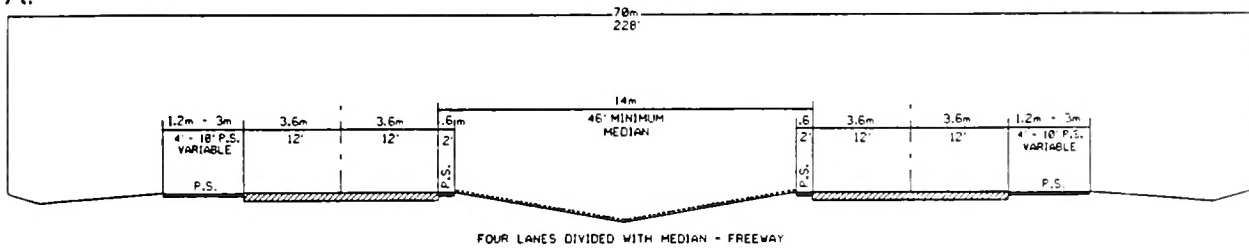
Street - Section Reference	LENGTH (mi.)	EXISTING RDWAY (Ft)	EXIST. ROW	PRACT. CAPACITY	1988 ADT	2010 ADT	RECOMMENDED CROSS-SECTION
THIRD STREET:							
NC 12 - US 158	0.25	24	--	12,000	2,300	-----	36-I 60
FIFTH STREET:							
NC 12 - US 158	0.20	24	--	12,000	---	-----	36-I 60
US 158 - BAY DRIVE	0.61	20	--	10,500	---	-----	24-L 60
AYCOCK STREET:							
NC 12 - US 158	0.20	20	--	10,500	---	-----	36-I 60
HELGA STREET:							
NC 12 - US 158	0.20	20	--	10,500	---	-----	36-I 60
US 158 - BAY DRIVE	0.36	20	--	10,500	---	-----	24-L 60
KITTY HAWK							
KITTY HAWK ROAD: SR 1206							
US 158 - WOODARD ROAD	0.10	24	--	12,000	4,900	25,300	60-D 90
WOODARD ROAD - WOODS ROAD	1.54	20	60	10,500	---	-----	36-I ADQ
WOODS ROAD: SR 1206							
US 158 - KITTY HAWK ROAD	2.06	20	60	10,500	700	11,000	36-I ADQ
TATEWAY ROAD:							
US 158 - RABBIT HOLLOW DRIVE	0.10	20	--	10,500	---	-----	36-I 60
RABBIT HOLLOW DRIVE - VELWAY ST	0.21	20	--	10,500	---	-----	24-L 60
VELWAY STREET:							
KILL DEVIL HILLS CITY LIMITS - TATEWAY ROAD	0.14	18	--	8,000	---	-----	24-L 60

Street - Section Reference	LENGTH (mi.)	EXISTING RDWAY (Ft)	EXIST. ROW	PRACT. CAPACITY	1988 ADT	2010 ADT	RECOMMENDED CROSS-SECTION
CAY TRAIL:							
(NEW LOCATION)							
NC 12 (LINDBERG AVE) - US 158	0.08	N/A	N/A	N/A	---	-----	36-I 60
BALCHEN STREET:							
NC 12 (LINDBERG AVE) - US 158	0.06	20	--	10,500	---	-----	36-I 60
ECKNER STREET:							
NC 12 (LINDBERG AVE) - US 158	0.06	20	--	10,500	---	-----	36-I 60
GRISSOM STREET:							
(NEW LOCATION)							
NC 12 (LINDBERG AVE) - US 158	0.06	N/A	N/A	N/A	---	-----	36-I 60
NAGS HEAD							
GULL STREET:							
NC 12 - US 158	0.09	20	--	10,500	---	-----	36-I 60
LAKE SIDE STREET:							
NC 12 - US 158	0.09	20	--	10,500	---	-----	36-I 60
SEACHASE DRIVE:							
NC 12 - US 158	0.06	20	--	10,500	---	-----	36-I 60
MALL DRIVE:							
NC 12 - US 158	0.09	20	--	10,500	2,800	-----	36-I 60
DANUBE AVENUE:							
NC 12 - US 158	0.09	20	--	10,500	1,850	-----	36-I 60

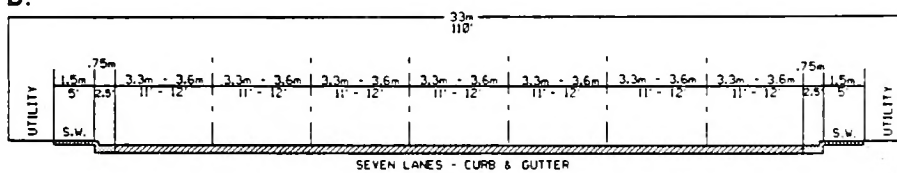
Street - Section Reference	LENGTH (mi.)	EXISTING RDWAY (Ft)	EXIST. ROW	PRACT. CAPACITY	1988 ADT	2010 ADT	RECOMMENDED CROSS-SECTION
DEERING STREET:							
(NEW LOCATION)							
NC 12 - US 158	0.09	N/A	N/A	N/A	---	-----	36-I 60
HOLLOWELL STREET:							
NC 12 - US 158	0.11	20	--	10,500	800	-----	36-I 60
VILLA DUNES DRIVE:							
(NEW LOCATION)							
BLADEN STREET - US 158	0.13	N/A	N/A	N/A	---	-----	36-I 60
BLADEN STREET:							
NC 12 - WRIGHTSVILLE AVENUE	0.13	18	--	8,000	---	-----	36-I 60
BARNES STREET:							
NC 12 - US 158	0.23	20	--	10,500	1,720	-----	36-I 60
WEST SATTERFIELD LANDING ROAD:							
(NEW LOCATION)							
NC 12 - US 158	0.19	N/A	N/A	N/A	---	-----	36-I 60
EIGHTH STREET:							
NC 12 - US 158	0.17	20	--	10,500	---	-----	36-I 60
SOUTHERN SHORES							
CHICAHAWK TRAIL:							
NC 12 - TRINITE TRAIL	0.59	18/20	--	9,000	---	-----	ADQ ADQ
TRINITE TRAIL:							
CHICAHAWK TRAIL - JUNIPER TRAIL	0.46	18/20	--	9,000	---	-----	ADQ ADQ
JUNIPER TRAIL:							
TRINITE TRAIL - US 158	0.91	18/20	--	9,000	---	-----	ADQ ADQ

TYPICAL THOROUGHFARE CROSS SECTIONS

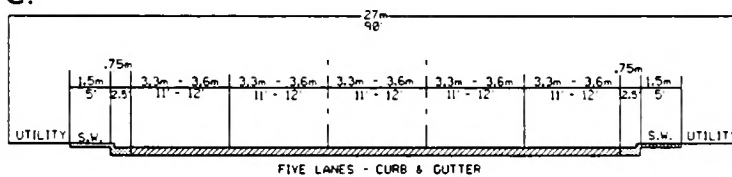
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B.



C.



D.

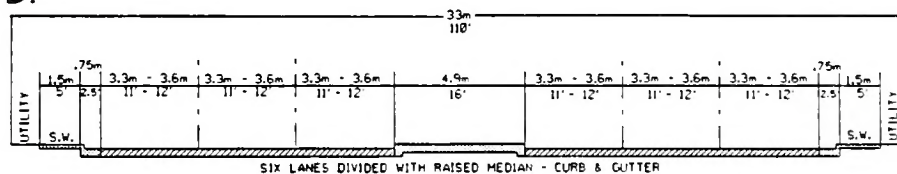
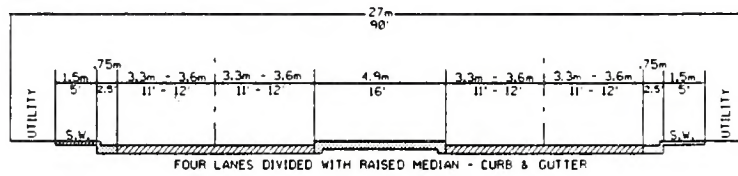


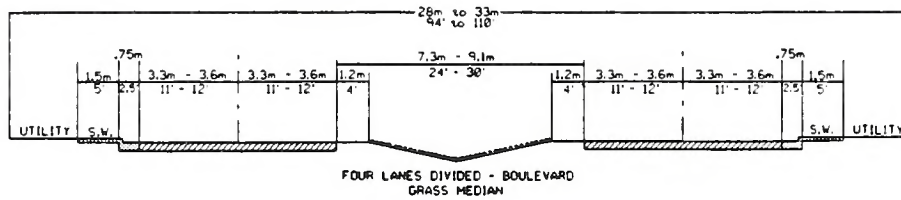
FIGURE E-1

TYPICAL THOROUGHFARE CROSS SECTIONS

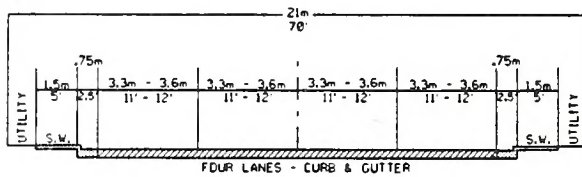
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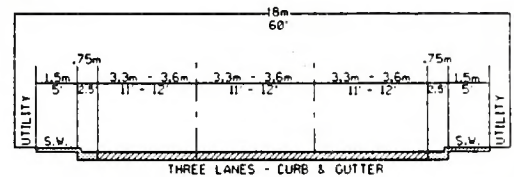
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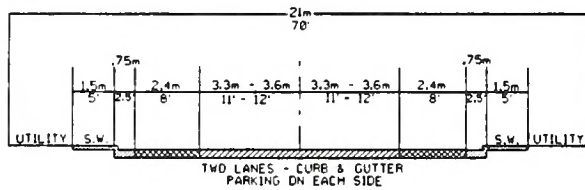
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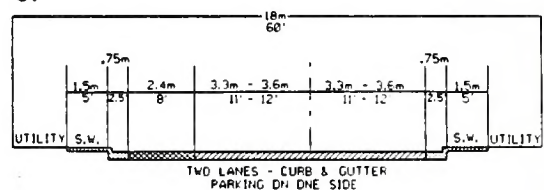
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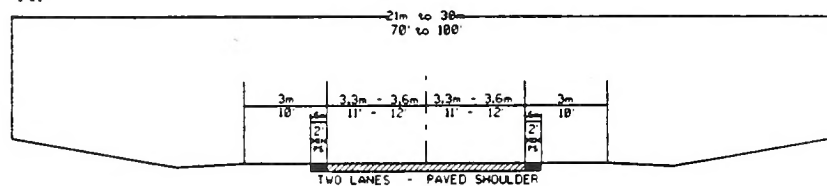
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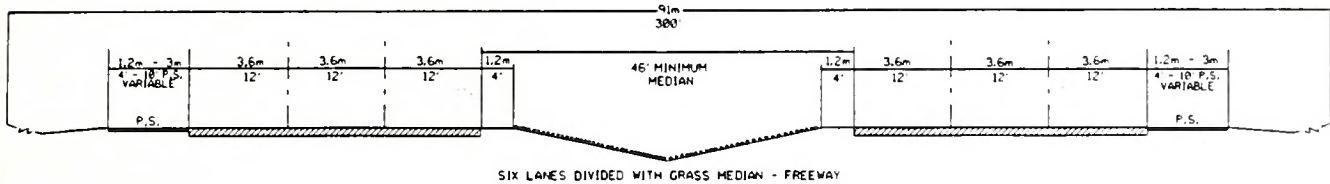


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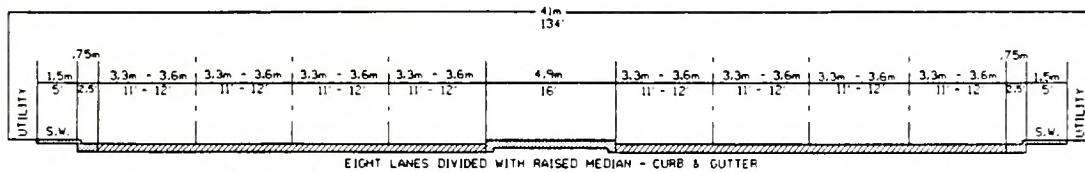


TYPICAL THOROUGHFARE CROSS SECTIONS

L.

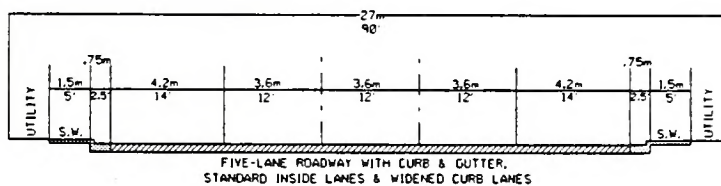


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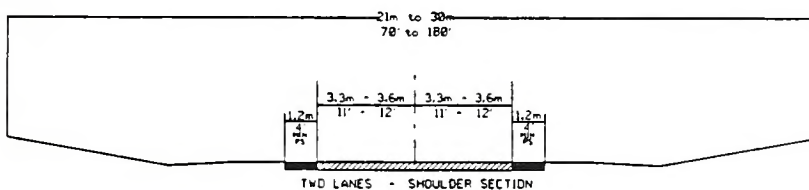


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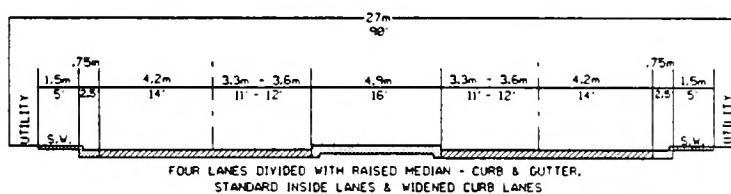
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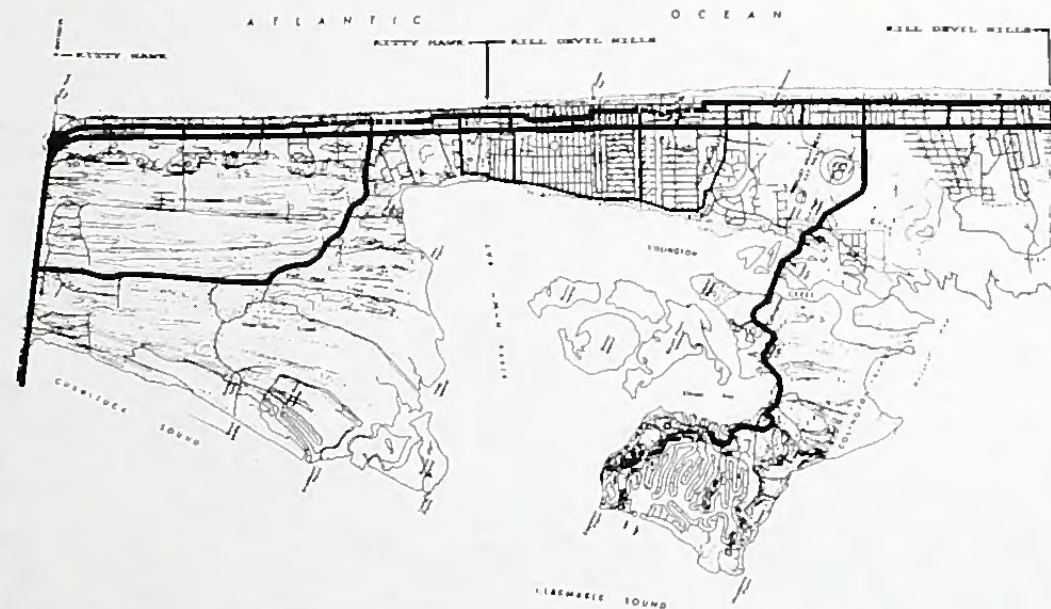
P.



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LEGEND:

EXISTING PROPOSED

MAJOR THOROUGHFARE

MINOR THOROUGHFARE

PARTIAL INTERCHANGE

REVISIONS:

DATE	LEFT	KILL DEVIL	SOUTHERN
REVISION	REVISION	REVISION	REVISION

ADOPTED BY:

TOWN OF NAGS HEAD

TOWN OF KITTY HAWK

TOWN OF KILL DEVIL HILLS

TOWN OF SOUTHERN SHORES

STATEWIDE PLANNING BRANCH

N.C. DEPARTMENT OF TRANSPORTATION

PUBLIC HEARING DATES

PROPOSED THOROUGHFARE PLAN FOR THE TOWNS OF NAGS HEAD, KILL DEVIL HILLS, KITTY HAWK AND SOUTHERN SHORES

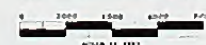
DARE COUNTY, NORTH CAROLINA

FEBRUARY, 1994

PREPARED BY THE NORTH CAROLINA DEPARTMENT OF TRANSPORTATION DIVISION OF HIGHWAYS-STATEWIDE PLANNING BRANCH

IN COOPERATION WITH

U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION



MAY 1992

FIGURE E-2

